

Précis of *The neuropsychology of anxiety: An enquiry into the functions of the septo-hippocampal system*

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Abstract: A model of the neuropsychology of anxiety is proposed. The model is based in the first instance upon an analysis of the behavioural effects of the antianxiety drugs (benzodiazepines, barbiturates, and alcohol) in animals. From such psychopharmacological experiments the concept of a “behavioural inhibition system” (BIS) has been developed. This system responds to novel stimuli or to those associated with punishment or nonreward by inhibiting ongoing behaviour and increasing arousal and attention to the environment. It is activity in the BIS that constitutes anxiety and that is reduced by antianxiety drugs. The effects of the antianxiety drugs in the brain also suggest hypotheses concerning the neural substrate of anxiety. Although the benzodiazepines and barbiturates facilitate the effects of γ -aminobutyrate, this is insufficient to explain their highly specific behavioural effects. Because of similarities between the behavioural effects of certain lesions and those of the antianxiety drugs, it is proposed that these drugs reduce anxiety by impairing the functioning of a widespread neural system including the septo-hippocampal system (SHS), the Papez circuit, the prefrontal cortex, and ascending monoaminergic and cholinergic pathways which innervate these forebrain structures. Analysis of the functions of this system (based on anatomical, physiological, and behavioural data) suggests that it acts as a comparator: it compares predicted to actual sensory events and activates the outputs of the BIS when there is a mismatch or when the predicted event is aversive. Suggestions are made as to the functions of particular pathways within this overall brain system. The resulting theory is applied to the symptoms and treatment of anxiety in man, its relations to depression, and the personality of individuals who are susceptible to anxiety or depression.

Keywords: antianxiety drugs; anxiety; arousal; ascending noradrenergic pathways; ascending serotonergic pathways; attention; comparator; depression; Papez circuit; personality; prefrontal cortex; septo-hippocampal system (SHS);

The neuropsychology of anxiety: An enquiry into the functions of the septo-hippocampal system (henceforth *Anxiety*; Gray 1982a) has a deliberately broken-backed title, held together by an ambiguous colon: it seeks to engage the attention of two largely separate populations – those interested in anxiety and in the septo-hippocampal system, respectively – and to convince them that they have major concerns in common. However, *Anxiety* does not claim that the septo-hippocampal system (SHS) mediates only anxiety, or that anxiety depends solely on that system. What it does claim is that the kinds of information-processing that go on in the SHS are of central importance in the state of anxiety; and that this hypothesis can offer a fresh and illuminating perspective on both the functions of the SHS and the nature of anxiety.

The major aim of the book is to delineate the psychology of anxiety and the neural basis on which it rests. It argues that this aim can be achieved by considering: (1) the behavioural changes caused in animals by drugs which are successfully used to combat anxiety in man (Rickels 1978); and (2) the likely neural route by which such antianxiety drugs cause these changes in behaviour. The drugs concerned are the benzodiazepines, the bar-

biturates, and alcohol; though chemically quite distinct from each other, these all produce similar changes in animal behaviour (Gray 1977). *Anxiety* assumes that a description of the psychological processes that are altered by these drugs is equivalent to a description of the psychology of anxiety; and that a description of the neural processes altered by them is equivalent to a description of the neurology of anxiety.

There are a number of important qualifications which must be attached to these assumptions. Here I shall mention only two. It is vital to distinguish those behavioural changes in animals (and their underlying neural processes) that reflect reduced anxiety from those that reflect other effects of the antianxiety drugs, for instance, their anticonvulsant, sedative, or muscle-relaxant properties (Tallman, Paul, Skolnick & Gallagher 1980); and it is vital to show that animals plausibly experience both (a) anxiety that is similar to human anxiety, and (b) relief of anxiety after treatment with antianxiety drugs.

It is possible along these lines to analyse the psychology and neurology of anxiety quite independently of each other, and indeed this has been the usual approach. But *Anxiety* is above all concerned with the illumination that

comes from harnessing the two types of analysis together. It tries to show, for example, that certain hypotheses which have emerged from purely neurochemical investigations are, in the light of known behavioural data, almost certainly too simplistic; and, conversely, that knowledge of the brain structures on which the antianxiety drugs act can greatly enrich psychological theories of anxiety derived from behavioural findings alone. In consequence, the theory developed in *Anxiety* is genuinely neuropsychological – not neural *and* psychological.

The behavioural effects of antianxiety drugs

The starting point of *Anxiety* is a purely psychological analysis of the effects of the antianxiety drugs. This part of the book is based on an extensive review of the behavioural effects of antianxiety drugs in experimental animals (Gray 1977). These effects are first analysed within the conceptual framework afforded by two-process learning theory (Gray 1975), which treats much of learning as deriving from the interaction between the two separate and fundamental processes of classical and instrumental conditioning. However, in part because it is difficult to fit certain patterns of drug effects into this framework, and in greater part in order to accommodate data from studies of the SHS, later parts of the book adopt a different theoretical approach. While the later approach is based upon different premises, it is compatible with two-process learning theory; and the perspective that this theory offers continues to be a valuable point of departure, since it allows a remarkably simple description of the behavioural effects produced by the antianxiety drugs.

This description is summarised in Figure 1. The diagram shows that the antianxiety drugs block the behavioural effects of three classes of stimuli: those associated with punishment, those associated with frustrative nonreward (Amsel 1962), and novel stimuli. All three classes of stimuli elicit three general kinds of behavioural change in undrugged animals: ongoing responses are inhibited, the level of arousal is increased (i.e., the next response is initiated with increased vigour), and there is increased attention to the environment. The antianxiety drugs block all these behavioural changes, irrespective of which class of stimulus elicits them. These conclusions are given detailed empirical justification in the book; here, as at similar later points in the *Précis*, there is no space to summarise the relevant review of the data.

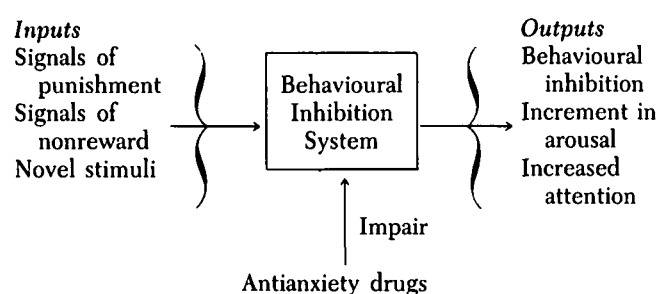


Figure 1. The behavioural inhibition system. This responds to any of its adequate inputs with all of its outputs, and comprises the hypothetical substrate on which the antianxiety drugs act to reduce anxiety.

These general findings lead to the postulation of a “behavioural inhibition system” (BIS), which (1) mediates the outputs shown to the right of Figure 1 upon receipt of any of the inputs listed to the left, and (2) upon which the antianxiety drugs act to reduce anxiety. *Ex hypothesi*, activity in the BIS constitutes anxiety. This is the model of anxiety proposed by Gray (1976; 1977). In later parts of the book, while this model is preserved, it is greatly expanded in the light of data from studies of the brain. It is important to note, however, that the BIS remains a *psychological* model, not itself to be directly identified with any brain structures or systems. Later it will be argued that activity in certain brain systems under certain conditions is equivalent to activity in the BIS; but there is only partial overlap between the two levels of theorising: these same brain systems, at other times and under other conditions, underlie psychological states other than anxiety.

The BIS model (Fig. 1) requires several comments.

1. Although entirely derived from animal experiments, it is plausible as a model of human anxiety. No one will be surprised to hear that anxiety is a response to threats of punishment, failure, or disappointment, or to exposure to a novel environment. This implies that animals do indeed experience anxiety, and that this state is alleviated by the antianxiety drugs.

2. The model appears to be equally applicable to all species in which the antianxiety drugs have been tested (from goldfish to chimpanzee). This implies, in agreement with data on the benzodiazepine receptor (Nielsen, Braestrup & Squires 1978), that anxiety is phylogenetically old and stable. Thus it is reasonable to search for the neural substrate of human anxiety in the brains of animals.

3. The model is equally applicable to all the major classes of antianxiety drugs. Thus any mode of physiological action which is applicable to one of these classes only cannot be an essential step in anxiety reduction.

4. Provided the dose is not too high, the behavioural effects of the antianxiety drugs are highly selective. For example, responses to punishment are unchanged, but responses to stimuli which warn of punishment are reduced; and similarly responses to nonreward are unchanged, but those to stimuli associated with nonreward are reduced (Feldon, Guillemon, Gray, De Wit & McNaughton 1979). This rules out any account of antianxiety drug action which assimilates this to sedation.

5. The action of the antianxiety drugs cannot be described in terms of a change in motor behaviour as such; depending on the direction of change in motor behaviour produced by anxiogenic stimuli (i.e., those listed to the left of Figure 1), these substances may either increase (e.g., Geller & Seifter 1960) or decrease (e.g., Wagner 1963) motor output. Nor can their action be described in terms of perceptual changes; for example, drugged animals correctly adjust their behaviour relative to the size of the reward, but they fail to show the characteristic undershoot (Crespi’s 1942 depression effect) when reward size is unexpectedly reduced (e.g., Rosen, Glass & Ison 1967). Finally, with the clinically important exception of learning to tolerate nonreward (Feldon et al. 1979) or punishment (Davis, Brookes, Gray & Rawlins 1981), learning is not affected by the antianxiety drugs, only performance. Furthermore, the responses whose expression is altered

by the antianxiety drugs are "learned" only in the sense that the animal must learn which stimuli to respond to in this way; the responses produced by the BIS do not themselves appear to be learned (Gray 1975). Thus, in agreement with clinical impression, the best description of the effect of the antianxiety drugs is that they temporarily change mood or disposition.

The search for the behavioural inhibition system in the brain

The antianxiety drugs provide not only the tool with which the BIS has been fashioned, but also the chief means of tracking its neural substrate in the brain. In describing this line of investigation I shall distinguish between two approaches, one direct and one indirect, to the problem of their mode of action.

The direct approach is favoured by biochemists and physiologists. They ask how the antianxiety drugs act on neurons: do they, for example, alter membranes, act on receptors, or interact with known transmitters?

This approach has recently been very productive, especially in the case of the benzodiazepines. It has been shown that these drugs bind strongly to a highly selective receptor which is located on neuronal membranes in the brain (Möhler & Okada 1977; Braestrup & Squires 1977). However, the significance of this demonstration is sharply limited by the fact that the benzodiazepine receptor does *not* bind barbiturates or alcohol (see point 3 in the previous section). The next step in the chain of events brings us onto stronger ground, however. Activation of the benzodiazepine receptor apparently affects a closely associated γ -aminobutyrate (GABA) receptor, increasing its affinity for GABA and thus facilitating the action of this inhibitory transmitter (Guidotti, Toffano & Costa 1978). Furthermore, the barbiturates share this tendency to facilitate the effects of GABA (Ransom & Barker 1976), perhaps by acting on a third ("picROTOXININ") receptor that is closely allied to the other two (Ticku & Olsen 1978). Thus the common action of the antianxiety drugs may lie in a facilitation of GABA-mediated inhibition in the brain (Costa & Greengard 1975).

The trouble with this hypothesis is that GABA, like the benzodiazepine receptor, is widely distributed in the brain and spinal cord. There is little reason to suppose that a general facilitation of its action would produce the very specific behavioural effects observed after treatment with antianxiety drugs (see point 4 in the previous section). On the contrary, such facilitation might be expected to produce sedative or anticonvulsant effects, both of which are observed after either benzodiazepine or barbiturate administration. Thus, though the antianxiety drugs may commence their action by facilitating GABA-ergic inhibition, it is necessary to find a more localised site at which their effects on brain function might be translated into reduced anxiety.

The indirect approach to the problem posed by the mode of action of the antianxiety drugs is complementary to the direct one. It looks for interventions in the brain which produce behavioural changes that are either systematically the same as, or systematically opposite to, those of the antianxiety drugs. This approach is central to the argument used in *Anxiety*. A key point in the argu-

ment is the similarity between the effects of the antianxiety drugs and those of lesions to the SHS (Gray 1970b).

Table 1 compares the common effects of the benzodiazepines, the barbiturates, and alcohol, on the one hand, and those of septal and hippocampal lesions, on the other (Gray & McNaughton 1982, in press). It can be seen, in the first instance, that the effects of septal and hippocampal lesions almost always resemble each other. Given the extensive anatomical and physiological connections between the two structures (Elliott & Whelan 1978), this is not surprising. What is surprising is that, wherever the effects of septal and hippocampal lesions are the same, those of the antianxiety drugs (if they are known) are (nearly always) likewise the same.

Many of the findings listed in Table 1 that concern the antianxiety drugs were used to construct the BIS model. Thus the similarities between the drug and lesion effects offer *prima facie* evidence that the SHS plays a part in mediating the functions of the BIS. However, a wider range of data has been gathered on septal and hippocampal animals than on animals treated with antianxiety drugs, and certain features of the septo-hippocampal syndrome may turn out to be absent from the drug syndrome. Indeed, there is already evidence that disruption of performance in the radial-arm maze, which is so profound after damage to the SHS (Olton 1978b), is only slight and transient after injection of chlordiazepoxide (Brookes, Rawlins, Preston, & Gray, unpublished observations). Thus the SHS has functions which may be spared after antianxiety drug treatment. Conversely, certain features of the drug syndrome are absent from the septo-hippocampal syndrome. Most important, the increased arousal elicited by anxiogenic stimuli in normal animals still appears to be present after damage to the SHS (Dickinson 1974), although not after drug treatment (Gray 1977).¹ We must conclude, then, that there is substantial but nonetheless only partial overlap between the BIS and the SHS (Gray, Feldon, Rawlins, Owen & McNaughton 1978): the SHS has other functions besides participation in the BIS, and the BIS depends on activity in other structures besides the SHS.

Theories of hippocampal function

If the SHS plays a key role in the BIS, we can gain insight into anxiety by asking what functions are discharged by the SHS. But at this point we come up against the problem that there is, to say the least, no agreement as to what these functions are. Table 2 summarises some of the different hypotheses that have been advanced. The sheer range of the psychological constructs in this table is remarkable, as is the fact that most of these hypotheses have been proposed largely in the light of only one corresponding source of data. More recently, attempts have been made to test theories against a wider range of data (e.g., O'Keefe & Nadel 1978; Olton, Becker & Handelmann 1979), and I have followed this welcome trend. [See also *BBS* multiple book review of O'Keefe & Nadel in *BBS* 4(2) 1979.]

In *Anxiety* I assume that each method of studying the functions of the SHS has some merit and can offer a guide, but only a limited guide, to the truth; correspondingly, most of the existing hypotheses have probably got some

Table 1. Comparison between the behavioural effects of septal (S) and hippocampal (H) lesions and antianxiety drugs

1 Task	2 Syndrome common to S and H lesions	3 Features in S, not H, syndrome	4 Features in H, not S, syndrome	5 Effects of antianxiety drugs
Rewarded running, CRF	0			0
Rewarded bar-pressing, CRF		+		0
Rewarded bar-pressing, intermittent reinforcement	+			+ ^a
Passive avoidance	—			—
Classical conditioning with aversive UCS	0			0
On-the-baseline conditioned suppression		—		—
Off-the-baseline conditioned suppression		?	?0	?0
Taste aversion		?	—	?
Agonistic escape	+			?
Skilled escape	0			0
One-way active avoidance		—		0
Non-spatial active avoidance	+			+
Two-way avoidance	+			+
Threshold of detection of shock	0			0
Movement elicited by shock	+			?
Movement elicited by novel stimuli	+			?
Distraction	—			?
General activity		?	+ ^b	?
Exploration of novel stimuli	—			—
Spontaneous alternation	— ^c			—
Open-field ambulation		—/+ ^d	+	+/— ^d
Rearing	—			—
Habituation rate	?—			?
Open-field defecation		—		?
Emergence time		—	?	—
Shock-induced aggression		+	—	? ^e
Hyperreactivity syndrome		+		0
Social aggression	—			?+ ^e
Social interaction		+	?	?+ ^f
Resistance to extinction	+			+
Partial reinforcement extinction effect	—			— ^g
Partial reinforcement acquisition effect		?0 ^h	?	—
Performance on DRL schedule	—			—
Fixed-interval scallop		—	?+	—
Simultaneous discrimination	0			0
Successive discrimination	— ⁱ			—
Single alternation		?	?—	—
Spatial discrimination	—			?
Reversal learning	—			—
Double runway frustration effect	0			0
Crespi depression effect		?	?—	—

Note: This comparison is based on the information in chapters 2 and 6 of *Anxiety* together with the more detailed reviews given by Gray (1977) and Gray and McNaughton (1982, in press): +, facilitation; —, impairment; 0, no consistent change; ?, insufficient data. If an entry in columns 3 or 4 is accompanied by a question mark in the other column, the difference between the septal and hippocampal syndromes cannot be established because of insufficient data.

^aAn exception is the VI schedule, on which rates are not increased by antianxiety drugs but are increased by both lesions. ^bThe apparent difference between the effects of the two lesions may be an artefact of the available data, since both lesions increase activity in the shuttlebox and decrease activity in the running wheel. ^cIt is possible that septal lesions increase stimulus perseveration and hippocampal lesions, response perseveration; but this apparent difference has been questioned by McNaughton and Feldon (1980). ^dSeptal lesions reduce ambulation upon early exposure to the open field but increase ambulation with continued exposure; the pattern of change with antianxiety drugs is the reverse. ^eIn this case the question mark indicates, not insufficient data, but major inconsistencies in the data (see Gray 1977). ^fThe antianxiety drugs increase social interaction only if this is suppressed by strong stimulation or environmental novelty (File 1980). ^gAntianxiety drugs are relatively ineffective with short intertrial intervals, but the two lesions have very large effects under these conditions (Feldon et al. 1979; Feldon & Gray 1979a; 1979b; Rawlins et al. 1980). ^hFeldon and Gray (1979a; 1979b). ⁱSeptal lesions improve successive discrimination with olfactory cues.

Table 2. *Hypotheses of hippocampal function and their relation to experimental method*

Method	Hypothesis
Anatomy	Emotion (Papez 1937)
Clinical	Memory consolidation (Milner 1970)
Lesion	Inhibition (McCleary 1966; Douglas 1967; Kimble 1969)
Theta	Attention (Grastyán, Lissak, Madarasz, & Donhoffer 1959); arousal (Green & Arduini 1954); movement (Vanderwolf 1971)
Single-unit	Orienting reflex (Vinogradova 1975); space (O'Keefe & Nadel 1978)
Drug analogy	Anxiety (Gray 1970b)
Stimulation	Memory consolidation (Landfield 1976; Jaffard, Destrade, Darkin, & Ebel 1979); anxiety (Gray 1972a)
Recent lesion studies	Working memory (Olton et al. 1979); recognition memory (Caffan 1972); contextual retrieval (Hirsh 1974; Winocur 1982)
Recent clinical studies	Interference (Weiskrantz & Warrington 1975); multidimensional stimulus analysis (Butters & Germaek 1975)

things right, but other things wrong. Thus, in analysing afresh the functions of the SHS, I have attempted (a) to encompass as wide a range of data as possible, and (b) to distil from each existing theory of septo-hippocampal function the part best supported by the data which gave rise to it.

To achieve this purpose it was necessary to review *in extenso* a number of areas of data: the anatomy and physiology of the SHS; the behavioural effects of septal and hippocampal lesions; the relation between electrical activity in the SHS and behaviour; the behavioural effects of electrical stimulation of the SHS; and the data, clinical and experimental, which have been seen as indicating a role for the hippocampus in memory. Certain parts of these data, especially those relating to the behavioural effects of lesions, have been reviewed extensively before. Nonetheless, since I was not always able to agree with former interpretations, it was necessary to go over this ground carefully again in *Anxiety*. Clearly it is not possible to do that in this Précis as well. I shall therefore limit myself to the major conclusions reached in the book, selecting in particular those "pieces and patches" (Nadel & O'Keefe 1974) which either bear closely on previous theoretical positions or have been used in the construction of my own.

Anxiety's pieces and patches

A. Anatomy. The description of septo-hippocampal anatomy in *Anxiety* relies largely on secondary sources (e.g., O'Keefe & Nadel 1978; Swanson 1978), though these have been brought up to date with more recent findings where necessary. Key points emphasised are as follows.

1. The SHS, and the Papez circuit (subicular cortex → mammillary bodies → anterior thalamus → cingulate cortex → subicular cortex) with which it is closely associated, are superbly well equipped for talking to themselves: loops abound, entrances and exits are few (Fig. 2).

2. The subicular area occupies a nodal point in the disposition of these loops (Fig. 2). It receives information from the temporal lobe, both directly and via the entorhinal cortex (Van Hoesen, Rosene & Mesulam 1979); and information from area CA1 of the hippocampus, which again starts out in the entorhinal cortex but first traverses the hippocampal circuit (entorhinal cortex $\rightarrow$ dentate gyrus $\rightarrow$ CA3 $\rightarrow$ CA1). It then sends this information out along the fornix, only to have it come back once again after a long trip round the Papez circuit.

3. Among these loops the oddest is the one through the hippocampus itself. If we leave aside the cholinergic afferents to the hippocampal formation from the medial septal area and the monoaminergic afferents from the locus coeruleus (noradrenergic) and raphe nuclei (serotonergic), all of which appear to discharge nonspecific functions, the information that goes to the subicular area from the entorhinal cortex appears to have nothing added to it on the way round the hippocampal circuit. Thus the dentate granule cells have no known afferents other than those from the entorhinal area and a return projection from CA3; CA3 similarly receives projections only from the dentate gyrus and entorhinal cortex, and CA1 only from CA3 and the entorhinal cortex. Given that the entorhinal cortex projects in any case directly to the subicular area, one wonders what function is served by the addition of this longer route (see section B4, below).

B. Physiology. The following points are emphasised in *Anxiety*:

1. Passage of information round the hippocampal circuit appears to involve a "gate," probably located at the dentate-to-CA3 step. This gate has been observed in purely electrophysiological experiments, in which, in response to stimulation of points earlier in the hippocampal circuit, the impulses recorded at later points may

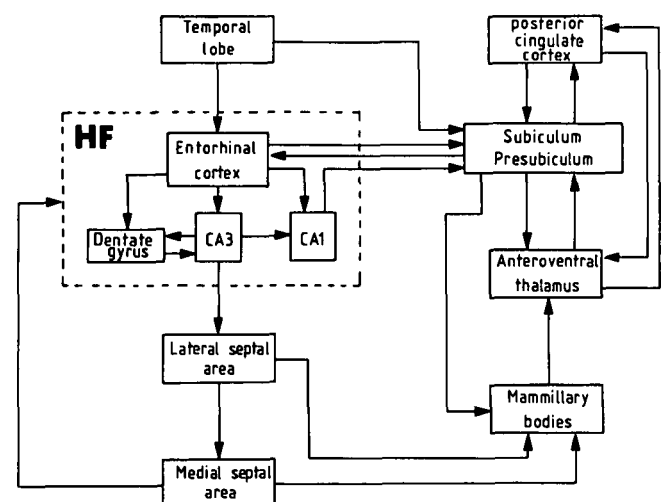


Figure 2. The loops that compose the septo-hippocampal system and the associated Papez circuit. The medial septal area projects to the whole of the hippocampal formation (HF), including the entorhinal cortex.

show either potentiation (Andersen 1978) or habituation (Alger & Teyler 1976), the former being favoured by higher frequencies of stimulation. More behaviourally oriented studies have shown that the response to a simple exteroceptive stimulus of no biological significance rapidly habituates (Vinogradova 1975), whereas the response to a stimulus that has been paired with reinforcement (positive or negative) augments (Segal 1977b; 1977d; Berger & Thompson 1978a).

2. Passage of information round the hippocampal circuit is facilitated by the noradrenergic and serotonergic innervation of the hippocampal formation (Segal 1977c; Assaf & Miller 1978; Assaf, Mason, & Miller 1979). This facilitation takes the form of an increased signal-to-noise ratio (McNaughton & Mason 1980; Segal 1980), that is, the spontaneous firing rate of hippocampal neurons is reduced but the response to inputs from the entorhinal cortex is increased. Segal's (1977b; 1977c; 1977d) experiments demonstrate that the influence of the two monoamines on the hippocampal response to stimulation closely resembles that of pairing a stimulus with reinforcement.

3. Vinogradova's (1975) and Thompson and Berger's (1978a; 1978b) experiments indicate that, after habituation or augmentation of unit responding has taken place in the hippocampus, the changed amplitude of response is passed on from CA3 to CA1 and the lateral septal area.

4. Vinogradova's (1975) experiments show that the unit response to simple sensory stimuli is less specific to the parameters of the stimulus in the dentate gyrus than in the entorhinal cortex. This adds to the puzzle mentioned in section A3, above: why should the entorhinal area send less specific information by a more circuitous route to the subicular cortex, to which it sends apparently more specific information directly?

5. Experiments in which single hippocampal neurons have been recorded in behaving animals have seemed to provide powerful support for O'Keefe and Nadel's (1978) spatial theory. However, a consideration of *all* the results reported in experiments of this kind leads to different conclusions (Gray 1979d). It is clear that place *can* be a powerful determinant of responses in hippocampal cells (O'Keefe & Dostrovsky 1971). But other experiments have demonstrated unit "fields" that can have nothing to do with space (e.g., Berger & Thompson 1978a). The only rule that will encompass all the data so far reported is that hippocampal units will pick up and fire to the regularities – *whatever they may be* – that the experimenter builds into the animal's environment. [See also BBS multiple book review of O'Keefe & Nadel's *The hippocampus as a cognitive map*, BBS 2(4) 1979.]

C. Hippocampal theta. It is now well established that the hippocampal theta rhythm is generated at two locations in the hippocampal formation (in CA1 and the dentate gyrus) (Winson 1976), and that these generators are under the control of pacemaker cells in the medial septal area (Petsche, Stumpf & Gogolak 1962). The functional significance of theta remains an enigma, however.

1. Studying the behavioural correlates of theta, Vanderwolf (e.g., Vanderwolf, Kramis & Robinson 1978) has shown that there are two types of theta. The first ("atropine-resistant") type is associated with movement; it is of relatively high frequency and is not affected by systemic

administration of anticholinergic drugs. The second ("atropine-sensitive") type is not associated with movement, is of relatively low frequency, and may be blocked by systemic anticholinergics. Graeff, Quintero, and Gray (1980) report that atropine-sensitive theta can be elicited by median raphe stimulation and that the animal freezes during this stimulation and shows clear signs of fear. Thus atropine-sensitive theta may be associated, not merely with absence of movement, but with active suppression of movement.

2. We have studied the pharmacology of theta electrically induced ("driven") by low-frequency stimulation of the septal pacemakers. Our research shows: (a) that there is a minimum threshold for septal driving of theta at 7.7 Hz; (b) that this minimum in the "theta-driving curve" is jointly dependent on the noradrenergic and serotonergic afferents to the SHS; and (c) that the 7.7 Hz minimum is eliminated by antianxiety drugs in a manner consistent with blockade of the noradrenergic input to the SHS (Gray, McNaughton, James & Kelly 1975; McNaughton, James, Stewart, Gray, Valero & Drewnowski 1977). In addition, Graeff et al. (1980) have found that the maximum frequency of atropine-sensitive theta elicited by raphe stimulation lies close to 7.7 Hz, and that the threshold for its elicitation is raised by antianxiety drugs. It is possible, therefore, that 7.7 Hz constitutes an upper limit of the frequency of atropine-sensitive theta.

3. Theta-driving septal stimulation usually produces behavioural effects opposite to – and theta-blocking (high-frequency) stimulation effects the same as – those of septal and hippocampal lesions. This pattern of results strongly suggests that theta is an active state of hippocampal function.

4. Vinogradova (1975) and O'Keefe and Nadel (1978) have suggested that the function of theta is to quantise time with respect to the passage of different items of information through the SHS. However, the time between successive corresponding phases of theta waves (rarely less than 80–100 ms) is about an order of magnitude too great if it is used just to quantise the passage of information through the four-synapse hippocampal circuit. An alternative hypothesis is that theta paces the passage of successive items of information round all the loops that make up the SHS and the Papez circuit (Fig. 2). Parmeggiani, Azzaroni, and Lenzi (1971) have estimated the conduction time for information to travel around the Papez circuit to be 50–60 ms; this value is about the right order of magnitude to be paced by the theta rhythm.

5. Rudell, Fox, and Ranck (1980) report that the size of hippocampal population spikes recorded in response to electrical stimulation of appropriate afferent pathways varies systematically with the phase of the simultaneously recorded theta rhythm. Furthermore, even though dentate and CA1 theta are 180° out of phase with each other (Winson 1976), the maximum response in both regions occurred at the same phase of dentate theta. Thus a stimulus reaching the hippocampal formation at the right time is likely to receive privileged transmission through all points in the hippocampal circuit. In this way theta could play an important role in the selection of stimuli for entry into the hippocampal circuit, as well as in timing the entry of successive stimuli.

D. Behavioural effects of septal and hippocampal lesions. Table 1 summarises these effects. The following points are of special interest.

1. There is a clear disinhibition of motor behaviour in a variety of different situations. This is in agreement with reports of inhibition of movement after both hippocampal (e.g., Kaada, Jansen & Andersen 1953) and septal (e.g., Brayley & Albert 1977) stimulation. This kind of observation accords well with theories that attribute to the SHS a role in direct motor inhibition (e.g., McCleary 1966; Altman, Brunner & Bayer 1973).

2. However, there are also findings which make no sense within this framework, but which are instead consistent with Douglas's (1967) version of behavioural inhibition theory, with its emphasis on the gating out of redundant stimuli. Examples of this kind are the loss of Kamin's blocking effect (Solomon 1977; Rickert, Bennett, Lane & French 1978) and the loss of latent inhibition (Weiss, Friedman & McGregor 1974; Solomon & Moore 1975).

3. Since there is no loss in the capacity for movement after damage to the SHS, the correlation between atropine-resistant theta and movement (C1, above) cannot reflect an executive role for theta in the production or control of movement (Vanderwolf 1971). But the fact that theta is an active state of hippocampal function (C3), and that it occurs when animals are in motion, rules out the proposal (Altman et al. 1973) that theta is in general concerned with the inhibition of movement (though this proposal may be correct for atropine-sensitive theta: C1).

4. On the spatial hypothesis (O'Keefe & Nadel 1978) the disruptive effects of septal and hippocampal lesions should be particularly apparent in tasks which require an extensive analysis of spatial information. O'Keefe and Nadel (1978) claim that this is indeed the case. But while it is true that spatially complex tasks do show up the lesioned animal's deficits very clearly, there are also many reports of profound impairment in tasks which contain no obvious spatial component (e.g., the deficit in responding on schedules of differential reinforcement of low rates of response, DRL; or the deficit in nonspatial reversals). It seems, therefore, that, while spatial tasks contain features which render them particularly difficult for the lesioned animals, these features are not unique to spatial tasks. The same conclusion has been reached by Olton et al. (1979) in their analysis of spatial and nonspatial problems in the radial-arm maze.

5. Animals with septal or hippocampal damage are able to store and retain new information over long periods. Such deficits in memory function as are observed are either due to special factors (e.g., interference from similar stimulus-response associations) or may relate to special forms of memory (e.g., recognition memory, Gaffan 1972; or working memory, Olton et al. 1979). This conclusion is in agreement with the results of single-unit recording experiments. These show that hippocampal units are readily *disposable*: they respond to complex features of the environment as soon as the animal is exposed to them (Hill 1978), and the same units respond to quite different features in different environments (Ranck, in Elliott & Whelan 1978, p. 310). This kind of observation is difficult to reconcile with a storage function.

6. I have already commented on the remarkable resemblance between the effects of septo-hippocampal damage and antianxiety drug treatment (Table 1). Whether or not the theory proposed in *Anxiety* is correct, this resemblance requires explanation.

E. Memory. It is still not clear whether the human amnesic syndrome, which first gave rise to the hypothesis that the hippocampus has a memory function (Milner 1970), genuinely results from damage to the hippocampus and related structures, or whether the involvement of the hippocampus has been incidental, the critical structures being located elsewhere (Victor, Adams & Collins 1971; Weiskrantz 1978; Horel 1978). This ambiguity presents obvious difficulties for the would-be theorist needing to know what he is to theorise about. The problem has been somewhat reduced, however, by recent evidence that allows some rapprochement between the hitherto sundered syndromes of human amnesia and hippocampal rat. This rapprochement has come from both sides.

1. It is now unlikely that the human impairment consists centrally in a failure to lay down and/or store long-term memories, as proposed by Milner (1970). Rather, the processes of recall and recognition in the human amnesic appear to be particularly susceptible to interference from similar material (Weiskrantz & Warrington 1975; Weiskrantz 1978). The reasons for this susceptibility to interference are still unclear. One proposal with some empirical support is that the amnesic fails to code stimuli adequately along multiple available dimensions (Butters & Cermak 1975; Oscar-Berman & Samuels 1977). A second proposal, not incompatible with the first, is that there is a failure to differentiate stimuli and/or responses according to context (Hirsh 1974; Winocur 1982).²

2. New paradigms have demonstrated that, under appropriate conditions, animals do display memory deficits, and that these, as in the human amnesic syndrome, depend on interference from similar stimuli and/or responses (Gaffan 1977a, 1977b; Olton 1978b; Olton et al. 1979). Gaffan (1977a; 1977b) believes that these deficits reflect the loss specifically of recognition memory. However, his own data show that, in a delayed matching-to-sample paradigm, fornix-lesioned monkeys were impaired in the recognition of a colour or a position only if two other stimuli had intervened between presentation of the sample and the test stimulus. Thus recognition memory per se was not affected. Olton's (1978a; 1978b) working-memory hypothesis explicitly acknowledges the role of interference, since the function of working memory is to hold information which is necessary for correct responding but which changes from trial to trial or moment to moment.

3. Though failure of memory has now been demonstrated in animals with hippocampal damage, it would be wrong to conclude from this that the hippocampus plays a role primarily in memory functions. First, animals with damage to the SHS can be shown to be excessively susceptible to interference in tasks in which memory does not play a critical role (e.g., Winocur & Mills 1970; Winocur & Breckenridge 1973; Dabrowska & Drzewiecka 1975; Sikorsky, Donovick, Burright & Chin 1977). Second, there are many impairments after septal

or hippocampal lesions which cannot plausibly be attributed to memory loss, for example, the impairment in DRL responding.³

A new theory of septo-hippocampal function

Assumptions. The “new” theory of septo-hippocampal function put forward in *Anxiety* is in many respects not new, since it borrows much from many previous theories. In trying to integrate these and their supporting data, I have made the following assumptions.

1. There is no long-term storage of information in the hippocampus.

2. The SHS plays no role in the initial learning of responses to new stimuli (provided there is no interference from similar stimuli or responses).

3. The SHS receives much information about behaviour which it does not at the time control. Support for this assumption comes from several sources. For example, the single-unit experiments show that stimuli which predict reinforcement elicit powerful hippocampal responses (section B1, above). Yet precisely those tasks in which such observations are made are immune to disruption by septal or hippocampal lesions. Similarly, voluntary movement is accompanied by a pronounced theta rhythm (an active state of hippocampal function; see section C3 above); yet destruction of the hippocampus or loss of theta leaves voluntary movement undisturbed.

4. Spatial tasks make powerful demands upon hippocampal function, but the SHS is not exclusively specialised to play a role in handling spatial information.

5. A successful theory must find functions not only for the SHS, but also for the major connections of this system, and especially the Papez circuit (see Vinogradova 1975; Gabriel, Foster, Orona, Saltwick & Stanton 1980).

6. The final and central assumption is one which commanded maximum support at a recent symposium (Elliott & Whelan 1978, pp. 418–19) on septo-hippocampal function: that the SHS acts as a comparator. This hypothesis has been developed most extensively by Vinogradova (1975), but it is a necessary feature of several other theories, including O’Keefe and Nadel’s (1978) spatial mapping model and my own concept of the BIS. According to Vinogradova (1975), area CA3 acts as a detector of novelty/familiarity. But the lesion evidence makes it clear that the hippocampus does both less and more than detect novelty: less, because the behavioural orienting reflex and its habituation are barely disrupted by hippocampal lesions; more, because most of the changes seen after hippocampal damage are not readily explained simply as altered responses to novelty. Furthermore, on Vinogradova’s own evidence, CA3 can perform only very crudely as a comparator. “The neural reactions in CA3 do not code (and consequently cannot transmit) information about the quality of their sensory input. Their activity appears to be a strong generalized regulatory signal, which may possibly exert modulatory effects on the output structures” (Vinogradova 1975, p. 10). The present theory takes this comment seriously and so, while it incorporates Vinogradova’s CA3 comparator, it treats this as part only of a more extensive system.

The theory. The central task of the overall system is to compare, quite generally, actual with expected stimuli. The system functions in two modes. If actual stimuli are successfully matched with expected ones, it functions in “checking mode,” and behavioural control remains with other (unspecified) brain systems. If there is discordance between actual and expected stimuli, or if the predicted stimulus is aversive (conditions jointly termed “mismatch”), it takes direct control over behaviour and now functions in “control mode.” Before we proceed further, it is worth asking in general terms what kinds of information processing are necessary for such a comparator to perform its functions.

Clearly, the comparator must have access to information both about current sensory events (“the world”) and about expected (“predicted”) events. These two classes of information must themselves be closely interrelated, however. Predictions can only be generated (a task for which a “generator of predictions” is needed) in the light of information about the world. Thus current sensory events must be transmitted to the generator of predictions. In addition, the latter must have access to information about past environmental regularities which, in conjunction with the current state of the world, determine the content of the next prediction. But this will be sufficient only if the environment is one over which the subject has no control. If, as in most cases, the subject’s behaviour affects the world, the generator of predictions must also have access to information about the next intended set of movements, that is, motor programmes or plans. Prediction will now depend on the conjunction of the present state of the world, stored past environmental regularities, the next intended step in the motor programme, and stored past relationships between such steps and changes in the environment.

Once made, the prediction must be tested against the world. For this to be possible, the right sensory inputs must be chosen. It follows that the input of sensory information must be selected in the light of what is predicted. Such selection can in principle be accomplished in one of two ways. The subject’s motor behaviour can be left under the control of other systems, with selection accomplished by choosing among the sensory events occurring anyway as a result of that motor behaviour; or the selection mechanism can take active control over motor behaviour and command appropriate exploratory action. This might range from simple adjustments of sensory organs (e.g., dilating the pupils) to complex patterns of locomotion.

It follows from these arguments that the major informational transactions for which we must seek neural correlates are those set out in Figure 3. This figure contains a simplifying assumption: that the same neural message that identifies the current state of the world, once it has been successfully matched with the relevant prediction, also serves as the information which initiates the generation of the next prediction. In this way, the occurrence of a mismatch has the reasonable consequence that it automatically brings to a halt the process of generating predictions.

As already indicated, the SHS takes active control of behaviour only when a mismatch is detected. In this control mode, two of the outputs of the BIS (Fig. 1), but not the increment in arousal (see below), are activated.

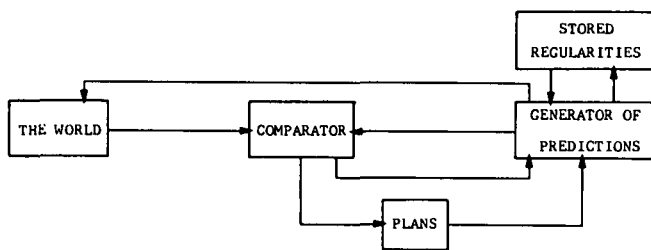


Figure 3. The kinds of information processing required for the successful functioning of the hypothetical comparator (see text for further explanation.)

First, there is an immediate inhibition of any motor programme which is in the course of execution. This inhibition is not envisaged as taking place at the level of individual motor commands, but rather as an interruption in the function of higher-level systems concerned with the planning and overall execution of motor programmes.

Second, the motor programme which was in the course of execution at the time that mismatch was detected is tagged with an indication which, in English, might read "faulty, needs checking." This has two further consequences. (1) On future occasions the relevant programme is executed with greater restraint (more slowly, more easily interrupted by hesitations for exploratory behaviour, more readily abandoned in favour of other programmes, etc.). (2) The tagged motor programme is given especially careful attention the next time it occurs; that is to say, the system exercises particularly carefully its basic function of checking predicted against actual events. It is supposed that in performing this function the system is able to subject each stimulus in the environment to an analysis that ranges over many dimensions (e.g., brightness, hue, position, size, relation to other stimuli, etc.). The dependencies between the subject's responses and behavioural outcomes can likewise be subjected to this type of multidimensional analysis (e.g., a turn in a rat's T-maze can be classified as left-going, white-approaching, alternating, moving along the longitudinal axis of the room that surrounds the maze, etc.).

Third, the system initiates specific exploratory and investigative behaviour designed to answer particular questions that arise from the operation of the comparator (e.g., is it left-going or white-approaching which gives rise to mismatch?); this proposal has much in common with Kimble's (1975) concept of "hypothesis testing." Like the inhibition of ongoing motor programmes, it is not supposed that this function is exercised directly at the point of motor output, but rather that the system modulates the control of exploratory behaviour carried out by other brain systems. Among the environmental stimuli for which the system commands a special search in this way, an important class is constituted by those which themselves predict disruption of the subject's plans, that is, stimuli which are associated with punishment, nonreward, or failure.

But the system is also continuously active at times when there is agreement between predicted and actual events; if it were not, mismatch would go undetected. Furthermore, it is supposed that the system is able to

identify certain stimuli as being particularly "important" and requiring especially carefully checking. In this connection, as we shall see, the role played by the ascending monoaminergic inputs to the hippocampus is of particular significance.

In attempting to match these psychological functions to the anatomy and physiology of the SHS and its associated structures, we first propose that sensory information enters the system via the entorhinal cortex. The diagram illustrated in Figure 3 prompts one to look for the comparator itself in a place where an incoming message which started out in the entorhinal cortex can both: (1) terminate a loop by identifying the current state of the world which has to be matched against a prediction that has circled round the loop; and (2) initiate a loop (to commence generation of the next prediction). The subicular area is ideally suited for this purpose. As we have seen (Fig. 2), information reaches it from CA1 after passing round the basic hippocampal circuit. It is then recirculated through the fornix to the mammillary bodies, the anteroventral thalamus, the cingulate cortex, and back to the subicular area. Thus a plausible hypothesis accords the status of comparator to the subicular area: the input from CA1 identifies the current state of the world; its recirculation round the Papez loop forms part of the process of generating a prediction and arrival of the input from the cingulate cortex (and/or the direct input from the anteroventral thalamus) in the subicular area constitutes the prediction.

If the subicular loop subserves in this way the generation of predictions, it must have access to stored regularities concerning the environment and the relations between the organism's responses and the environment. It is plausible to consider the anteroventral thalamus as a point of access for response-independent environmental regularities. Where response-dependent environmental regularities are concerned, it is possible that the cingulate cortex plays this role. A generator of predictions must have information, not only about past dependencies between responses and outcomes, but also about current and intended responses. This information might also be available in the cingulate cortex, where it could function as part of the Papez loop. Alternatively, this region or the prefrontal cortex might send relevant motor information to the entorhinal area, to which both project. In that case information about current and intended movements would form part of the description of the world. Probably both routes are used, current movements entering into the entorhinal description of the world, intended movements taking part in the generation of predictions as the Papez loop sweeps past the cingulate cortex.

I have described the anteroventral thalamus and cingulate cortex as "points of access" to stored regularities. This phrase is compatible with two interpretations. According to the first, the relevant information is stored locally in these regions; according to the second, it is stored elsewhere and is simply activated by messages from the thalamus and cingulate cortex. The second alternative is the more probable, structures in the temporal lobe constituting the most likely sites of storage. Note that the temporal lobe projects directly to the subicular area (Van Hoesen et al. 1979). Thus a prediction arriving in the subicular area might be made up of

elements arriving from the thalamus, the cingulate cortex, the temporal lobe and the prefrontal cortex, in any of several possible combinations.

The part of the theory relating to the Papez circuit is the most frankly speculative, and there are few data against which to test it. However, observations of single-unit firing in general agree with the notion of a predictive function for this circuit. Thus cingulate units fire during the delay period of a delayed-response paradigm in relation to an absent cue or the impending direction of response (Niki & Watanabe 1976); cingulate and anteroventral thalamic units, recorded at the start of reversal training, fire in a manner appropriate to the original significance of the now-reversed stimuli (Gabriel, Miller & Saltwick 1977); and mammillary and anteroventral thalamic units extrapolate features of repeated stimuli and fire even when these are omitted from their expected position in a regular sequence (Vinogradova 1975).

We next ask how the system effects selection of the sensory inputs used both to verify and to generate predictions. The foregoing arguments constrain the choice of appropriate pathway to the projection from the subicular area to the entorhinal cortex (Köhler, Shipley, Srebro & Harkmark 1978; Segal 1977a); this corresponds with the connection from "the generator of predictions" to "the world" in Figure 3. Thus the subiculo-entorhinal projection is held to be responsible for the selection of sensory input to the SHS.

An important feature of this model is so far not well motivated. Why should the neural message representing the current state of the world be sent all round the hippocampal circuit to the subicular area, when it can just as well take the direct entorhinal-subicular route (Fig. 2)? In response to this question the theory holds that the hippocampal circuit acts as a further part of the selection process, ensuring that the subicular comparator receives only those items of information about the world which are important to its functioning.

On this hypothesis – and in agreement with Vinogradova's (1975) single-unit observations (point B4, above) – the entorhinal area sends to the subicular area detailed sensory information via its direct projection and, at the same time, a nonspecific message via the hippocampal circuit. The detailed sensory information is used by the subicular area for matching against predicted sensory input and for generating the next prediction only if this region also receives the nonspecific message sent from CA1. The latter therefore serves as an "enabling" signal – Vinogradova's (1975) "generalized regulatory signal." This enabling signal is sent on only in connection with stimuli that are of importance to the animal; otherwise they are habituated during their passage through the dentate-CA3 gate. Passage through this gate is augmented for stimuli associated with reinforcement (point B1, above) and is facilitated by the noradrenergic and serotonergic afferents to the SHS (point B2, above). In short, the direct input to the subicular area from the entorhinal cortex describes the current state of the world; the input via the hippocampus determines whether this description is treated by the subicular comparator as important; and the monoaminergic inputs to the SHS increase the probability that stimuli

entering from the entorhinal cortex will indeed be tagged as important.

This is the heart of the theory of septo-hippocampal function developed in *Anxiety*. There are many other specific suggestions which we do not have space to develop here. Some important points I have not yet discussed are as follows.

1. The routes by which the SHS affects motor behaviour are unclear. Behavioural inhibition in the narrow sense is perhaps effected by way of the projection from the subicular area to the cingulate cortex (Meibach & Siegel 1977; Swanson 1978) and/or by way of the inhibitory pathway from the lateral septal area to the hypothalamus described by Albert (e.g., Albert, Brayley & Milner 1978). The subiculo-cingulate projection may also mediate the influence that the SHS assumes over exploratory behaviour when it operates in control mode.

2. Although the SHS has no long-term storage capacity of its own, it is able to modify information stored elsewhere after stimulus/response analysis and hypothesis testing. These modifications are most probably mediated by the outputs of the SHS to the temporal lobe via the entorhinal cortex (see Hirsh 1974).

3. The function of the theta rhythm is to pace and quantise the circulation of information around both the SHS and the Papez circuit, thus ensuring that the corresponding sensory events and predictions arrive together in the subicular comparator. Vanderwolf's two types of theta correspond to the two modes of hippocampal functioning: atropine-resistant theta to checking mode, atropine-sensitive theta to control mode. [See also Vanderwolf & Robinson: "Reticulo-Cortical Activity and

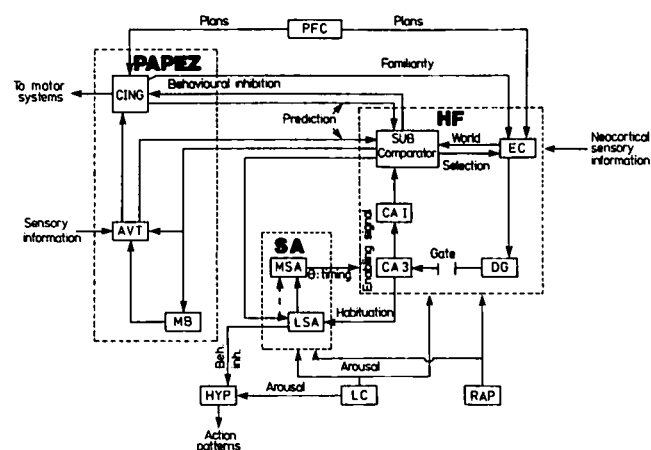


Figure 4. A summary of the theory developed in *Anxiety*. The three major building blocks are shown in heavy print: HF, the hippocampal formation, made up of the entorhinal cortex, EC, the dentate gyrus, DG, CA3, CA1, and the subicular area, SUB; SA, the septal area, containing the medial and lateral septal areas, MSA and LSA; and the Papez circuit, which receives projections from and returns them to the subicular area via the mammillary bodies, MB, anteroventral thalamus, AVT, and cingulate cortex, CING. Other structures shown are the hypothalamus, HYP, the locus coeruleus, LC, the raphe nuclei, RAP, and the prefrontal cortex, PFC. Arrows show direction of projection; broken arrow within SA indicates a projection without anatomical confirmation. Words in lower case show postulated functions; beh. inh., behavioural inhibition. (For further explanation, see text.)

Behavior" *BBS* 4(3) 1981.] The frequency of atropine-resistant theta varies with the rate of processing of sensory events and predictions, giving rise to the observed correlation between rate of movement and frequency of theta (cf. O'Keefe & Nadel 1978). The lower frequency of atropine-sensitive than atropine-resistant theta arises because the control mode requires more complex processing of information than the checking mode. The special significance of 7.7 Hz theta (C2 above) may be that it is the highest rate at which information can comfortably be transmitted while the system remains in control mode. It is also possible that different frequencies of theta allow the Papez circuit to access different locations at which relevant information is stored (Parmeggiani, Lenzi & Azzaroni 1974).

A summary of the main features of the theory (including some not described here) is presented in Figure 4.

Relationship to previous theories

The theory incorporates many previous concepts. It shares similarities with O'Keefe and Nadel's (1978) cognitive mapping system, but with two major differences: it allots no special role to space, and no storage to the hippocampus. From Kimble (1975) it takes the notion of hypothesis testing, and from Hirsh (1974) the idea that, once the SHS has sorted a problem out, it adds an appropriate contextual tag to material stored elsewhere. Its indebtedness to Vinogradova's (1975) concept of the comparator is obvious, as to Douglas's (1967) closely related view that the hippocampus gates out redundant and/or unimportant stimuli. But while the model adopts many of these cognitive aspects of previous theories of hippocampal function, it does not neglect the clear role (attested by both lesion and stimulation experiments) which the SHS plays in straightforward motor inhibition, as postulated by McCleary (1966) and Altman et al. (1973), among others. Contrary to an apparently widespread view (Elliott & Whelan 1978, p. 306), there is no contradiction here: it is possible, often essential, to stop and think.

The model is also compatible with more recent views of the role of the hippocampus in memory, especially Olton's (1978a; 1978b) working memory hypothesis – although, to cope with the data from experiments in the radial-arm maze, as well as our own findings in experiments on the partial reinforcement extinction effect (e.g., Feldon & Gray 1979a; 1979b), it is necessary to complicate still further the neural machinery described above (see *Anxiety*, discussion of Fig. 10-6). As for the human amnesic syndrome – on the assumption that this indeed turns out to depend on damage to the hippocampus or closely related structures – both the empirically established role of interference (Weiskrantz 1978) and the suggestion (Butters & Cermak 1975) that this may arise from a failure of multidimensional stimulus analysis are compatible with the new theory. It is probable, however, that an adequate account of hippocampal function in working memory tasks requires a greater role to be assigned to descending neocortical controls over the activities of the SHS. An increased role of this kind is in any case necessary to account for certain features of

human anxiety and its treatment (*Anxiety*, chapter 14). Such neocortical controls are probably of greater importance in primates, including man, than in subprimates; this assumption is supported by anatomical evidence of relations between the SHS and the neocortex in primates that have not been reported in other species (e.g., Leichnetz & Astruc 1976; Tanaka & Goldman 1976).

The relationship between the present theory of SHS function and the earlier BIS model for anxiety (Fig. 1) should be clear. The SHS is sensitive to the same stimuli which operate the BIS, and it produces two of its three outputs: inhibition of ongoing behaviour, and what in the earlier model was called simply "increased attention to environmental stimuli." The latter function, however, has now been greatly amplified. In order to do this, it has been necessary to enter the black box and to speculate on its inner workings, both psychological and neural. The third output of the BIS – an increased level of arousal – is not attributed to the SHS itself, but rather, as we shall see, to the ascending monoaminergic fibres which innervate not only the SHS, but also other structures.

Some may count the fact that the new theory incorporates so many features of earlier theories an advantage. It would, after all, be something of a disaster if so much previous investment of thought and labour had yielded no insights worthy of survival. Others will count it as mere eclecticism. Since the theory is a patchwork quilt, it is supported by existing data to the extent that each patch is itself supported. Only future experimental tests directly addressed to its specific new postulates will allow one to judge whether the whole is greater than the sum of its parts, or the quilt the weaker for its seaming. I believe that the theory has at least been specified in sufficient detail to permit such testing to be carried out.

How do the antianxiety drugs reduce anxiety?

Although much of *Anxiety* is concerned with septo-hippocampal function, its chief aim is to elucidate the neuropsychology of anxiety. The direct and indirect approaches to the question, "Where in the brain do the antianxiety drugs act?" have provided two answers: "everywhere" (by facilitating the action of GABA), and "in the SHS" (or at any rate somewhere closely related to the SHS). The gap between these two answers is not as great as it may seem, however, since the direct approach has concerned not only the antianxiety action of the drugs but also their other effects (sedation, muscle relaxation, etc.). Nonetheless, there is still some way to go before the two answers can be joined into one.

If the system described in the previous section has something to do with anxiety, how would we increase the intensity of this emotion? It is clear, I think, that we would do so by opening the dentate-CA3 gate wider. This would have the effect that the system treats more stimuli as "important, needing checking." The way to open the dentate gate wider is to increase the noradrenergic and/or serotonergic inputs to the SHS (section B2 above). This inference is consistent with both major current theories of the roles of monoamine trans-

mitters in anxiety, one of which holds that the critical pathway is serotonergic (Stein, Wise & Berger 1973), while the other favours the dorsal ascending noradrenergic bundle (DANB) (Lidbrink, Corrodi, Fuxe & Olson 1973; Gray et al. 1975; Redmond 1979).

It is unlikely, however, that these two transmitters simply duplicate the same function. Lesion experiments suggest that the DANB is particularly closely related to the behavioural effects of stimuli associated with non-reward (Mason & Iversen 1975; McNaughton & Mason 1980; Owen, Boarder, Gray & Fillenz 1982) and to the effects of novelty (Owen 1979); and that central serotonergic pathways are similarly related to the effects of stimuli associated with punishment (e.g., Tye, Everitt & Iversen 1977). However, this parcelling of functions is almost certainly too simple. We have found that DANB lesions reliably disinhibit bar-pressing suppressed by stimuli associated with punishment (Tsaltas & Gray, in preparation), and others have found improved shuttlebox avoidance after such lesions (Ögren & Fuxe 1977; Mason & Fibiger 1979d). In addition, there are reports of impaired responses to nonreward after treatment with drugs that block serotonergic transmission (Thornton & Goudie 1978; Beninger & Phillips 1979).

One possibility which is consistent with most of the data is that the noradrenergic input to the SHS qualifies concurrent sensory input simply as "important," while the serotonergic input adds to this the specification "aversive." Another possibility, not incompatible with the first, allots to the noradrenergic input the role of boosting the information-processing capacity of the SHS and to the serotonergic input the role of facilitating motor inhibition. It is consistent with this hypothesis that motor disinhibition is not seen after DANB lesions (McNaughton & Mason 1980; *Anxiety*, chapter 11), but clearly occurs after damage to the serotonergic afferents to the SHS (Jacobs, Trimbach, Eubanks & Trulson 1975; Williams & Azmitia 1981). On this view (which can encompass much of the data) the likelihood of observing a lesion effect after destruction of one or another of the ascending monoamine pathways would depend on the requirements for motor inhibition (usually greater when punishment is employed) and stimulus-response analysis respectively.

The hypothesis that the ascending serotonergic pathways play a role in mediating responses to anxiogenic stimuli has been proposed several times before (e.g., Graeff & Schoenfeld 1970; Stein et al. 1973) and is unlikely to encounter major objections. The functions of the DANB, in contrast, are highly controversial (McNaughton & Mason 1980). Two major alternative theories of DANB function are discussed in *Anxiety*. The hypothesis that this pathway mediates the effects of reward or reinforcement (Stein 1968; Crow 1973) is rejected on grounds that have prompted others to the same conclusion (Mason & Iversen 1979; McNaughton & Mason 1980), namely that DANB lesions leave conventionally rewarded behaviour intact. The second hypothesis holds that the DANB is responsible for the selection of task-relevant cues (Mason & Iversen 1979). This view has points in common with the present theory, with the important difference that the latter holds that the DANB selects appropriate cues for input to the SHS. In spite of their similarities, in certain cases the two hypotheses

make different predictions about the effects of DANB lesions. One such difference concerns the partial reinforcement extinction effect. Mason & Iversen (1979) predict that this will be attenuated exclusively because of a rise in resistance to extinction in continuously reinforced animals, whereas the present hypothesis predicts in addition a fall in resistance to extinction in animals trained on partial reinforcement. The data uphold the latter prediction (Owen, Boarder & Gray 1977; Owen et al. 1982).

It will be clear that *Anxiety* does not choose between noradrenaline and serotonin. It holds, rather, that the two monoamine pathways function jointly under conditions of stress. This hypothesis is consistent with the evidence that stress increases the turnover of both noradrenaline and serotonin in the forebrain, and that in both cases this is reversed by treatment with the anti-anxiety drugs (Libdrink et al. 1973; Stone 1975). Dopaminergic pathways do not in general appear to be as affected by stress as noradrenergic or serotonergic pathways (Stone 1975). However, Glowinski's group has demonstrated a selective increase, reversed by benzodiazepine treatment, in the turnover of dopamine in the prefrontal and cingulate cortices (Lavielle, Tassin, Thierry, Blanc, Hervé, Barthelemy & Glowinski 1978). These regions receive their dopaminergic innervation from nucleus A 10 in the ventral tegmental area (Divac, Björklund, Lindvall & Passingham 1978; Emson & Koob 1978), which in turn receives afferents from the locus coeruleus (Simon, Le Moal, Stinus & Calas 1979); and Glowinski (pers. comm. 1980) has evidence suggesting that the effects of stress (and perhaps also those of anti-anxiety drugs) on dopamine turnover in the prefrontal cortex are mediated by this projection from the locus coeruleus. Behaviourally, the projection from nucleus A 10 to the prefrontal and cingulate cortices appears to be involved in motor inhibition under conditions of stress (Blanc 1980). This, then, may be one route by which the anti-anxiety drugs disinhibit suppressed motor behaviour.

In addition to these ascending monoamine pathways, it is necessary to take into account an ascending cholinergic system. There has long been evidence of cholinergic involvement in behavioural inhibition (Carlton 1963). Warburton's (1977) analysis leads him to conclude that a cholinergic system mediates the selection of relevant stimuli to which to respond or inhibit responding. This formulation is similar to the description of septo-hippocampal function adumbrated above. This similarity may reflect the participation in Warburton's system of the cholinergic projection from the medial septal area to the hippocampal formation, and/or an ascending cholinergic system, afferent to the septal area, which is responsible for Vanderwolf's atropine-sensitive theta. An important site of co-ordination between the ascending serotonergic and cholinergic systems is perhaps represented by the median raphe nucleus, which contains both serotonergic (Azmitia & Segal 1978) and probably cholinergic (Shute & Lewis 1967) cell-bodies. Stimulation here simultaneously elicits behavioural inhibition and low-frequency theta, the former being blocked by antiserotonergic drugs, the latter by anticholinergics, and both by anti-anxiety drugs (Robinson & Vanderwolf 1978; Graeff & Silveira Filho 1978; Graeff et al. 1980).

The most likely sites of the specific anti-anxiety effects of

the antianxiety drugs, then, are the ascending noradrenergic and serotonergic pathways, the ascending dopaminergic pathway to the prefrontal and cingulate cortices (perhaps secondarily to the projection from the locus coeruleus to nucleus A 10), and (much less certainly) the ascending cholinergic pathway to the SHS.

It remains to link these conclusions to the established effects of the antianxiety drugs on GABA-ergic transmission. There are three major possibilities, not mutually exclusive. The first is that the antianxiety drugs increase GABA-ergic inhibition of neurons in the locus coeruleus and raphe nuclei themselves (Iversen & Schon 1973; Fuxe, Agnati, Bolme, Hökfelt, Lidbrink, Ljungdahl, Perez de la Mora & Ögren 1975; Gallager 1978; Belin, Aguera, Tappaz, McRae-Deguerce, Bobillier & Pujol 1979; Guyenet & Aghajanian 1979). The second is that they increase GABA-mediated, bicuculline-insensitive, presynaptic inhibition of the release of noradrenaline and serotonin within the SHS (Bowerly, Hill, Hudson, Doble, Middlemiss & Turnbull 1980; Mellanby, Gray, Quintero, Holt & McNaughton 1981). The third is that they increase GABA-mediated recurrent inhibition within the hippocampal formation (Haefely 1981).

Noradrenaline, anxiety, and depression

So far we have concentrated on the role of the ascending monoamine afferents to the SHS. However, the effects of stress, especially on central noradrenergic systems, are more widespread than this (Stone 1975).

The anatomy and physiology of the DANB are inconsistent with the transmission of specific information to its target organs. In accordance with this constraint we have supposed that the noradrenergic input to the SHS acts as a booster, facilitating activities which normally go on in that system anyway. *Anxiety* makes similar assumptions with regard to the noradrenergic input (travelling in part in the DANB and in part in the ventral noradrenergic bundle) to the hypothalamus. This region is seen as integrating fixed action patterns, of which the most important under conditions of stress are those of escape, defence, and aggression ("fight-flight"). Stress appears to increase the activity of all ascending noradrenergic neurons (Stone 1975). This will have the consequence, if the argument so far is correct, that diencephalic fight-flight mechanisms will be simultaneously primed (via the direct noradrenergic input to the hypothalamus) and inhibited (via the SHS, as described earlier).

These postulates have the effect of making the ascending noradrenergic systems equivalent to the "increment arousal" output of the BIS (Fig. 1; Gray & Smith 1969), for which, it will be recalled, no corresponding function could be found within the SHS. This equivalence is supported by evidence that disruption of central noradrenergic function eliminates the potentiated startle response (Davis, Redmond & Baraban 1979) and the partial reinforcement acquisition effect (Owen et al. 1982); both of these changes can be attributed to a blockade of the arousal component of anxiety, and both are also produced by the antianxiety drugs (Gray 1977). Essentially the same hypothesis has been proposed by Redmond (1979), who supports it with a wealth of pharmacological and other evidence. Redmond and his col-

leagues (e.g. Gold, Redmond & Kleber 1978) have in addition suggested that the symptoms of opiate withdrawal (which resemble those of a severe anxiety attack) are due to a massive rebound activation of locus coeruleus neurons. If so, the autonomic signs of anxiety may be due in large measure to activation of the fibres that descend from this nucleus into the spinal cord.

Under conditions of prolonged stress there are biphasic changes in the functioning of noradrenergic neurons, a temporary exhaustion being followed by restored or even increased functional capacity (Stone 1975; Fillenz 1977). The exhaustion phase apparently underlies at least some instances of the "helplessness" (a putative analogue of human clinical depression: Maier & Seligman 1976) that follows exposure to uncontrollable aversive events (e.g., Weiss, Glazer, Pohorecky, Brick & Miller 1975); conversely, the later phase of restored function may account for the disappearance of helplessness after multiple exposure to uncontrollable aversive events, that is, for what Miller (1976) has termed "toughening up." If we accept helplessness as an analogue for depression, this condition may therefore be seen as reflecting a fall in the functional capacity of central noradrenergic neurons (Weiss et al. 1975). It follows from this hypothesis, together with the treatment of anxiety outlined above, that anxiety and depression are, neurochemically speaking, opposed states.

This conclusion runs into difficulties, however, when we take into account recent research on the mode of action of antidepressant drugs. It used to be thought that these work by increasing central monoaminergic activity (Schildkraut 1965). This view is supported by observations on both major classes of antidepressants, the tricyclics, which block the re-uptake of monoamine transmitters after release, and the monoamine oxidase inhibitors (MAOI), which prevent their intraneuronal inactivation (van Praag 1978). These actions are immediate, however, while the clinical effect of the antidepressants is not usually seen for several weeks. Recent research has therefore investigated the possibility that more long-term changes in the functioning of central monoamine systems might underlie the therapeutic effects of these drugs. This research has demonstrated changes in noradrenergic postsynaptic receptors which agree quite well with the time-course of therapeutic action (Sulser 1978; Stone 1979). But these changes take a form consistent with a *reduction* of central noradrenergic function by antidepressant drugs. This in turn implies that depression, like anxiety, is related to an increase in the activity of noradrenergic neurons. Thus, whereas on Schildkraut's (1965) hypothesis depression can easily be seen as opposed to anxiety, on the newer evidence (Sulser 1978) it becomes similar to anxiety.

Anxiety suggests that this dilemma can be resolved by the recognition that there are, in the extreme, two kinds of depression, neurotic and psychotic, although these are the poles of a continuum rather than two discrete categories (Roth 1979). Psychotic depression is characterised by the type of symptom for which the phenomena of helplessness in animals provide a reasonable analogue, especially motor retardation and weight loss. Neurotic depression, on the other hand, is difficult to distinguish from anxiety (Roth 1979), usually co-exists with anxiety, and responds to treatment with antianxiety drugs (e.g.,

Johnstone, Owens, Frith, McPherson, Dowie, Riley & Gold 1980). There are also differences in the personality of individuals susceptible to the two kinds of depression. In terms of Eysenck and Eysenck's (1969; 1976) dimensional scheme, neurotic depression is seen chiefly in neurotic introverts with low P scores while psychotic depression is seen in stable extraverts with elevated P scores (e.g., Paykel 1971; Eysenck & Eysenck 1976).

To account for these data it is proposed that there is a dimension of individual differences which, in psychiatric terms, encompasses successively: susceptibility to anxiety, neurotic depression, and psychotic depression. In neurochemical terms, this reflects, at the anxiety pole, a tendency to respond to stress with heightened noradrenergic activity and, at the psychotic depression pole, a tendency for noradrenergic neurons to become functionally exhausted. Both behavioural and neurochemical observations in experiments on helplessness suggest that functional exhaustion is likely to occur sooner in the diencephalon than in the forebrain (*Anxiety*, chapter 12). Thus, mixed states of anxiety and depression may arise when there is simultaneously a strong noradrenergic input to the SHS together with partial functional exhaustion in the noradrenergic input to the hypothalamus. Furthermore, as stress is prolonged, functional exhaustion should replace heightened noradrenergic activity, that is, depression should replace anxiety; Seligman (1975) and Engel and Schmale (1972) have made similar suggestions.

On this view the effective treatments for depression should differ depending on where the patient and his illness are located on the anxiety–depression continuum. In agreement with this deduction, anxious and neurotically depressed patients benefit from treatment either with antianxiety drugs or with MAOI (Johnstone et al. 1980; Paykel, Parker, Penrose & Rassaby 1979); and both these types of patient also benefit from therapeutic lesions to the prefrontal or cingulate cortices (Powell 1979). At the other end of the continuum, patients with psychotic depression benefit from electroconvulsive therapy (ECT), but not from antianxiety drugs or MAOI (Roth 1979). The tricyclic antidepressants are more ambiguous: they are apparently most effective in psychotic depression, but have some lesser effects in the neurotic variety (Roth 1979).

If the argument so far is correct, therefore, we would expect the MAOI, like the antianxiety drugs, to act principally to reduce noradrenergic activity, and perhaps especially so in the forebrain, while the tricyclics would reverse diminished noradrenergic activity, especially, perhaps, in the hypothalamus. There are no data relevant to these predictions, but they are open to experimental test. We would also expect ECT principally to increase noradrenergic function. The evidence shows that this treatment increases the activity of tyrosine hydroxylase (Modigh 1976), the rate-limiting enzyme in the synthesis of noradrenaline, and also the postsynaptic response to dopamine (Modigh 1975) and serotonin (Green & Deakin 1980); the latter two changes are not seen in animals in which the dorsal and ventral noradrenergic bundles are first destroyed (Green & Deakin 1980). Thus the therapeutic effect of ECT may be due to the development of enhanced monoaminergic transmission rather gener-

ally, with the increased activity in noradrenergic systems observed by Modigh (1976) playing a key role in this development.

Application of the theory to man

Applied to human clinical syndromes, this theory of anxiety can explain most naturally the symptoms of obsessive–compulsive neurosis (Rachman & Hodgson 1979), especially those that involve excessive checking of potential environmental hazards. In this respect it contrasts with earlier theories of anxiety based on animal experiments, which have usually found their most natural application to phobic symptoms. However, phobic symptoms can also readily be understood within the theory, which attributes them to the inhibition of motor programmes, as described above.

One important question, however, concerns the nature of the stimuli which give rise to phobias. Earlier theories which treated phobias as Pavlovian conditioned reflexes have been severely criticised, largely on the grounds that phobic stimuli are highly selective (spiders and snakes rather than cars and electric plugs). Seligman (1971) and Eysenck (1979) have tried to solve this problem by speaking of such phobic stimuli as “prepared” (by Darwinian evolution) to enter into a conditioned association with aversive unconditioned stimuli. While accepting the role of evolution, I find the notion of preparedness confused (Gray 1979c) and the evidence cited in its support (e.g., Öhman 1979) less convincing than it has seemed (*Anxiety*, chapter 14). Instead, it is proposed simply that the majority of phobic stimuli are stimuli which innately activate the BIS. This requires that we add to the list of anxiogenic stimuli listed to the left of Figure 1 the category of “innate fear stimuli.” Among these a particularly important role is played (e.g., in agoraphobia and social phobia) by stimuli which arise during the course of social interaction.

Pharmacotherapy is seen as depending essentially on a reduction in the intensity of the noradrenergic and serotonergic signals reaching the SHS; this, given the general theory, would diminish the subject's tendency to check the environment for threat. The effectiveness of behaviour therapy in cases of anxiety (Mathews 1978), in contrast, is thought to derive from manipulations of schedules of presentation of anxiogenic stimuli which maximise the rate of habituation (Watts 1979). This difference between the modes of action of the two kinds of therapy may account for the temporary effects of the antianxiety drugs, as contrasted with the enduring effects of behavioural treatment. The methods of behaviour therapy may also be regarded as a form of toughening up (Miller 1976). Given that, in certain experimental paradigms, toughening up is impaired by damage to the SHS (e.g., Rawlins, Feldon & Gray 1980) or the DANB (Owen et al. 1982), or by treatment with antianxiety drugs (e.g., Davis et al. 1981), one might expect that pharmacotherapy could interfere with the long-term success of behaviour therapy. Data reported by Hafner and Marks (1976) support this deduction, but much more work is required to test it properly.

The success of behaviour therapy apparently depends

on the extent to which the patient is exposed to the stimuli he fears (Teasdale 1977). But this poses a dilemma, since such exposure presumably gives rise to the patient's symptoms in the first place. Watts (1979) attempts to resolve this dilemma by reference to the opposing processes of sensitisation and habituation, which have been shown experimentally to occur when an initially novel stimulus is repeatedly presented (Groves & Thompson 1970). His analysis allows him to propose specific combinations of stimulation parameters which should aid in the therapy of anxiety; and these turn out to correspond to the very different methods of systematic desensitisation (Wolpe 1978) and flooding (Levis & Hare 1977). Furthermore, the processes he describes are very similar to those which, on quite independent grounds, have been built into the present theory of anxiety. Thus, sensitisation corresponds quite closely to the alerting function attributed to the ascending noradrenergic pathways, and habituation to the generation and confirmation of predictions by the SHS and Papez circuit.

In man a particularly important role in anxiety is apparently played by the prefrontal and cingulate regions of the neocortex. Destruction of each of these areas has been successfully used in the treatment of anxiety (Powell 1979). Within the present theory the role of these regions is twofold. First, they supply to the SHS information about the subject's own ongoing motor programmes; this information is essential if the SHS (together with the Papez circuit) is to generate adequate predictions of the next expected event to match against actual events. Second, they afford a route by which language systems in the neocortex can control the activities of the limbic structures which are the chief neural substrate of anxiety. In turn, these structures, via subicular and hippocampal projections to the entorhinal cortex, are able to scan verbally coded stores of information when performing the functions allotted to them in the theory outlined above. In this way, human anxiety can be triggered by stimuli that are largely verbal (relatively independently of ascending monoaminergic influences) and verbally coded search strategies can be used to cope with perceived threats. It is for this reason, if the theory is correct, that lesions to the prefrontal and cingulate cortices are effective in cases of anxiety that are resistant to drug therapy (Powell 1979).

The theory developed in *Anxiety* has a natural extension to the personality attributes of individuals susceptible to anxiety: these are treated simply as having a highly reactive BIS. To allow this description, the axes of Eysenck and Eysenck's (1969) personality space are rotated so that one dimension runs through high trait anxiety (Gray 1970a). This rotation has a number of advantages; in particular, it is better able to account for the effects of reward and punishment on learning in introverts and extraverts respectively (Gray 1981). It is possible, in addition, to derive numerous predictions from the theory as to the behaviour of individuals high on trait anxiety relative to those low on this trait. However, these predictions are only indicated in a very general manner in *Anxiety*. One important question that is left unresolved concerns the opposite pole of the dimension of trait anxiety. This might be located among the individuals described by Hare and Cox (1978) as primary psychopaths

(Gray, Owen, Davis & Tsaltas, in press). The fact that primary psychopaths are particularly deficient at learning passive avoidance (Hare & Cox 1978) is in agreement with this hypothesis. An alternative possibility has been outlined in the previous section, namely, that there is a dimension running from susceptibility-to-anxiety to susceptibility-to-psychotic-depression.

Conclusion

Let me conclude, as I began, by emphasising that the behavioural inhibition and septo-hippocampal systems only partially overlap. The functions subserved by the SHS and the Papez circuit are, I believe, always the same. But they can be brought into play in at least three different ways. The first is largely under the influence of the ascending monoamine pathways, which function, as Redmond (1979) has stressed, as a general alarm bell. This is anxiety at its most basic. The second way is also, according to the theory, experienced as anxiety, but it depends on descending neocortical influences, especially those originating in the prefrontal cortex. This kind of anxiety is particularly important in man. The third way is also probably under neocortical control, but it differs from the first two in that it is relatively emotion-free. It occurs, say, when a human subject is required to recognise items from lists of words, or a rodent has to select a previously unentered arm in a radial maze (Olton et al. 1979). However, even under those circumstances, one is only *relatively* free of emotion: the making of errors (which it is the business of the SHS to detect) is, I think, never entirely without emotional effect. But then neither is anxiety ever entirely free of thought. And if *Anxiety* has blurred the lines between thinking and feeling, that is no accident, and no bad thing.

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NOTES

1. The arguments that lead to this inference are indirect and require more detailed consideration of the data reported in certain experiments than it is possible to give here (see *Anxiety*, end of chapter 6).
2. Winocur and Hirsh see their two theories as opposed to each other: however, careful analysis shows that they are essentially the same (*Anxiety*, chapter 9).
3. The attempt by Olton et al. (1979) to account for this in terms of a loss of working memory is *not* plausible. What

changes from moment to moment in a DRL task is not what the animal must remember, but the requirement to respond or not to respond.

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Anxiety: Dysfunction of transmission or modulation?

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Recent findings suggest that altered neuroregulator functions involving both neurotransmission and neuromodulation can change behavior and may sometimes contribute to behavioral abnormalities. While neurotransmitters convey nerve signals directly from pre- to postsynaptic elements, neuromodulators do not act in a transsynaptic way but alter the effectiveness of neurotransmitters or affect a large number of neurons at some distance from the site of release (Barchas, Akil, Elliott, Holman & Watson 1978). [See also Dismukes: "New Concepts of Molecular Communication Between Neurons" *BBS* 2(3) 1979.]

Gray's theory suggestively combines his own view and that of others concerning the functional significance of the septo-hippocampal system with recent information about the role of noradrenergic and serotonergic transmission in the control of the forebrain, impressively leading us to a novel interpretation of anxiety. But while Gray handles with ease the most advanced approaches related to neurotransmission, he surprisingly neglects the concept of neuromodulation. The neuromodulators, hormones or fragments of hormones, are not unknown to Gray. He has tried to build hormonal effects into the hypothesis of a functional identity between punishment and reward. But pituitary-adrenal hormones failed to fit in the hypothesis (Gray & Garrud 1977). It now seems that he has selected the wrong hormones. The hypothalamic neurosecretory hormones, vasopressin in particular, would fit excellently in Gray's modulation-less theory.

Vasopressin exerts long-term influences on behavior by facilitating noradrenergic transmission in the ascending dorsal noradrenergic bundle system, probably through an action at the presynaptic level (Kovács, Bohus & Versteeg 1979). Modulation of serotonergic transmission in the ascending 5-HTP system, originating from the dorsal raphe nucleus also plays a role in the behavioral actions of vasopressin (Bohus, Conti, Kovács & Versteeg 1982). Alterations in the activity of these two systems is a crucial issue in Gray's theory. It is of particular interest that one of the loci of action of vasopressin is the dentate gyrus of the hippocampus, that is, at the level of the "gate" of Gray's comparator circuit. The dentate gyrus appears to be not only a behaviorally active site, but the peptide also facilitates noradrenaline turnover rate in situ, following local injection (Kovács, Bohus, Versteeg, de Kloet & de Wied 1979). It is also remarkable that in the brain region of rats classified as good avoiders on the basis of their performance in passive avoidance retention tests the noradrenaline turnover rate was, in the dentate gyrus of the hippocampus only, substantially higher

than in the same brain region of rats classified as poor avoiders (Kovács, Versteeg, de Kloet & Bohus 1981). Vasopressin–noradrenaline interactions in the brain have one more interesting aspect. Vasopressin injection into behaviorally active sites such as the dentate gyrus of the hippocampus, dorso-lateral septum, and the dorsal raphe is invariably followed by facilitated catecholamine turnover in the red nucleus, which is considered to be involved in motor-control processes. It may well be that the catecholamine changes in this nucleus represent the neurochemical basis of motor output of behavior, and eventually that of behavioral inhibition.

Including the concept of neuromodulation obviously does not fill in all the gaps in Gray's hypothesis. However, it should be recalled that the behavioral actions of vasopressin in animal and man have been placed in a cognitive framework, primarily that of memory (de Wied & Bohus 1979; Bohus 1981).

Cognitive changes represent a major issue in the obsessive symptoms of anxiety. If we accept Gray's proposal that phobias may be considered as passive avoidance, while obsessional rituals are active avoidance, it is then less speculative to suggest the involvement of vasopressin in these processes. In the rat, vasopressin causes long-term retention of passive avoidance responses and resistance to extinction of active avoidance behavior. These behavioral effects can be observed days or weeks following one or several injections of the peptide (de Wied, van Wimersma Greidanus, Bohus, Urban & Gispen 1976). Unfortunately, it has not yet been investigated whether anxiolytics, preferably chronically administered, block these long-term behavioral actions of vasopressin, but the experiments can be easily done. Along this line, it is of interest that Gold, Reuss and Goodwin (1978) have hypothesized that central vasopressin function might be diminished in depression and facilitated in mania. Evidence supporting this hypothesis is not yet firm, and the therapeutic value of vasopressin in depressive patients remains to be established. However, our own findings and those of others suggest that changes in the modulation of noradrenergic transmission by a dysfunctioning central vasopressinergic system may be an important etiological factor in pathological brain processes.

Inhibition, attention, and the hippocampus

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Many subscribe to the idea of a thoroughgoing biopsychology; few are able to show how it might be accomplished. Gray's *The neuropsychology of anxiety* is a pioneering effort in this direction and one that future biopsychological models will have to take into account.

Gray's monograph is remarkable for its attempt to demonstrate homologies among three phenomena at widely different levels of analysis: (1) the hypothetical construct of the behavioral inhibition system (BIS); (2) the anatomically defined septo-hippocampal system (SHS); and (3) clinically defined anxiety disorders. We find Gray's separate discussions of each of these phenomena to be illuminating advances over analogous past formulations. Gray's concept of the BIS is more attractive than simple aversive conditioning models of anxiety, his view of SHS function is more adequate than O'Keefe and Nadel's (1978) spatial mapping hypothesis, and his description of clinical anxiety is much more sophisticated than earlier behavioral approaches. [See also *BBS* multiple book review of O'Keefe & Nadel; *BBS* 2(4) 1979.]

In general, we agree with Gray's analysis of hippocampal function, which is consistent with the present consensus (e.g., Elliott & Whelan 1978; Moore & Solomon 1980; Olton, Becker & Handelman 1979). Gray is careful not to argue that the SHS

is the anatomical locus of the BIS but rather is part of many neural systems and is therefore involved in a variety of behaviors. The one-structure/one-function approach, such as the spatial mapping hypothesis of hippocampal function, has been difficult to defend.

Yet we are somewhat dubious regarding Gray's success in linking SHS function to the BIS on the one hand and to clinical anxiety on the other. Specifically, we question whether the concept of inhibition adequately links the SHS-BIS relationship and whether the concept of attention adequately links the SHS-clinical anxiety relationship.

The view that the SHS is involved in behavioral inhibition is not novel (e.g., Douglas 1967; Kimble 1968), nor has it gone uncriticized (e.g., Olton et al. 1979; Solomon 1979). As we understand it, Gray defines inhibition as a BIS output that causes an interruption of ongoing behavior and is accompanied by increased attention to environmental stimuli. Yet learning theorists distinguish among a variety of forms of behavioral inhibition and their controlling processes (see Boakes & Halliday 1972). For example, one line of research that produces a problem for Gray's view concerns the role of the SHS in Pavlovian conditioned inhibition. In the conditioned inhibition paradigm, CS_A (conditional stimulus A) is always followed by a reinforcing UCS (unconditional stimulus) except when it occurs in a compound with CS_X . In this task the animal learns to respond in the presence of CS_A but not in the presence of the compound $CS_A + X$. Because CS_X is a signal for UCS nonoccurrence, one would expect it to trigger BIS-SHS activation, according to Gray's argument. Yet hippocampal lesions do not disrupt the animal's ability to inhibit responding in the presence of CS_X (Solomon 1977). Gray's theory would not seem to account adequately for this form of behavioral inhibition.

Similarly, Gray attempts to link SHS function to clinical anxiety via the concept of attention. Specifically, he suggests that opening the hippocampal gate should have the effect of rendering more stimuli as "important, needing checking," and that these attentional anomalies are central to clinical anxiety. This formulation raises two questions: (1) What is the nature of any attentional dysfunction in clinical anxiety? and (2) Does this correspond to the current understanding of the attentional aspects of hippocampal function? Unfortunately, the experimental psychopathology of attentional dysfunction in anxiety is incomplete, so that Gray's formulation is at best only intuitively appealing. The one solid empirical generalization in this area appears to be that anxious patients show delayed habituation of the electrodermal OR (orienting response), an anomaly that can be reversed with sedative drugs (Lader & Wing 1966). As Gray points out, however, the relationship between OR habituation and SHS function is tenuous. Hippocampal damage has little effect on peripheral OR parameters.

Our view, based on work with the latent inhibition and blocking paradigms, is that the hippocampus functions in its attentional aspect to ignore or "tune out" stimuli that are informationally redundant or motivationally irrelevant. Disrupted hippocampal function causes a failure to ignore irrelevant stimuli, leading to attentional perseveration (Solomon 1979; Solomon, Crider, Winkelman, Turi, Kamer & Kaplan 1981; Crider, Solomon & McMahon, in press). This position is compatible with Gray's treatment of the SHS, but it is both conceptually and empirically dissociable from the known autonomic OR anomalies among anxious patients. As we have pointed out, our view of hippocampal function is not inconsistent with descriptions of selective attention dysfunctions seen among schizophrenic patients (Chapman & Chapman 1973; Crider 1979). It is entirely possible that the selective attention deficit seen in schizophrenia will also be found among the anxiety disorders. At this point, however, there is little if any empirical justification for linking the known functions of the SHS to attentional anomalies in clinical anxiety.

In sum, we find Gray's monograph a stimulating excursion

into biopsychology. On the other hand, the concepts of inhibition and attention may not be strong enough to carry the theoretical burden Gray places on them.

Homunculus in the subiculum

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Orientation and arousal, tuning out of irrelevant stimuli, spatial mapping, memory consolidation, behavioral inhibition, contextual labelling, control of voluntary movement, formation of engrams for classical conditioning: Although not complete, this list illustrates the great diversity of the ideas that people have had about hippocampal function. The diversity has produced some cynicism about the theoretical enterprise in regard to the hippocampus, and even those individuals brave (or foolhardy) enough to attempt recently to add additional theories to the list have approached this apologetically, often declaring (as does Gray 1982a, p. 295) that what is to be presented is not really a new theory, but just a new twist which borrows heavily from the past.

Yet it is not at all clear that apologies for new theories are needed. Several of the older theories, although haloed by many citations, are obsolete because they were made too early to be influenced by critical pieces of data, or because they were made with too little of the extant data in mind. Also, as Gray points out (1982a p. 257) with his characteristic eloquence, "several [theories] appear to shade rather easily into one another: is sensitivity to interference different from loss of contextual labelling, non-reward from novelty, detection of familiarity from recognition memory? Perhaps we are dealing with a master of disguises." Thus new theories are needed if they (1) parsimoniously subsume the old and the new facts, and if they (2) reduce to common denominators sets of older theories that are superficially diverse but substantively equivalent. Gray's idea (that the hippocampal system performs detection of mismatch between incoming and expected stimuli, and behavioral suppression in the service of reprogramming to meet the new circumstances) is important because it does both of these things.

The theory is truly remarkable and perhaps unsurpassed in providing a framework for review and integration of a tremendous range of data, including major portions of the psychopharmacology of anxiety, limbic system neuroanatomy, neurophysiology, and neuropsychology, and the experimental psychology of animal learning and motivation. It also provides a way to combine the merits and thus to reduce to a common denominator several of the more cogent earlier theories, including the ones which stressed the hippocampus's role in behavioral inhibition, tuning out of irrelevant stimuli, and working memory. It avoids major gaps present in other theories, such as the spatial mapping theories' inability to account for disruption, due to hippocampal damage, of performance in tasks (e.g., the blocking paradigm) devoid of any spatial component. Also, the theory provides a way to resolve major empirical paradoxes (not known when many of the earlier theories were constructed) such as the failure of hippocampal lesions to impair the performance of voluntary behavior despite the striking correlation between such behavior and the hippocampal EEG, or the failure of lesions to alter the acquisition of the classically conditioned nictitating membrane response, despite the very dramatic hippocampal neuronal correlates of that acquisition demonstrated by Berger and Thompson (1978a) and their colleagues. Finally, the theory is exceptional in pointedly incorporating a salient contribution of hippocampal functioning shown repeatedly in lesioning studies (but ignored in most of the other theories), that is, the inability to suppress well-established

patterns of learned behavior in response to changed environmental contingencies.

A problem with the book is that Gray's fanciful, literary style sometimes hinders communication. Thus the poetic definition of a working memory task as an "ever-changing list from which the animal must choose the correct alternative, itself ever-changing" leaves the present reviewer hung up in a loop.

Our own studies have demonstrated that neuronal discharges in the cingulate cortex and in the reciprocally interconnected anteroventral (AV) thalamic nucleus differentiate between the auditory S+ and S- used for discriminative avoidance conditioning in rabbits (Gabriel, Foster, Orona, Slatwick, & Stanton 1980). Gray's statement about these results, that the differentiated activity represents an aspect of the "prediction," to be compared by the hippocampal mismatch detection system to incoming stimuli, is interesting and challenging. However, his subsequent assertion that the AV nucleus "is a point of access for response-independent environmental regularities," whereas the cingulate cortex "which has close relations with motor systems" (p. 269) plays a role in response-dependent environmental regularities, seems at best unclear and at worst to be the reverse of what it should be. This is so because the cingulate cortical differentiated response develops early in training prior to the advent of significant discriminative behavior, and it declines in the late acquisition stages when the behavioral response is well learned. In contrast, the differentiated activity of the AV nucleus develops in the late training stages and persists through overtraining in parallel with the performance of the discriminative behavior. From these data one would be inclined to argue that the AV nuclear activity is the more response-dependent, and the cingulate cortical activity, the more response-independent.

During reversal training the cingulate cortical neurons showed an enhancement of the differentiated discharge appropriate to original acquisition (the old learning) and although this activity declined during the course of reversal learning, no reverse neuronal differentiation developed in this structure as the rabbits acquired the reverse behavioral discrimination. Certainly, this absence of correlation with the behavior cannot favor Gray's view of the cingulate cortex as having close relations with motor systems.

One final problem is the occasional tendency of Gray to attribute homuncular properties (whole sets of cognitive functions) to single components of the system. This is illustrated by his statement (p. 282) that

when the mismatch is important . . . the subicular comparator will ensure that the 'faulty' programme is executed with increasing hesitation, that exploratory activities are undertaken, and that behaviour is varied in the attempt to light upon a more effective programme. If alternative response strategies are possible (e.g. taking a different turn in a maze), the comparator will instruct them to be tried out.

It is not impossible that the subiculum can do all of these things, but Gray's theory gives no insight into the ways in which they are accomplished.

"Antianxiety and opiates"

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The septal-hippocampal circuit has been the object of intense study and scrutiny over the years. Various hypotheses have been put forward concerning the function of this area of the brain, most of which have been limited by their methodological specificity. With an eye to this serious difficulty in the literature, Gray has done an impressive job integrating a number of diverse methodologies and hypotheses. The line of argument which he follows throughout is scholarly and well presented. The ambition to build connecting links between the psychological phe-

nomenon of anxiety and an underlying neurobiological substrate is an exciting model of scientific endeavor, not only for scientists in this area, but also for researchers and thinkers alike concerned with other realms of animal and human behavior. The conclusion or point of view which Gray reaches is that the septal-hippocampal circuit is involved in the detection of novelty that carries a fundamental importance for the subsequent behavior of the animal. In looking for a plausible neurobiological site for the phenomenon of anxiety in animals and, by inference, in man, he argues that anxiety involves the operation of those areas of the brain susceptible to the influence of antianxiety drugs. He then proceeds to build a strong case for the striking similarities seen in conditioned behavior as a result of antianxiety drugs and septal-hippocampal lesions. Consequently, anxiety is seen as one aspect of the response of the organism to the detection of novelty or nonreward in the environment or in the animal's intended behavioral program, mediated by the septal-hippocampal system.

Gray also carefully reviews the role of noradrenergic input from the locus coeruleus and serotonergic input from the raphe nucleus to the septal-hippocampal area. He concludes that both are involved in the mediation of anxiety. In doing so, he makes reference to our work which suggests that the opiate withdrawal syndrome is an intensified anxiety or panic state (Gold, Pottash, Sweeney & Kleber 1979). Opiates have anxiolytic effects which relate to their abuse by physicians, musicians, and other performers. In keeping with this, the most consistently reported effect of opiates in a group of naltrexone-blocked and placebo-treated individuals is not euphoria but antianxiety (Mirrin, Meyer & McNamee 1976). We have presented phenomenological and behavioral data which have demonstrated marked similarities between human opiate withdrawal and naturally occurring panic states (Gold, Pottash, Sweeney & Kleber 1980; Sweeney, Gold, Ryan & Pottash 1980). We have suggested that this may reflect a common central noradrenergic (NE) hyperactivity (Gold, Byck, Sweeney & Kleber 1979; Gold & Kleber 1979). This locus coeruleus (LC) hyperactivity hypothesis is also supported by our primate studies, which have demonstrated striking behavioral similarities between opiate withdrawal, electrical stimulation of the LC, pharmacological activation of the central noradrenergic system by piperoxane or yohimbine, and reversal of these anxiety-panic states with clonidine, morphine, or endorphins (Redmond, Hwang, Baulu & Gold 1979; Gold & Redmond 1977; Redmond, Gold & Ywang 1978; Gold & Pottash 1981).

The similarities in signs and symptoms exhibited by patients withdrawing from opiates and by patients experiencing spontaneous attacks of panic anxiety are also seen after abrupt discontinuation of chronic clonidine administration, further supporting the hypothesis of LC hyperactivity. These similarities suggest a possible common endorphin-noradrenergic mechanism mediating the syndromes of opiate withdrawal and panic anxiety. Thus, naturally occurring panic states might result from a functional deficit in endorphin-mediated inhibition of ascending NE activity (Gold, Redmond & Kleber 1979; Gold, Pottash, Extein & Kleber 1980; Gold, Pottash, Sweeney, & Kleber 1980; Gold & Pottash 1981).

It may be of interest that our work has demonstrated clonidine, a potent alpha-adrenergic agonist, to be successful in blocking the symptoms of opiate withdrawal only in a time-limited manner. After fourteen or more days there can be episodic reoccurrence of signs of withdrawal or panic attacks in individuals who remain opiate free as a result of protracted abstinence. This suggests that the regulation of anxiety via noradrenergic input to the septal-hippocampal circuit as proposed by Gray is perhaps too simple a model to account for these data. We have proposed catecholamine-peptide hypotheses to explain them.

Along similar lines, why do some people experience anxiety as reinforcing? They seem to seek out states of intense anxiety in

order to have them end and they experience this termination as reinforcing. Examples include sky-diving, dare-devil stunts, and automobile racing. Experienced in this way, anxiety most probably represents less the detection of novelty and more a state of anticipation of novelty or nonreward which is deliberately solicited. One possibility which Gray does not elaborate is that the septal-hippocampal system may "turn itself off" through activation of endorphin-mediated pathways arising in the hypothalamus which terminate on the LC. Endorphin termination of catecholamine and other neural system hyperactivity may be self-reinforcing.

Finally, we wonder about the role of cardiovascular mechanisms in generating an anxiety attack. The condition known as mitral-valve prolapse is frequently an underlying cause of severe panic attacks accompanied by anxiety. Such panic attacks are virtually indistinguishable from naturally occurring panic attacks. Treatment with inderal results in the disappearance of the attacks. This is a condition in which the peripheral components of autonomic nervous system reactivity may result in a subsequent elaboration by central neurobiological and psychological mechanisms of an anxiety state. A phenomenon of this sort makes it important for any theory of central mechanisms underlying anxiety to address the crucial question of central-peripheral interactions. We welcome the results of future studies, which will benefit greatly from the fruits of Gray's integrative efforts and provocative reformulations exemplified in this volume.

Integrating the literature on anxiety, memory, and the hippocampus

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For those who admire Jeffrey Gray's theoretical acumen, his new book will be a disappointment. The first chapter offers a précis of his formulation of a behavioural inhibition system (BIS) which was derived from a two-process theory of learning and developed in considerable depth in his book of 1975.

In chapters 2–9, we find excellent up-to-date reviews of the anatomy, electrophysiology, and lesion/behaviour studies relevant to the septo-hippocampal system. Much of this data has been reviewed before, if not by Gray, in the symposium volumes on hippocampus which have appeared in profusion in recent years. From this body of data different theoretical positions emerge depending on the emphasis placed on the particular kind of experiments and the results they yield. The unit-recording work and anatomical considerations were influential in the theory developed by O'Keefe and Nadel (1978) and the relationship between theta rhythm and voluntary movement in Black's writings (see Black 1975); the clinical evidence of amnesia associated with hippocampal pathology gave impetus to the memory theory. It is results relating to the drug and lesion effects that have influenced Gray most in the formulation of his theory that the septo-hippocampal system mediates the response to anxiety.

In chapter 10 he rapidly disposes of the alternative views of hippocampal function with the statement "we can reject then, memory consolidation, voluntary movement and spatial analysis as accounts of septo-hippocampal function" and goes on to present his own interpretation of the data. This points to a central role for the septo-hippocampal system in the expression of anxiety, which is generated as a consequence of activation of the behavioural inhibition system by certain classes of stimuli. The effects of lesions and drugs on paradigms which generate behavioural inhibition are the important sources of information for the formulation of this theory. The quasi-physiological theo-

ry of septo-hippocampal function proposed draws both on anatomical data and (heavily) on the electrophysiological studies of Vinogradova (1975). Its central assumption is that the septo-hippocampal system acts as a comparator. The theory not only accommodates Gray's own experimental findings but also those of Caffan and Olton that hippocampal disruption impairs the familiarity judgements germane to recognition memory (Caffan 1976) and working memory (Olton, Becker & Handelmann 1979).

As far as Gray's own work is concerned the validation of the theory lies in the demonstration that the hippocampus, together with its afferent inputs, is indeed the substrate through which anxiolytic drugs, at least the benzodiazepines, have their effect on behavioural inhibition. For a number of years he and his collaborators have tenaciously pursued this problem. They now have a substantial amount of positive support for their proposition that the ascending serotonin (5HT) and noradrenaline (NA) inputs to the septo-hippocampal system are essential for the actions of anxiolytics. These results, and complementary studies from other laboratories, are reviewed in depth in chapter 11. Any theorist worth his salt will pursue a parsimonious unitary theory until it has to be dropped. I, personally, am not yet convinced that the pharmacological manipulations which influence the response to nonreward and punishment (the two key stimulus inputs to the behavioural inhibition system) are as similar as the theory demands. Whereas 5HT and NA manipulations induced with drugs or lesions have the predicted effects on nonreward, such that a reduction in their level of activity has an effect equivalent to that of benzodiazepines on behaviour, this is not true of their effects on responding controlled by punishment contingencies. Benzodiazepines produce a massive release of punished responding which is not matched by a lesion to the NA innervation of hippocampus. 5HT lesions, on the other hand, do release punished responding. As a correspondence of nonreward and punishment is central to Gray's theory, these results may prove to be a real challenge to parsimony. However, the great virtue of the theory is that it makes many predictions and our task is now to test them.

In the final chapters of the book Gray relates the experimental findings to the clinical phenomenon of anxiety. He is at his best when discussing these issues in relation to biological psychiatry, which accounts for the enthusiasm many clinicians feel for his earlier book *The psychology of fear and stress* (1971a). He raises largely the same issues in the present text, although it is brought up to date with a discussion of the interesting phenomenon of learned helplessness. I think Gray's speculation that some subgroups of depressed patients may indeed have a neurochemical dysfunction overlapping with that proposed to exist in anxiety is a provocative and interesting one. It is certainly the case that some depressed patients respond favourably to anxiolytics.

Many experimentalists and clinicians will find Gray's new book useful, not because it raises many new issues, or indeed presents much new data, but because it aims to integrate a vast neurobiological literature and focus it on a dimension of behaviour. It is difficult to think of any other area of research which has reached the same level of integration. Many will disagree with this, claiming that the level of knowledge is equally great in other areas such as sensory processing and perception, but that one has been spared there from over-synthesis. Gray lays himself open to this kind of criticism by including in this book aspects of forebrain function not previously included in his neuropsychological model of anxiety. The chapters on memory and frontal lobes stand out as being largely unintegrated, at either the theoretical or experimental level. One has the feeling that Gray has accepted the established facts that the hippocampus is in some way related to memory processes and that the frontal lobes are anatomically related to the hippocampal system and thus concludes that it must surely be possible to integrate these facts into his model of the septo-hippocampal system and

anxiety. On neither count is he convincing and these chapters seem to me to spoil the otherwise coherent theme of the book. I think it is premature to claim that the septo-hippocampal system serves a unitary function which will account for the mediation of anxiety and memory. It may serve both by independent use of its neural connections. Whether Gray is right or wrong, his efforts to integrate our present level of knowledge will be an important impetus to the continuing efforts to define the functions of this intriguing brain system.

The evolution of hesitation, doubt, and map-making

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Gray paints upon a large canvas, some areas of which are intensely detailed while others are (with no disrespect intended) relatively gray. Sometimes the effect is disconcerting, as when we contrast the detailed exposition of the precise effects of anti-anxiety drugs on the behaviour of animals with the inevitably grosser behavioural analysis of the symptoms of anxiety in man. Sometimes, too, it is possible to cavil at what is said, or left unsaid, in the areas of detailed analysis: for example, even though it is mentioned that different species use differing modes of exploration, that is not a full explanation of why hippocampal theta frequencies should vary so widely across species when hippocampal anatomy appears relatively constant. Nevertheless, in addressing the important topic of hippocampal function, which forms the linchpin of *The neuropsychology of anxiety*, Gray has meticulously presented a theory that takes care of an extraordinarily high proportion of the available data. In this commentary I wish to enlarge the canvas still further by suggesting briefly a possible evolutionary background to this "organ of hesitation and doubt" (Simonov 1974). I do so, not by adding any original ideas, alas, but by pointing to a marked parallel between the primary function of the hippocampus, as proposed by Gray, and the intellectual development that Jerison (1973) considered to be the key to the survival of the early mammals in an age dominated by reptilian species.

Jerison's data are casts (endocasts) moulded from the internal surfaces of fossilised crania. These, he contends, provide a reliable indication of the gross anatomy of the brains that were once housed in those skulls. Although there is some debate concerning the significance of the brain/body size ratio as an indicator of brain processing power (Rensch 1955; Passingham 1975), it is a good enough measure for the purposes of this necessarily brief and simplified account. From these data it is clear that the emergence of mammals (and, to a lesser extent, birds) was associated with a dramatic increase in brain/body volume ratio. How did this increase in brain capacity relate to their fitness, in a Darwinian sense? Jerison argues that the earliest mammals (or their immediate reptilian ancestors) were unable to compete with the ruling giant reptiles for possession of the lush, warm pastures and swamps of the middle Mesozoic period. Already equipped with a thermoregulatory mechanism, the mammals were able to survive in the uncrowded but cooler nocturnal niches. The well-developed visual system that these creatures probably inherited from their reptilian ancestors would not have been much use to nocturnal species: how, then, could these animals gain the sensory impression of spatial relationships in their environment that is crucial to the survival of an active, mobile organism?

The early mammals probably inherited an impressive olfactory sense and a rudimentary auditory sense. These senses may be used to locate the sources of distant stimuli, but only by coding distance and direction temporally. This would be achieved by

storing data concerning successive stimuli and comparing them. Changes in stimulus strength and quality over time, together with information concerning the organism's own movements, provide the raw material from which a sort of map may be constructed. (This process, it will be noticed, serves the same ends as that ascribed to the hippocampus by O'Keefe & Nadel [1978] but by a somewhat different means.) [See also BBS multiple book review of O'Keefe & Nadel, *BBS* 2(4) 1979.] The enlarged brain of early mammals, Jerison contends, is associated with the evolution of a system capable of carrying out these functions. At the heart of this system lies the comparator mechanism which enables constancies and inconstancies between previously sampled and currently impinging stimuli to be recognised. Jerison makes the point that even if such a device were initially dedicated to the service of a single sensory modality (e.g., olfaction), it would be in accordance with the general pattern of evolution if it were found to be involved in achieving other related functions at a later stage. Thus he sees the system as exploited to its best advantage when, following the climatic calamity that wiped out the giant reptiles, mammals were able to fill the newly vacated daytime habitats. It is likely that those nocturnal mammals tending towards daytime activity would succeed according to the development of a sophisticated visual system that could cope with the rapidly shifting patterns of relatively high intensity to which, owing to their agility and their forest habitats, they were now exposed.

If Jerison's hypothesis is accepted, a mechanism capable of imposing the necessary constancy on this complex world had already evolved in the form of integrative, comparator, and memory capabilities subserving (at least) the senses of olfaction and audition. The system would achieve this end by enabling, through successive comparisons, generalisations and distinctions to be made between patterns of stimuli. In the case of the mammalian visual system, it is to be expected that the information to be stored and processed would be complex, perhaps along the lines delineated by Sutherland (1968) for solving just this kind of perceptual problem. Again, however, a comparator mechanism would lie at the heart of the process. Gray's theory would strongly suggest that the development of the capacity to compare information from two sources – a memory store and a sensory area of the brain, say, or perhaps two memories – was contingent upon the evolution of the paleocortical hippocampus.

Jerison's (1973) account of the crucially important functions of a comparator at its earliest stages of evolution is not without significance for the notion of the hippocampus as comparator. In the first place, if such was indeed its origin, it could account for those close anatomical ties of the septo-hippocampal system with the olfactory system that resulted in its early description as a part of the rhinencephalon. Secondly, it may be that "cognitive mapping" of a sort was indeed the prime function of the hippocampus, though its involvement was (and is) as a comparator of stimuli past and present, rather than as the site of the resulting "map." As other areas of the brain evolved, if they did so in a piecemeal fashion, the hippocampus would be available to them, too, as a unit specialising in the single process of comparison of its varied inputs. In humans one may speculate that the hippocampus is involved in functions ranging from those as simple as orientation towards the source of a regular stimulus that has ceased (Gray 1975), to those as complex as the comparison, during rapid eye movement (dreaming) sleep, of the day's events with one's existing "world view" (resulting, perhaps, in the revision of the one to compensate for apparent inconsistencies with the other; Hartmann 1973).

There is not space here to rehearse all the implications of this view. Many of them could be (and some, indirectly, have been) tested by comparative and developmental studies of anatomy and of behaviour. But it is interesting to consider that, while all evolution may be a series of accidents, the functions of the mammalian hippocampus as outlined in *The neuropsychology of*

anxiety may be no more than a spin-off from a system that once increased the survival chances of creatures condemned to, but ill-equipped for, a life of darkness.

Putting anxiety in its place?

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Jeffrey Gray's is a rich and provocative book, packed with experimental data and theoretical ideas. Because of its amount of detail, those concerned with anxiety and septo-hippocampal functions will probably want to read it for themselves. Readers familiar with O'Keefe and Nadel's recent book, *The Hippocampus as a Cognitive Map* (1978), will see a somewhat parallel structure in Gray's book. There are chapters in both devoted to the anatomy of the septo-hippocampal (SH) system, the electrophysiology of the system, lesion effects, and so on. It is very interesting that Gray comes to quite different conclusions regarding the probable functions of the septo-hippocampal system than do O'Keefe and Nadel, after an analysis of virtually the same body of data. [See also BBS multiple book review of O'Keefe & Nadel 1978: *BBS* 2(4) 1979.]

Although Gray's theoretical approach to the functions of the septo-hippocampal system is not easily captured in a few words, the flavor of his conception can be appreciated by a few quotations: Gray views the SH system as a "comparator," the function of which is "to compare, quite generally, actual with expected stimuli." It is postulated to have two general modes of operation: "just checking" (not controlling behavior) and "control" mode (directly controlling behavior). In my view, this distinction, while plausible, tends to weaken the predictive value of the theory, as it is not always apparent before the fact in which mode the SH system will be operating, although Gray does try to grapple with this problem in a series of "scenarios" in which the theory is applied to some available data. But with the experimental tools at present at our disposal, it is difficult to generate predictions from Gray's theory which cannot also be fairly readily generated from other, quite different theories. It would be intriguing to see a list of differential predictions concerning, for example, the behavior of hippocampal-lesioned animals, which would be offered by Gray on the one hand, and O'Keefe and Nadel on the other. I count it as somewhat of a disappointment that the present volume, while clearly able to handle much of the available data with one or another aspect of the proposed theory, does not clearly lend itself to unique predictions. This is, I acknowledge, an extremely tough challenge to any theory, and one which has not yet been met in a satisfactory manner. For example, when a hippocampus-lesioned rat makes repeated errors in a spatial maze immediately after running an errorless trial 30 seconds earlier, is it because the rat has momentarily lost its "cognitive map," because it is deficient in its "working memory system" or its "reference" memory system, or because it has difficulty in inhibiting an erroneous or "run-away" motor program which has somehow gained momentary control of its behavior?

Gray opts for the latter explanation, viewing the SH system as "labelling" or "red-flagging" nonadaptive motor programs: "The tagged motor program is given enhanced attention in the subicular comparator. This is the heart of the theory. For the most important function it attributes to the septo-hippocampal system is that of scrutinizing motor programs that have gone wrong." Again, while this is an interesting idea, operational tests of how this idea could be proven wrong are not obvious.

In terms of the neural circuitry proposed to underlie the functions of the septo-hippocampal comparator, I also find it difficult to conceive of how to test the elaborate wiring proposed by Gray. For example, if a unit in the hippocampus responds to a

particular form of sensory stimulation, how does one judge that that unit is "just checking" or performing some other more "controlling" function? Theories must be testable to be useful, and while Gray's ideas here are often plausible, I suspect some of them are not testable in their present form. Gray's Figure 4 summarizes the theory developed in the book.

It is certainly easier to criticize an effort of this magnitude than to accomplish it, and Gray deserves great credit for the synthesis of theoretical ideas and experimental data which comprise this book. It is certainly well worth the effort to read, and will undoubtedly provide ideas for many workers in a variety of areas. If the theory falls somewhat short of one's ideal in terms of clarity and testability, that is perhaps a reflection of the potential scope of Gray's ideas, which is considerable indeed.

Conditioned suppression and behavioural inhibition

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The "face validity" of a theory of anxiety depends in part on its providing an account of those sorts of human behaviour that we label as anxious. If, as in this case, the theory rests mainly on animal evidence, then it should also encompass data from experimental paradigms that are thought to generate anxious or fearful behaviour in animals. In the latter case there is a clear discrepancy between Gray's theory and conventional wisdom. I believe that Gray's theory would be strengthened and brought into closer contact with the vast literature on conditioned fear if this discrepancy were removed.

The discrepancy is this: while signals of punishment and innate fear stimuli are said to activate the behavioural inhibition system, and thus cause anxiety, conditioned aversive stimuli do not, according to Gray, necessarily have this effect. This is at variance with the view that the presentation of a conditioned stimulus (CS) previously paired with an aversive unconditioned stimulus (US) elicits fear or anxiety. In particular, it has often been proposed that the conditioned suppression of positively reinforced operant behaviour reliably elicited by an aversive CS (Estes & Skinner 1941) results from, or is associated with, fear. Independent evidence for this comes from at least two sources: anxiolytic drugs generally reduce conditioned suppression (Millenson & Leslie 1974), and fear responses such as freezing are seen during conditioned suppression (Brady & Hunt 1955) and are sometimes correlated with the amount of suppression (Hoffman & Barrett 1971). Gray, however, rejects the idea that conditioned suppression is an example of fear, in the sense of directly activating the behavioural inhibition system, because of the effects of benzodiazepines, barbiturates, and alcohol on conditioned suppression. Having marshalled evidence from many sources to support the view that these drugs affect the behavioural inhibition system, Gray concludes that they do not systematically affect conditioned suppression. I believe that this conclusion is erroneous and will briefly review the relevant studies. (Most of these studies are discussed in more detail by Millenson and Leslie 1974, and by Gray 1977.)

The experiments can be divided into three groups according to the procedures used: (a) those using the original Estes-Skinner procedure in which CS and US are repeatedly presented while the operant response is being performed ("on-the-baseline"); (b) those using the Estes-Skinner procedure but with "off-the-baseline" CS-US pairings arranged to occur at a time or a place where the operant response cannot be performed; (c) those using variants of a procedure devised by Leaf and Muller (1965) who measured the time taken for a consum-

matory response to recover after one or a few trials in which a CS (sometimes just apparatus cues) had been paired with a shock in an off-the-baseline procedure. In category (a) there are numerous studies showing that acute doses of benzodiazepines or barbiturates, administered after suppression has been established, reduce conditioned suppression (e.g., Miczek 1973; Miller 1961). There are fewer studies in category (b), but Cicala and Hartley (1967) with chlórdiazepoxide, and Miller (1961) with sodium amylobarbitone, found that drug treatment reduced conditioned suppression previously established by off-the-baseline CS-US pairings. Category (c) is problematical because the two most carefully conducted studies produced conflicting results. In a series of experiments, Tenen (1967) found that chlórdiazepoxide and other benzodiazepines given during testing reduced suppression, while Scobie and Garske (1970) found a marked effect of chlórdiazepoxide only when it was given during CS-shock pairings. While the case is thus undemonstrated for category (c), we may safely conclude that the Estes-Skinner procedure, with either on-the-baseline or off-the-baseline CS-US pairings, produces conditioned suppression that is reduced by benzodiazepines and barbiturates. (There are too few studies with alcohol or septal or hippocampal lesions to draw conclusions from them.) Gray (1977) arrived at a different conclusion by lumping categories (b) and (c) together as off-the-baseline procedures. He has gone on, in the volume now under review, to suggest that there are fundamental differences in the outcome of off-the-baseline and on-the-baseline procedures. The present analysis suggests that this is unlikely to be true. Indeed, there is no evidence for it from other conditioned suppression studies where on- vs. off-the-baseline comparisons are possible (e.g., Leslie & Garrud 1976).

Several benefits accrue if we accept that aversive CSs that elicit conditioned suppression act on the behavioural inhibition system. In general, it allows us to retain the plausible assumption that both such CSs and signals of punishment have many of their common effects by acting on the same system. More specifically, it enables us to start tracing connections between the behavioural inhibition system and conditioned suppression. On the basis of drug studies, Gray decided that the behavioural inhibition system is involved in the early stages of two-way active avoidance learning, but not later on. Kamin, Brimer, and Black (1963) trained rats on a two-way avoidance task and then tested to see whether the discriminative stimulus for the avoidance response would, when presented in another apparatus, suppress positively reinforced lever pressing – a category (b) “off-the-baseline” procedure. They found more conditioned suppression to this stimulus early in avoidance training (after 9 successful avoidances) than later on (after 27 successful avoidances). The parallel between behavioural inhibition and conditioned suppression seems compelling.

Some questions of strategy in neuropsychological research on anxiety

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Gray's book shows that the author has assimilated a massive amount of research about the septo-hippocampal system and anxiety, and it distils a great number of his own experimental investigations in the same area: both labours command admiration. So it is against this general background of approval that I wish to pose some theoretical questions.

At the beginning of his book, Gray (1982a, p. 3) suggests that “a description of the psychological processes which are altered by the anti-anxiety drugs should be a description of the psychology of anxiety . . . [and] that a description of the neural pro-

cesses which are altered by these drugs is a description of the neurology of anxiety.” I am going to suggest that this central two-pronged argument is invalid and that the consequences are far from trivial.

1. The neurological prong of the argument. Let us take the second prong of this argument or, to be more exact, the second of what are two arguments conjoined. It is of the following logical form:

If X (antianxiety drugs) reduce Y (anxiety) by altering region of the brain Z (the septo-hippocampal system, SHS),
Then Y amounts, neurologically, to Z.

That this is invalid reasoning can be seen by transposing the same argument into another context so that its conclusion will be shown transparently not to follow from its premises.

If X (a potato in the exhaust) reduces or disrupts Y (the functioning of an internal combustion engine) by altering Z (the free outflow of gases along the exhaust system),
Then Y amounts, mechanically, to Z.

Of course, the functioning of an internal combustion engine is not fully explained in terms of the functioning of its exhaust system, even though the latter may be an integral part of the former. It is invalid to argue that if so-and-so disrupts something, then a description of the workings of the particular part that was disrupted will amount necessarily to a description of the mechanisms of the whole or even of the part in question. Of course some disruptions may be so complete that a description of what they disrupt and how they disrupt will include a description of all the essential workings of what they disrupt. But not all disruptions are like this, and Gray's strategy, in implicitly presuming that they are, is invalid.

Now this point is not a trivial logic-chopping “philosopher's point”; it has enormous ramifications for the research in question. For, when Gray's strategy is laid bare in this way, it follows, even if he has shown correctly that antianxiety drugs reduce anxiety by radically altering the functioning of the SHS, that he has not discovered *thereby* what is the fundamental neurological mechanism of anxiety. He may have laid bare only the exhaust system, so to speak, and failed to note the workings of the fuel and filter systems, the ignition system, the combustion chambers, the piston mechanism, and so on.

Gray's position is not protected by pointing out, as he does, that by triangulation with reference to a variety of antianxiety drugs he has pinpointed the specific antianxiety neurological effects of the drugs in question and avoided confusing them with, in the context, their side effects. For my point is not to deny that his strategy corners neurological mechanisms essential to anxiety but to deny that with his strategy he can be sure he has cornered *all* or even most of what is peculiar, neurologically, to anxiety rather than, say, common to grief or joy or surprise as well as to anxiety. The SHS may be important to the functioning of several emotions yet not be a factor which is distinctive of any of them, just as an exhaust system is integral to both conventional and turbo engines yet is not the factor which differentiates them (though it is not part of the functioning of electric engines, so that reference to it will serve to separate electric engines from the former).

To draw out the ramifications further, those of us who give an unambiguously cognitive account of the psychology of emotions, such that the factor which differentiates one emotion from another is a fairly sophisticated cognitive function, such as an evaluation, will be surprised if it does turn out that this sophisticated process is *wholly* accommodated by the SHS and the Papez circuit. But such surprise is admittedly still supported by little concrete evidence, for no one has as yet isolated an evaluative centre (or circuit or system) in the brain, though Gray's “comparator” may be related to what cognitivists have in mind. So all that I really wish to argue here is that, while Gray may have isolated the fundamental neurological basis of anxiety,

his strategy does not, as he seems to think, ensure that he must have done so; and the difference between must and may could be massive. Indeed, in chapter 10 of *The Neuropsychology of Anxiety* (henceforth *Anxiety*), when reviewing the evidence concerning the function of the SHS, Gray himself is much more modest in the claims he makes for the results of his strategy, admitting that "anxiety may be based on structures that are more widespread in the brain than the septo-hippocampal system alone." Indeed if, as Gray suggests, the basic function of the SHS is to act as a "comparator between expected and actual stimulation" for some animal, then it would seem unlikely that the SHS can by itself explain the difference between anxiety and many other emotions. Fairly obviously, joy and surprise, to take just two examples, can result from the "comparator" comparing expected and actual stimulation. Slender hopes unexpectedly fulfilled by actual stimulation – that is, a realised gap between expected and actual stimulation – can lead to surprise or joy and, probably, a number of other emotions, including those at the other end of those spectra. But more on this in the next section.

2. The psychological prong of the argument. Now it is time to look at the first, psychological prong of Gray's basic argument. It is of the following logical form:

If X (anxiolytic drugs) in reducing or suppressing Y (anxiety) suppresses psychological process P,
Then P amounts to a psychological profile of Y.

This argument, like the previous one about the neurology of anxiety, is formally invalid and for similar reasons. One cannot conclude from the given premises that P is necessarily the psychological profile of Y, or indeed that it even captures most of the essentials. The psychological processes suppressed by anti-anxiety drugs, and so the psychological profile that results from Gray's strategy, which is parasitic on the above argument, in fact looks decidedly thin as a profile of the psychology of anxiety when examined carefully.

In *Anxiety* (p. 459), Gray outlines this profile in the following way: "Activity in this [behavioural inhibition] system constitutes anxiety," and this system is in turn to be described in terms of its major inputs, namely "stimuli that warn of punishment or non-reward, novel stimuli, and innate fear stimuli," and its major outputs, "inhibition of ongoing motor behaviour, increased level of arousal and increased attention to the environment, and especially to novel elements in the environment." In this profile, what is described under input is, strictly speaking, the input as processed by the putative "comparator," which compares input with expectation of input so that it then labels the input as novel or nonrewarding or some such.

Now the thinness of this as a profile of anxiety can be seen by the fact that a number of situations which satisfy the above model would not satisfy many people that they are examples of anxiety. Or, more strongly, a number of emotions which are clearly not anxiety would easily satisfy Gray's model of anxiety which results from his research. For example, a person receives a stimulus and pronounces it to be novel; this in turn inhibits his ongoing motor behaviour and causes him to experience increased levels of arousal and increased levels of attention towards the novel stimulus. While this profile fulfills Gray's conditions for the person being in a state of anxiety, it also fits with that person being in a state of awe or excitement or admiration. The person might be a birdwatcher who has just seen his first golden eagle in full flight; he stands still gazing upward, his attention riveted, his excitement intense at seeing something so unexpected and novel. Or the person, of deprived background but with an innate liking for music, quite by chance and completely unexpectedly strays into a free open-air concert with Ashkenazy playing Mozart.

I would like to say much about the psychology of anxiety which is, I think, quite complex and subtle, but given the confines of this review, I shall have to be content with a few brief remarks. Anxiety is not, I think, characteristically about the

present, novel or otherwise, but about the future. In particular, it is about some future event whose outcome from the vantage point of the present, at least to the person concerned, is uncertain. Further, the outcome matters a great deal to the person concerned and may be such as to demand a great deal of him or her in the way of future or present action. (I am inclined to think that a greater insight into the nature of anxiety is to be gained from clinical work; see, for example, Rycroft 1970, ch. 1.) [See also Panksepp: "Toward a General Psychobiological Theory of Emotions" *BBS* 5(3) 1982.]

It seems to me that Gray's "comparator," while on the right track, goes only a very little way along the road – the evaluative road – which I believe must be traversed by those who wish to give an adequate taxonomy of the emotions. Let me illustrate this. The anxious person does not merely believe there is a disparity between expectations and present or future events; he believes that future events are demanding (of him or his "quasi-self," friends, relations, etc.), uncertain, yet of great importance, and can lead to failure or some similar "falling short" on his part. The anxious person evaluates future events, or rather likely or predicted future events, according to a very complicated set of rating or evaluative scales – the certain/uncertain, demanding/undemanding, and important/unimportant scales for a start – and certainly not just according to an anticipated/unanticipated or aversive/nonaversive scale. (For my fuller speculations about the evaluative component in emotions, see Lyons 1980, chs. 3, 4.)

Noradrenaline: Attention or anxiety?

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Gray's *Neuropsychology of Anxiety* is truly a magnificent opus. The synthesis of behavioural, neurochemical, and clinical data is a major advance in our study of the brain. It seems almost churlish to quibble with even the slightest point in its all-embracing compass. However, the role ascribed by Gray to the noradrenergic input to the hippocampal-septal system (SHS) is based on few direct experiments, fails to explain a lot of data relating to central noradrenaline (NA) function, and is directly contradicted by a few further experiments.

Gray suggests that the dorsal noradrenergic bundle (DNAB) acts to tag stimuli as "important" by serving as a general alarm bell. I have suggested (Mason 1981), on the contrary, that the DNAB tags stimuli as "unimportant, irrelevant" and is part of a Broadbent (1958) sensory filter which acts to exclude stimuli impinging on the organism that are irrelevant to the task in hand. By incorporating a role for the DNAB in fear and anxiety as suggested initially by Redmond and associates (Redmond, Huang, Snyder & Maas 1976) Gray ignores the evidence that in the rat 6-hydroxydopamine (6-OHDA) lesions to the NA input to the SHS fail to impair behaviour motivated by fear in situations such as one-way active avoidance (Fibiger & Mason 1978), two-way active avoidance (Mason & Fibiger 1979d), passive avoidance (Roberts & Fibiger 1977; Mason & Fibiger 1978c; Mason & Fibiger 1979a), Sidman avoidance or a conditioned emotional response (Mason & Fibiger 1979b). None of these paradigms shows a decrease in fear motivation such as the Gray model would have to suggest if the DNAB were acting as a general alarm bell. Further, electrolytic or 6-OHDA lesion to the source nucleus of the DNAB, the locus coeruleus (LC), fails to affect one-way active avoidance (Sessions, Kant & Koob 1976) or alter a social-interaction measure of anxiety (Crow, Deakin, File, Longden & Wendlandt 1978). This has been reviewed more extensively elsewhere (Mason & Fibiger 1979a) and on balance the evidence would seem to contradict a role of the DNAB in fear or anxiety.

What about nonreward? While Gray is right on the behavioural level, that NA depletion from the SHS leads to resistance to extinction – this is called the dorsal bundle extinction effect (DBEE) by Mason and Iversen (1979) – he is not necessarily right to go on to argue that this indicates a role of the DNAB in nonreward processes *per se*.

Resistance to extinction after 6-OHDA lesion to the DNAB is now well established in numerous behavioural situations (see Mason & Iversen 1979) since its initial demonstration in a runway paradigm (Mason & Iversen 1975). However, the parameters of this effect are such as to rule out its being due to the mechanism suggested by Gray. Thus, the DBEE occurs after negatively reinforced behaviour as much as after positively reinforced learning. It has been seen in extinction of one-way active avoidance (Fibiger & Mason 1978; Ashford & Jones 1976), in extinction of two-way active avoidance (Mason & Fibiger 1979d; Plaznik & Kostowski 1980), and sundry other negatively reinforced tasks (see Mason 1980; 1981). While Gray's theory of the DNAB can explain increased resistance to extinction after omission of an expected reward such as a food pellet, it cannot explain increased resistance to extinction after negatively reinforced behaviour.

Further, the role of the DNAB suggested by Gray cannot explain why the lesion must be present during the acquisition learning of a continuously reinforced (CRF) task in order to cause resistance to extinction when the animal is subsequently placed into extinction testing (Mason & Fibiger 1979b; Mason 1979). Lesions of the DNAB after acquisition of CRF but prior to extinction do *not* cause resistance to extinction. An explanation in terms of selective attention (Mason 1980; 1981; Mason & Iversen 1979) is successful in explaining these parameters of the DBEE. Thus, Gray is not entitled to cite the dorsal bundle extinction effect as evidence for a role of the DNAB in nonreward, since the parameters of such resistance to extinction are incompatible with the predictions of the nonreward theory and the effect can best be explained on an attentional theory. Thus, the main, and perhaps sole, line of evidence for a role of the DNAB in nonreward processes is no evidence at all.

Gray's theory also ignores a wealth of data on the effects of NA depletion, which, while not directly contrary to the nonreward theory, is not at all explained by it. Thus, why are DNAB lesioned rats more distractible by irrelevant stimuli than controls (Koob, Kelley & Mason 1978; Roberts, Price & Fibiger 1976; Mason & Fibiger 1978a; Oke & Adams 1978)? Why does a lesion to the DNAB impair acquisition of visual and auditory discriminations (Mason & Iversen 1977; 1978; Mason & Fibiger 1979f; Mason & Lin 1980) and their reversals (Mason & Iversen 1978b) but not acquisition of spatial discriminations (Amaral & Foss 1975; Koob et al. 1978; Roberts et al. 1976) or their reversals (Roberts et al. 1976)? Why do NA depleted rats show better nonreversal shifts than controls (Mason & Fibiger 1979f; Mason & Lin 1980) and why do they show an attenuation of the blocking effect (Lorden, Rickert, Dawson & Pellemounter 1980) and attenuation of the latent inhibition effect (Mason & Lin 1980; Mason & Fibiger 1979f)? All of these can be explained, and indeed were predicted in advance of their discovery, by the attentional theory (Mason & Iversen 1977). Thus, Gray's theory leaves a lot of the behavioural effects of NA depletion unexplained, while its rival, the attentional theory, can encompass these.

Finally, some experiments run directly counter to the specific predictions of Gray's theory. Mason and Iversen (1978a) tested the ability of DNAB lesioned rats to associate stimuli with nonreward – something central to Gray's theory and for which DNAB lesions are predicted to yield an impairment. They used an escape from the S-minus paradigm described by Rilling, Askew, Ahlsgog, and Kramer (1969) in which during a successive visual discrimination task a second lever is introduced into the chamber during the S–, which has the sole effect of turning off the S– for a brief period. Normal rats learn to press this lever

since this has the consequence of terminating an aversive situation (the S– which is associated with nonreward). So too do DNAB rats, indicating that they are every bit as able as controls to associate stimuli with nonreward, despite the contrary prediction of Gray's theory.

Finally, as described as Gray, the attentional theory and the nonreward theory make different predictions as to the effect of DNAB lesion on partial reinforcement schedules. Gray's theory suggests an attenuation of the partial reinforcement extinction effect (PREE) such that partially reinforced DNAB rats are *less* resistant to extinction than partially reinforced control animals. The attentional theory predicts that, unlike continuously reinforced training where DNAB rats are *more* resistant to extinction than continuously reinforced controls, following partially reinforced training both DNAB and control rats will be *equally* resistant to extinction.

There are three distinct experiments in the literature dealing with this question. One, performed by Owen, Boarder, Gray, and Fillenz (1982) and described earlier by Owen, Boarder, and Gray (1977) in abstract form, finds that a DNAB lesion reduces the resistance to extinction of partially reinforced rats trained in a straight runway for 50 acquisition trials. The second study, by Owen, Boarder, Feldon, Gray, and Fillenz (1979), and comprising Experiment 3 in Owen et al. (1982), used 100 acquisition trials and found that despite better than 90% NA depletions no change in the extinction of partially reinforced DNAB rats occurred relative to partially reinforced controls. Finally, Mason and Fibiger (1978b) found that after a variable ratio schedule in a Skinner box (not the classic situation used to obtain a PREE but nonetheless an operationally defined partial reinforcement situation), DNAB and control rats extinguish identically. Thus, it would seem that two out of the three experiments available in the literature to date support the prediction of the attentional hypothesis and fail to confirm that of the nonreward theory.

In summary, some direct experimental tests of the nonreward role ascribed to the DNAB seem to go against it. However, the direct tests are few, and some of the experimental results are uncertain. The most useful function of Gray's book, even if its global theories turn out to be wrong, would be to stimulate research interest and motivate experimenters to return to the laboratory and find the answers to those clear-cut predictions of the rival theories.

Gray's Neuropsychology of anxiety: An enquiry into the functions of septo-hippocampal theories

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Gray's book has great breadth in the range of topics it covers and depth in the detail with which many of these topics have been reviewed. The theory of hippocampal function he has presented has the virtue that it covers anatomy, physiology, and behaviour in a way attempted by few other theories. We can judge it in at least three ways: in relation to the only previous large-scale theory of this type (O'Keefe & Nadel 1978); for its consistency with the data on which it is based; and for its capacity to cope with the details of individual studies.

In terms of the workings of the hippocampus itself, Gray's theory is very similar in form to O'Keefe and Nadel's and hence in many cases it makes essentially similar predictions. The omission of specific spatial and storage functions for the hippocampus improves the fit to some of the data. However, the theory also shares a fault with that of O'Keefe and Nadel. In both cases predictions depend not only on well-specified machinery

within the hippocampus but also on less well-specified machinery outside it. The *exact* nature of "predictions," "behavioural inhibition," "familiarity," and "plans" and the nature of the possible interactions between them are likely to play a role in the present scheme similar to that played by "taxon hypotheses" and "place" in the previous one. The ambiguities of such terms would be less important if the presentation of the theory included a rigorous derivation of the predictions for all the commonly used behavioural tests, especially in relation to any critical behavioural parameters. Unfortunately we are left to derive such predictions for ourselves. [See also *BBS* multiple book review of O'Keefe & Nadel. *BBS* 2(4) 1979.]

This is not to say that Gray's theory fails to make a number of unequivocal predictions. In fact, at least one of these is incompatible with the data he reviews. The hippocampal theta rhythm is assigned the task of coordinating (timing or quantising) *all* of the functions of the hippocampus and its interactions with, for example, the Papez circuit. It follows that the theory predicts that lesions of the medial septal nucleus, which abolish theta throughout the hippocampus, should reproduce all of the behavioural effects of hippocampal lesions. Unfortunately for the theory, they do not (Gray 1982a, pp. 286–87). This requires an immediate modification of Gray's theory, albeit one that its basic structure can probably survive.

Some more specific points can be made about Gray's treatment of theta rhythm and especially of the septal elicitation of theta. The threshold for the septal driving of hippocampal theta rhythm in the free-moving male rat shows a minimum in the middle of the normal theta frequency range. This minimum occurs as part of a *symmetrical* parabolic function (James, McNaughton, Rawlins, Feldon & Gray 1977) and hence, far from being an obvious dividing line between two types of theta, it is quite good evidence for the homogeneity of at least some theta mechanisms in relation to frequency. On this basis, Gray's statement that the dorsal bundle facilitates 7.7 Hz theta should more properly be taken as saying that it facilitates *all but extreme* frequencies of theta (i.e., 6 and 10 Hz).

A more important inaccuracy in this statement about the dorsal bundle (and equivalent statements about the minor tranquillisers, etc.) is the implicit assumption of a relation between the threshold for theta driving and the actual production of theta under normal circumstances.

There is no frequency-specificity in the production of theta rhythm by reticular stimulation, the function relating intensity of stimulation to frequency being linear (McNaughton & Sedgwick 1978). Further, manipulation of noradrenergic systems does not affect the frequency of theta, whether this is elicited by reticular stimulation (McNaughton & Sedgwick 1978) or occurs spontaneously (Robinson, Vanderwolf & Pappas 1977). The minor tranquilliser sodium amylobarbitone does reduce the frequency of theta elicited by reticular stimulation (McNaughton & Sedgwick 1978), but unlike the effects it has on septal elicitation (McNaughton, James, Stewart, Gray, Valero & Drewnowski 1977), it has no special effects in relation to 7.7 Hz theta. There is therefore no reason to relate changes in the threshold curve for septal elicitation to changes in theta frequency in the free-moving animal. Even when one might wish to attribute an action of the minor tranquillisers to an effect on theta mechanisms, there is no necessity to relate this to changes in noradrenergic systems. Further, there is no reason to suppose that the noradrenergic control of theta is particularly evident in the awake animal; the behavioural effects of noradrenergic lesions are just as likely to be due to an interference with theta-generating mechanisms during sleep (Gray 1982a, p. 355).

Gray's interpretation of some of my behavioural data also differs from my own. There is strong reason to suppose that septal and hippocampal lesions affect two separable components of spontaneous alternation (Dalland 1970). I have presented data which suggest that this difference may be more

quantitative than qualitative (McNaughton & Feldon 1980) and on the basis of this Gray has equated the effects of septal lesions, hippocampal lesions, and minor tranquillisers on this task. However, I also presented evidence showing that the effects of the minor tranquilliser sodium amylobarbitone are more selective than those of septal lesions. So, whether or not we can equate the effects of septal lesions with those of hippocampal lesions on this task, we cannot extend the parallel to minor tranquillisers.

When fitting the results with spontaneous alternation into the theory, we should also note the nature of the two types of alternation. Spontaneous alternation does not appear to be a reaction to specific stimuli within a novel environment (O'Connell 1964) but rather it is composed of odour trail avoidance and to a lesser extent alternation of body turns (Douglas 1966; McNaughton & Feldon 1980). It is difficult to see how this, and especially the specific action of amylobarbitone on body turning as opposed to odour trail alternation, fits into the model shown in Figure 4 of Gray's accompanying Précis.

I have presented the above simply as examples of the kind of detail with which modern theories should cope. Gray's is by no means unique in failing to do so. Few hypotheses in this area even attempt to deal with the available range of data on the hippocampus. The theory of O'Keefe and Nadel (1978), like Gray's, does make this attempt. Like Gray's theory, it assigns a central role to the theta rhythm while giving no good account of the data on the septal elicitation of theta. Similarly, it glosses over the nature of spontaneous alternation (an odour trail is not, after all, a place).

I have focused on the above simply because they relate to my own experiments. There are many other cases where the assessment of treatment effects should take into account the detailed structure of behaviour: open field behaviour (Whimbey & Dennenberg 1967), two-way avoidance (Calderazzo-Filho, Moschorakis & Izquierdo 1977), and the partial reinforcement extinction effect (Gray 1982a, pp. 39–40) being only some examples.

However, the relative success of these two detailed theories of hippocampal function suggests that their construction is a useful exercise. There are certainly sufficient data to be accounted for. In fact, rather than asking which of the theories is the better we can note that neither can cope with all the data in any straightforward fashion. While both theories are sufficiently detailed to be wrong, neither is sufficiently formal to be unambiguous. Even Gray's book, which starts with a discussion of learning theory, finishes with a theory couched in terms the behavioural definitions of which are clearly open to question. It is to be hoped that future theories will use more ostensive terms, embedded in a more formal structure. The difficult job of the construction of such theories should be considerably eased by the degree of order brought into the data by O'Keefe and Nadel and by Gray.

On novelty, places, and the septo-hippocampal system

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This book will be read and discussed by many, from quite diverse perspectives. In our commentary we limit ourselves to Gray's position on the functions of the hippocampus. And, basically, we come to praise Gray, not to harass him. There are many things about this book that we like. It is a genuine piece of scholarship, an attempt to bring together diverse data within a coherent theoretical framework. It speaks to important issues in

psychology and psychobiology. In contrast to most other accounts of hippocampal (or septo-hippocampal) function it speaks to data from a wide variety of paradigms and approaches. On this ground alone it deserves our respect. Further, we find ourselves in general agreement with Gray's claim that the hippocampus is a generator of predictions and evaluator of reality – a mismatch detector – though we prefer to see these functions as being executed within a spatial mapping framework, at least as regards exploration. And we are in strong agreement that the key to understanding the inhibitory defect after septo-hippocampal damage lies in the specification of the normal function as a consequence of which this system is involved in inhibitory processes. This is far better than saying it is an "inhibitory" structure and leaving it at that.

One of us (L.N.) has written about a special "spatial mapping" function for the system Gray is considering (O'Keefe and Nadel 1978; 1979), and feels that Gray has dismissed this position somewhat prematurely. But this is not the place to take up the argument in detail, as it depends on careful analyses of many lesion studies and their particulars. We leave it to others to decide on the relative merits of the two cases. In order to do so, they will need to read this book. When they do so, they might find the following comments useful.

1. In his discussion of lesion effects Gray, in contrast to O'Keefe and Nadel (1978), does not include studies with damage limited to fimbria and fornix, though he does mention such studies once in a while. This strategy, he claims, is necessitated by the fact that the partial damage caused by such lesions (owing to the recently discovered alternative pathway out of the hippocampus via the subiculum) is not appropriate for studying a neural system. There is some truth to this, but not when such a study demonstrates deficits. If a partial lesion elicits a defect, are we not justified in assuming that a total lesion would do at least as much? By such exclusions he comes to the view that the hippocampus is *not* involved in original learning, a position we find indefensible. In fact, there are a number of studies involving hippocampal, as well as fimbria/fornix lesions, demonstrating impairments in the acquisition of place-learning tasks in rats. We are unconvinced by Gray's attempt to explain away these deficits, which he cites, in terms of interference effects (see below for further elaboration of this point).

2. Though there are some indications to the contrary, it appears as if Gray does not quite understand the implications of the notion of space at the center of cognitive map theory. When local cues are available to an organism, it can easily learn where to go, or where not to go, on the basis of these cues. Without such cues, getting to a location *at a distance* requires some internally represented notion of places. This abstract quality of place is well demonstrated by the ability of rats to swim in a water tank directly to the *place* where a platform lies submerged and invisible in the milky water. Morris (1981; Morris, Garrud & Rawlins 1981) shows that this learning ability is absent in rats with hippocampal damage. The ability to approach a place defined only by the cues distant from the tank (and, crucially, by their relations) is quite different from the ability to approach or avoid an object/stimulus within the immediate grasp of the perceptual systems. This occasional confusion in Gray's discussion of the predictions of the spatial map theory should be borne in mind by those comparing the two approaches.

3. While Gray understands that one cannot fob off the inhibitory defect caused by hippocampal damage by simply postulating that this area is involved in some simple inhibitory function (as we noted above), he fails to see that the same logic must apply to what appears to be an increased susceptibility to interference after such damage. Thus, he offers as an "explanatory" alternative to the spatial theory the idea that certain deficits are due to the fact that the tasks employed were "sensitive to interference effects." Of course, this merely redescribes the defect. Spatial theory, right or wrong, offers an explanation of these effects in terms of differences between place and other kinds of learning in

this domain. Place learning is less susceptible to interference because of the maplike structure within which place information is represented (cognitive structure, not physical, is implied here), and the possibility of coding by spatial context, discussed below. Other kinds of learning are more susceptible to interference because the information on which they are based is stored in other kinds of representations, ones which do not have the properties of maps. While recent studies of intact rats offer strong support for this view (e.g., Maki, Brokofsky & Berg 1979), it remains an empirical matter. Our point here is that Gray does not address the issue. By virtue of what property or capacity of the septo-hippocampal system, as he sees it, is the organism protected from interference effects? It should be added that sidestepping this issue is common practice in the hippocampus literature. Some even go so far as to imply that this neural structure comes into play in situations requiring protection from interference, without bothering to tell us how the hippocampus "knows" which situations (or arbitrary laboratory tasks) require such protection.

4. Many feel, as does Gray, that hippocampal involvement in classical conditioning (e.g., Thompson 1976; Solomon 1977) finds no natural home in spatial-map theory. However, it is becoming increasingly clear that spatial context plays an important role in a variety of learning tasks which are themselves not spatial (e.g., Dweck & Wagner 1970; and many others). Such involvement, though poorly understood at present, could provide the basis for the emerging role of a mapping system in classical conditioning (cf. Nadel & Willner 1980). Some exciting recent data reported by Jim Ranck and John Kubie (at Neurosciences, 1981; and the Asilomar Conference, 1981; Kubie & Ranck 1981) suggest that spatial cells in the hippocampus code both specific places (with bursts of high-frequency activity when the animal is in the place) and overall contexts (with background "tonic" firing rates specific to individual contexts, and seen when the rat is outside the place field). These results are preliminary, and precise understanding of the role of the hippocampal system in spatial context effects remains a task for the future. The point is, however, that spatial map theory has a very natural application to a wide variety of conditioning studies which seem to bear little relation to anxiety theory.

5. Finally, some minor points. In discussing the use of drugs in the study of central nervous function, Gray states that it is reasonable to assume that such drugs impair the functioning of the neural structures they affect. Yet, he argues that electrical stimulation, surely a less specific CNS manipulation, could either impair, or mimic, normal neural function. This seems odd. Could it have anything to do with the kinds of studies Gray himself has done, and the sorts of interpretations he wishes to put on them? And Gray talks (in the book) about a study by Isaacson, Olton, Bauer, and Swart (1966) in which rats run from a shaking elevated maze to descend into shock. Somehow he deems such running *not* an escape behavior. We disagree. Later, Gray fails to see that prehabitation to a conditioned stimulus might have a different effect in one-way and two-way active avoidance. Within spatial map theory such a difference develops naturally from the differing roles played by place, and other, strategies in the two situations.

But enough of this quibbling. Gray's book is an honest attempt to wade through the maddening confusion that is the literature on the septo-hippocampal system, and should be treated as such – right or wrong.

Functions of the septo-hippocampal system

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In order to clarify some points, I have an experiment to propose. I ask Gray to predict the outcome he expects, and to interpret

the alternative outcomes. The experimental procedure involves two different discriminations. One is very sensitive to damage in the septo-hippocampal system, the other is not. If I understand Gray's position correctly, then (a) the processes of behavioral inhibition should be involved in the discrimination which is impaired following lesions of the septo-hippocampal system but not in the discrimination which is unimpaired, and (b) antianxiety drugs should produce the same dissociation as the lesions.

The experimental procedure is based on the one used by Olton and Papas (1979). In that experiment, the apparatus was a radial arm maze with two sets of arms. At the beginning of each test session, the arms in the *baited set* had one pellet of food placed at the end of each of them; the optimal strategy for the rat was to go to each of these arms once, and then not to return to them (i.e., to inhibit all responses after the first). The arms in the other, *unbaited set*, never had food on them; the optimal strategy for the rat was never to go to these arms (i.e., to inhibit all responses to them).

When first placed on the maze, normal rats responded to all arms, making both types of errors (returning to baited arms, and going to unbaited arms). With training, however, they learned to eliminate both types of errors, and eventually chose almost perfectly. Presumably, then, behavioral inhibition was involved in learning to avoid both types of errors.

The rats were then given bilateral lesions of the fimbria-fornix and retested on the discriminations. At the end of postoperative testing, a striking dissociation was present. In the unbaited set of arms, the rats chose as accurately as they had preoperatively, and almost never made a mistake by going to one of these arms. In the baited set of arms, however, the rats made many errors. Indeed, with complete lesions of the fimbria-fornix, rats responded no better than expected by chance in the baited arms, returning to arms they had already chosen as often as they went to unchosen arms. Thus, behavioral inhibition was completely intact for the unbaited arms, but completely eliminated for the baited arms.

I have the following questions. First, what concepts would Gray use to describe such a dissociation? If the concept of behavioral inhibition is used, what considerations lead to the expectation that such a process is involved in qualitatively different ways in the two sets of arms?¹ Alternatively, if the concept of behavioral inhibition is not used to describe this dissociation, what other concepts would Gray propose?

Second, what does Gray predict about the effects of antianxiety drugs on performance in this type of procedure? Should these drugs produce the same dissociation as seen following the lesions?

Third, if this experiment with the antianxiety drugs were carried out, how would Gray interpret the various possible results? Four patterns of performance are possible. Two of them show no dissociation: either no effect of the drug on performance in either set of arms, or an equivalent effect in both sets. The other two results do show a dissociation: either a relatively greater effect on performance in the baited set of arms as compared to that in the unbaited set, or a relatively greater effect on performance in the unbaited set of arms as compared to that in the baited set.

The procedures used by Gaffan (1974) and by Olton (Olton 1978b; Olton, Becker & Handelman 1979) are indeed very similar (Gray 1982b, p. 245). But I don't think that should lead one to be "suspicious" of the concepts guiding these research projects. At a practical level, rats learn discrimination procedures based on recognition memory considerably faster than those based on associative memory. Thus, if one is interested in assessing memory functions, recognition procedures provide a practical advantage as compared to associative procedures. The concepts of working and reference memory are orthogonal to those of recognition and associative memory, however, resulting in a 2 × 2 table of experimental procedures. Thus, each experiment by Gaffan and by Olton can be placed into one of the

four quadrants of the table (such as working memory and associative memory). (Indeed, to test the relative merits of the two positions with respect to describing the behavioral effects of lesions in the septo-hippocampal system, we are currently in the process of testing rats with fimbria-fornix lesions in a task that requires both working memory and associative memory. In such a task the rats have a severe impairment of choice accuracy. Thus, the poor choice accuracy of rats with septo-hippocampal lesions in the recognition procedures that Olton typically uses is due to the fact that they require working memory, and not because they can be solved on the basis of recognition memory.)

Various therapies have been used to treat anxiety (Gray 1982a, ch. 15, pp. 445–52). In addition to the two discussed by Gray, therapies using biofeedback have also been effective (Olton & Noonberg 1980). The physiological effects following biofeedback are similar to those seen in relaxation, but the means used to obtain these effects are markedly different. Does Gray see biofeedback as a third way to treat anxiety, differing in important ways from the other two? And more generally, what role does he see for biofeedback treatment, both at a practical level in helping patients, and at a theoretical level, in helping to understand the brain mechanisms involving anxiety?

One way of approaching the question of whether different brain structures are involved in the same functional system is to use a disconnection analysis (Geschwind 1965; Horel & Keating 1969; Mishkin 1958; 1966), which combines unilateral lesions and transections of interhemispheric commissures. When applied to the septo-hippocampal system, this kind of analysis does show that the projections to the septum are important in memory (Olton, Walker & Wolf, in press), a result which is consistent with those of more traditional analyses (Mitchell, Rawlins, Steward & Olton, in press). Thus, I think that the septum should be considered as being important in memory processing, a point that has apparently been overlooked in previous research (Gray 1982a, p. 233).

NOTE

1. Certainly the reinforcement contingency for the baited arms is different from that for the unbaited arms, and can be described accurately as a partial schedule (the first response is reinforced, all subsequent ones are not). But an important point is that this partial schedule, unlike that used in the alley way experiments, is not random, and normal rats are able to predict when a baited arm will have food and when it will not. This predictable pattern of reinforcement in the baited set of arms should not produce the same type of frustration as the unpredictable pattern in the alley way.

Anxiety viewed from the upper brain stem: Though panic and fear yield trepidation, should both be called anxiety?

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Gray's book is rich in fact, theory, and speculation. Much, indeed most, of the book can be appreciated without trying to fathom how it all relates to brain mechanisms of anxiety. I shall nevertheless restrict my comments to that cardinal, though often seemingly secondary, issue. The difficulties encountered in trying to understand anxiety are apparent in the scarcity of definitive psychobiological work in the area, and Gray proposes to attack the recalcitrant problem by unraveling how minor tranquilizers and behavioral inhibitory systems control brain functions and behavior. Reasonable enough. But the question immediately arises: do better strategies exist for establishing solid scientific contact with brain processes subsumed by the concept of anxiety? As summarized in my own target article in this issue. [See Panksepp: "Toward a General Psychobiological

Theory of Emotions" *BBS* 5(3) 1982.] I favor another approach – an approach to which Gray is surely no stranger but which is largely disregarded in his book

In my estimation, the simplest, most straightforward, and concrete reference point for the psychobiological analysis of anxiety is the study of brain circuits from which fear (e.g., flight) behaviors can be evoked by localized electrical stimulation. Why this alternative was not developed in Gray's splendid analysis of anxiety is perplexing. Although every approach has limitations, and complex problems need to be addressed from diverse but converging perspectives, during these early stages of inquiry into the biological nature of anxiety, it may behoove us to invest more of our efforts in approaches which have substantial face validity. Overall, Gray's approach is comparatively deficient in this desirable ingredient, if for no other reason than the difficulty of ascribing functional specificity to global pharmacological and general behavioral (e.g., inhibitory) approaches.

If we are to understand emotions, we need to capitalize on the simplest available strategies. Though nothing in the brain is simple, emotive behaviors evoked by diencephalic stimulation are a most compelling place to start a systematic analysis. It is doubtful that higher approaches will yield major breakthroughs until more definitive knowledge concerning lower mechanisms is achieved. There appear to be several reasons why such "simple" approaches have not been used more extensively.

First, the largely unfounded opinion that such states reflect activities of affectively vacuous motor-patterning mechanisms as opposed to key integrative mechanisms continues to influence thinking on the problem unduly. Humans do experience trepidation when stimulated in these brain areas, and animals will attempt to escape such stimulation (and it is noteworthy that the escape is attenuated by minor tranquilizers (Panksepp, Gandelman & Trowill 1970)).

Second, there has been a substantial retreat from this area of research since the revelation that behaviors elicited by hypothalamic stimulation exhibit considerable malleability (Valenstein, Cox & Kakolewski 1970). Not only has that finding remained perplexing for a decade, but it deflated hopes of analyzing motivation-specific brain circuits through the simple strategy of brain stimulation. However, with some tolerance of ambiguity, those variable behavioral effects may be a key to understanding the fundamental nature of emotions. Gray's approach is deficient at this level of brain organization. It is seeking to understand anxiety at higher reaches of the brain before the more fundamental neural foundations have been adequately explicated.

In attempting to study emotions with "a view from above" rather than "the view from below," one is inevitably confronted by complex brain decision-making processes, and all manner of quicksilver attributes of mind which, despite Gray's impressive attempt, remain controversial when applied to animal research. The "view from below," though requiring some mental analysis to stake out conceptual territories, can sustain a simplified, and hence testable, perspective more easily. The difficulty of the "view from above" is highlighted by how flexibly the concept of anxiety can be applied at that level. At lower levels of neural organization, anxiety can be restricted to mean the conditional activity of neural circuits which mediate fear (e.g., flight), while at higher levels one is confronted by experiences which range from general nervousness to panic. And so, the possibility of injudiciously mixing distinct brain processes increases.

For instance, are anxiety and panic only moderately different species within the same emotional class or fundamentally distinct phenomena? Gray treats the two as manifestations of a single central process, but it seems just as likely that they reflect activity in distinct, though synergistic, limbic circuits – anxiety arising from brain systems that elaborate fear, and panic from those that mediate separation distress. Pharmacological treatment of anxiety and panic attacks bears out the distinction.

While anxiety is reduced markedly by minor tranquilizers, panic attacks are not. Conversely, tricyclic antidepressants fail to blunt anticipatory anxiety, while markedly reducing the incidence of panic attacks in humans (Klein 1981), and separation distress in puppies (Scott 1974) and monkeys (Suomi, Seaman, Lewis, DeLizio & McKinney 1978). Minor tranquilizers are relatively ineffective in reducing separation distress, while opiates and clonidine (a drug figuring heavily in Gray's conclusion that increased brain norepinephrine activity is important in anxiety) are remarkably potent alleviators of separation distress (Panksepp, Meeker & Bean 1980).

If fear and panic are empirically distinguishable as fundamental processes, an amalgamation of their manifestations under the single concept of anxiety could yield inextricably tangled conclusions. For instance, if obsessive-compulsive disorders are related more to brain systems that mediate panic than to those mediating fear, then much of Gray's discussion in the last few chapters loses focus. Of course, one could use the term anxiety to refer to both of these entities, as did Klein (1964) earlier, but it now seems that conceptual distinctions need to be made between such processes in order to make sense of diverse forms of trepidation (Klein 1981).

Of course, we remain on the near shore of any substantive understanding of how the brain elaborates emotions. Different perspectives abound. Surely, many roads can lead to substantial advances in understanding such phenomena, and Gray's book provides a provocative sounding board for further clarification of essential issues.

The anatomy of anxiety?

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Gray's *The neuropsychology of anxiety* (henceforth *Anxiety*) is both a welcome book and an exasperating one. It is welcome because it counters O'Keefe & Nadel's (1978) "spatial" theory of hippocampal function with a wealth of detailed experimental evidence, mostly obtained (as is O'Keefe & Nadel's) with rats, and fits this evidence to an "anxiety" theory which is essentially identical with our "effort" theory (Pribram & McGuinness 1975). The book is exasperating in that it focuses so exclusively on the rat and practically ignores data on nonhuman primates, makes a somewhat superficial reading of those data when they are mentioned, and in a similar fashion does not quite penetrate the issues regarding hippocampal function in humans. In other words, we are so impressed with the positive aspects of the book that the omissions and superficialities become especially annoying. We will therefore concentrate on the omissions in order to alert the reader to them so that the full value of *Anxiety* can be appreciated.

For example, a set of nonhuman primate studies by Spevack and Pribram (1973) is dismissed as irrelevant (p. 163) because both amygdala and hippocampus were removed, but this fact is never mentioned with respect to the patient H. M. (Milner 1970) and *all* human subjects who are used as evidence for the effects of hippocampal damage. Gray ignores a whole series of studies that were performed on nonhuman primates in order to analyze the effects of medial temporal resections, and though some work involved replicating the entire lesions made in humans, literally dozens of studies were accomplished in which amygdalectomy or hippocampectomy was separately performed.

Another example is the "theory" attributed to Douglas (1967) by Gray – that the amygdala is involved in the processing of reinforcers while the hippocampus is involved in processing the currently nonreinforced aspects of a task. This is not a theory but a consistently obtained research finding. When Gray is dealing

with evidence obtained on rats, the experimental procedures and data obtained are carefully described and analyzed. Why should primate data be given such short shrift?

A similar criticism can be made regarding theory construction. Although the Douglas (1967), Kimble (1969; 1975), and Hirsh (1974) theories which emanated from our laboratory group are thoroughly aired, why is there no mention of the Isaacson and Pribram (1975) and Pribram and McGuinness (1975) formulations? Or those of Isaacson (1974), in whose laboratory Kimble and Douglas were also trained? Or the clinically based framework developed by Talland (1965; and *vide infra*)? These omissions would not be critical except for the fact that they anticipate and, in many respects, come up with formulations identical to those tendered by Gray. For example, Pribram and McGuinness point out that what they have labelled "effort" in their review is often called "chronic arousal" in the psychophysiological literature and "stress" and "anxiety" in the clinical literature. *Anxiety* would be strengthened, not weakened, by showing its affinity to earlier formulations arrived at from a different data base.

Now for a more important problem. In addition to the "anxiety" hypothesis, Gray continues to support a "behavioral inhibition" theory: "The safest conclusion for the moment, then, is that the septo-hippocampal system among its functions includes participation in the behavioral inhibition system" (p. 299). In fact, anxiety is defined in terms of "behavioral inhibition." Gray does suggest, however, that perhaps it is not "behavior" per se that is "inhibited" by the system but "motor programs" or "plans." Talland (1968), by contrast, had suggested that the hippocampus is involved in the *active* construction of plans. Several pieces of evidence support Talland against Gray. The first is the fact that whether hippocampectomized subjects are more distractible than control subjects depends on whether they are already engaged in carrying out a task (p. 278). If response inhibition refers to inhibition of distraction, then in at least some situations the subjects with hippocampal resections show an *increase* in response inhibition. Even more clearcut is the finding in nonhuman primates (Pribram, unpublished; Mahut 1971) and other species that hippocampectomized subjects, while deficient in performing spatial alternation, show absolutely no deficiency whatsoever in go/no-go alternation. They show no increase in errors on no-go trials, which of all task responses demand behavioral inhibition in its clearest form. Finally, there is Richard Thompson's (Berger & Thompson 1978a) electrophysiological evidence, which shows that a neuronal model of the nictitating response of rabbits is formed *prior* to the occurrence (not the inhibition) of that response.

The question then arises of how to relate the Talland "construction of plans" to "anxiety" or "effort" (and also to memory) rather than an "inhibition of plans." This problem is dealt with in a step-by-step fashion (the steps are taken from one set of data to the next) by Pribram and Isaacson (1975). Critical to this analysis is a view of the hippocampus as a comparator between "attentional" (input-regulating) and "intentional" (behavior-regulating) systems. Comparison may take effort as in "paying" attention; "thoughtful" retrievals from memory; or "intentional" inhibition of an impulsive response to distractors.

Perhaps, as shown by these examples, the major fault in *Anxiety* lies in its narrowness of focus. As Pribram and McGuinness (1975) detailed, the hippocampal system cannot be understood without reference to two systems represented in the forebrain by several basal ganglia: the amygdala of the limbic formations and the striatum (caudate/putamen). It is these systems which are primarily involved in orienting reactions (distraction) and their habituation – the stop mechanisms which Gray attributes to the hippocampus. The hippocampus, by contrast, mediates between stop and go.

Gray's analysis considers the septal nuclei as part and parcel of the same system as the hippocampus, but as a close review of the data show, the effects of septal and of hippocampal lesions are

not always the same. This is perhaps most dramatically shown in the effects on aggressive behavior (pp. 153–55). As Gray points out in his anatomical review (pp. 59–63), the septal region merges with the basal ganglia in the region of the nucleus accumbens. Recently, Isaacson (personal communication) has been able to counteract several of the effects of hippocampectomy with infusions of dopamine agonists into the nucleus accumbens.

Thus, although it is fashionable today, on the basis of electrophysiological and fornix anatomical connectivities, to consider the septo-hippocampal system as a unit, it might be profitable for some purposes to return to an earlier classification in which the septal/accumbens region and the amygdala were shown to share the same levels of processing. Pribram and Kruger (1954) classified amygdala and septal/accumbens together on the basis that they are only one synapse removed from the olfactory bulb and that they have both basal ganglia and more superficial ("cortical") nuclei in their makeup. The amygdala as well as the septal region has critical inputs to the hippocampus as shown by the chemical stimulation studies of Pribram and MacLean (1953).

When we turn to the neurochemical level there is the sense that this same narrowness interferes with theory construction. As already noted, "stop" and "go" systems are central to both the Gray and the Pribram/McGuinness theories. The critical data are the following:

Lindsley (Macader, Chalupa, & Lindsley, 1974) in keeping with many other recent publications (e.g., Fibiger et al., 1973; Ungerstedt, 1974) has dissociated two systems of neurons that influence the hippocampal circuit. One system originates in the median raphe and associated structures of the mesencephalic reticular formation, the other originates more laterally in the locus ceruleus and other portions of the periaqueductal grey. We have come to know these two systems in the present review as a serotonergic (indole amine) "stop" mechanism associated with arousal, and a dopaminergic and norepinephrinergic catechol amine "go" mechanism associated with readiness and activation. Lindsley's findings were obtained by electrical stimulations of the appropriate structures in the mesencephalic reticular formation. Such stimulations of the stop mechanisms produced hippocampal desynchronization and at the same time a synchronization of the amygdala circuits. (Pribram & McGuinness 1975, p. 130)

The key concepts (stop and go) relate to Gray's discussion of the two theories regarding monoamine transmitters involving serotonin and the dorsal ascending noradrenergic bundle. We suggested that the properties of these two systems may best be conceptualized as an interrupt or stop system mediated via the serotonergic pathway and an activating go system mediated by norepinephrine and dopamine. These designators fit the data reviewed by Gray on page 38 of the review (*Anxiety*, pp. 15–59) and are in accord with his conclusions. However, the anatomical and physiological data point to two separate systems which converge on the hippocampus rather than some unitary septo-hippocampal behavioral-inhibition function.

In addition, newer data show that the hippocampus, more than any other structure in the brain, contains receptors for adrenocortical hormones. While the stop and go functions are primarily indole (serotonergic) and catechol (norepinephrinergic and dopaminergic) aminergic and so are neuroregulatory, the hippocampus is seen as playing a critical role in the pituitary/adrenal "stress" system which modulates, by way of ACTH and other peptides derived from β lipotropin (such as the B enkephalins), the aminergic neuroregulatory mechanisms. Because of this higher order of modulation the functions of the hippocampal system are difficult to disentangle from those it modulates. This, we contend, is the reason why all the data reviewed by Gray do not fit a theory based solely on the septo-hippocampal system, a problem which he notes. Conversely, we find that the data *do* fit our three-tiered model, and find little in *Anxiety* to contradict it.

Thus, we do welcome Gray's detailed review of data and issues, and there are many commendable aspects of the volume not covered in our commentary. For example, one of the most important aspects of Gray's analysis relates to individual differences in response to stress, where he discusses an effect akin to learned helplessness produced in some individuals, but high anxiety states in others. Part of this difference may be due to the facility with which individuals generate endogenous ACTH and endorphins which, as noted in experiments by Maier, Davies, Grau, Jackson, Morrison, Moye, Madden, and Barchas (1980), leads to analgesic effects in some experimental animals but not in others. This effect could mimic learned helplessness in appearance, or could account for the rebound effect observed by Stone (1975). In yoked control animals experiencing inescapable shock an effect was noted which relates to the biphasic process discussed by Gray. Here the mediating agent was β -endorphin released from the pituitary. This agent produced an analgesic effect as measured by large increases in the threshold for pain.

Roth, Mefford, and Barchas (in press) have shown that the final product of the catechol chain is epinephrine (adrenaline) and that the most highly stress-linked neuropharmacological effect is depletion of epinephrine – falling in acute stress to 50% of base-line levels. Antidepressants appear to work *only* on animals with lowered base levels of epinephrine due to stress and not on normal controls. Similar depletions of epinephrine were found in depressed human patients.

It is therefore of some interest that human females do not show elevated levels of epinephrine in urine following mild stress whereas males do (Frankenhaeuser 1972) and that a great number of Gray's findings apply only to male rodents. Thus, while Eysenck's schema based on psychoticism (which Gray relates to learned helplessness and low catechols) and neuroticism (which Gray relates to anxiety and high catechols) scores is interesting and may be valid (Eysenck & Eysenck 1976), sex effects as well as endorphin release might add further dimensions, especially as learned helplessness is a more socially acceptable pattern for females.

These further examples indicate the richness of the fabric which Gray has woven into *The neuropsychology of anxiety*. The volume is certainly thoughtfully crafted and thought-provoking.

The relationship between memory and anxiety

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Recent attempts to describe the functions of the septo-hippocampal system have tended to concentrate on a role for the system in particular kinds of memory processing. At first sight, the title of Gray's book suggests that his views are in opposition to this trend: memory and anxiety seem a long way apart. In order to bridge this apparent gulf, Gray has undertaken a comparison of the behavioural effects of anti-anxiety drugs and of lesions in the septo-hippocampal system, with the intention of demonstrating that the two kinds of intervention have very similar consequences.

As very few of the data are new, much depends on the ability to reinterpret a body of lesion data almost identical to that interpreted by O'Keefe and Nadel (1978) as showing that the hippocampus is the neural basis of cognitive mapping. It is therefore rather critical for Gray's theory that his comparison between drug and lesion effects should show very close parallels between the two treatments: where this is not so, explanation is required.

If the two treatments produce similar effects, but to slightly differing degrees, discrepancies are not necessarily cause for alarm; if, however, the treatments produce similar effects, but in apparently different ways, then the similarity may be no more than coincidence. This would be cause for alarm. A case in point is the influence of inter-trial interval (ITI) on the treatment-induced blockade of the partial reinforcement extinction effect (PREE). At a 24-hour ITI, both the drug treatment and lesions in the septo-hippocampal system have clear effects on the PREE, in accordance with Gray's predictions (Feldon & Gray 1979a; Feldon, Guillemon, Gray, de Wit & McNaughton 1979). However, if the ITI is reduced to approximately 5 minutes – even if the number of acquisition trials is increased to 40 or 48 respectively – the drug effect is reduced or abolished (Ison & Pennes 1969), while the lesion effects are still entirely clear (Henke 1974; Rawlins, Feldon & Gray 1980). Increasing the number of acquisition trials to 64 at this ITI apparently re-establishes the drug-induced blockade (Gray 1969), while increasing the number of acquisition trials to 96 apparently eliminates the lesion-induced blockade (Henke 1974). Such apparent inconsistencies would be less important if there were no good reason for believing that, as ITI and the number of acquisition trials are changed, the PREE depends increasingly on different kinds of processes (Mackintosh 1974). The drug and lesion treatments might therefore be affecting behaviour through different psychological mechanisms.

One problem here, however, is that of "partial overlap": it is not clear to what extent the data derived from drug and lesion studies must agree in order for the theory to stand. The theory needs to be made more specific here.

It is equally the case that the neuroanatomical basis of the theory is hard to test without a good deal of more specificity being attributed to the various components and their neural links. But in the present state of our knowledge all that may be possible is to make rather arbitrary assumptions based on the slight information we have. Given the large number of loops in the system, it is very hard to predict exactly what should happen if, for example, the direct connection from the subicular comparator to the cingulum is severed. Would this abolish behavioural inhibition, or would the existing alternative routes (e.g., via the antero-ventral thalamus) preserve the function of the sectioned direct projection? This kind of problem can only be solved following a more precise description of the kind of information transformation which takes place at each stage of the circuits. At the moment, such a description has been attempted only for the hippocampal formation itself.

If we therefore concentrate upon the hippocampal formation itself, does the psychological description provided represent an improvement over previous descriptions? In that it has tried to incorporate much of the data, and some of the theoretical views, of its predecessors, it does have some advantages. For example, by emphasising a role for the septo-hippocampal system where responses need to be changed, it gets around the problem faced by the working memory theory (Olton, Becker & Handelmann 1979) which in its simplest form does not predict perseverative responding when reference memory needs to be changed (for example, when a previously consistently rewarded response is now extinguished). Again, Gray's theory handles the data from reversals of nonspatial discriminations more readily than can O'Keefe and Nadel's (1978) theory. However, in making a take-over bid for the data, Gray inevitably acquires some problems as well. For instance, Gray points out that the results of Rawlins et al.'s (1980) experiment were contrary to some of O'Keefe and Nadel's (1978) specific predictions, but later on points out that the new theory makes the same predictions as O'Keefe and Nadel did. [See also *BBS* 2(4) 1979.]

At the more general level, in reinterpreting both the data that led to the spatial theory and those that led to the various memory hypotheses, Gray relies heavily on the concept of interference. However, normal animals learning at least some

spatial working memory tasks (e.g., a discrete-trial, conditional alternation task on the T-maze) do not show any obvious signs of interference. Large lesions in the septo-hippocampal system do, however, produce a severe and lasting impairment of performance on such tasks. This deficit would be predicted by almost all theories of hippocampal function (including Gray's, at a general level), but in view of the excellent performance of the normal rats it is hard to consider it as a task which generates very high levels of interference, without some empirical demonstration in normal animals. The view that spatial learning per se is very prone to interference is highly appealing, but may not be shared by rats.

Has Gray, then, satisfactorily related anxiety to memory, in his reinterpretation of the animal experiments? The proposed link between the two relies on the way in which the animal treats uncertainty. Since in memory tasks uncertainty is held to result from interference, or from some ambiguity in the information provided, the theory must manage to define these (themselves uncertain) factors before the gulf is satisfactorily bridged. These problems do not affect the whole theory – many other sections of the book can stand or fall on their own – but if the functions of the septo-hippocampal system are the core, then they need further attention.

The dynamics of action and the neuropsychology of anxiety

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On the basis of effects on animals of drugs that alleviate anxiety in humans, Jeffrey Gray (1982a) proposes a neuropsychological model of anxiety. Fundamental to his model is the relationship between the behavioral inhibition system (BIS) and the septal-hippocampal system (SHS). Gray's central assumption is that the SHS acts as a comparator between actual and expected stimuli. Mismatches between predicted and observed stimuli activate the BIS which then inhibits ongoing behavior, increases arousal, and constitutes anxiety. Whether or not the physiology is correct, it is interesting to relate this model to the problem of human anxiety.

Gray suggests that his theory contradicts more conventional models of human anxiety (e.g., Spence & Spence 1966) and reports that there have been very few tests of his model in humans. While that is true, his model can be related to that of J. W. Atkinson and his associates who have developed a model that also conflicts with the conventional theories of anxiety, but which has been tested. It is useful to consider the parallels between Gray's model and the dynamics of action (DOA) of Atkinson and Birch (1970; 1978), for Gray gives a physiological basis for some of DOA and the evidence for DOA provides some support for the generalization to humans of Gray's model.

Atkinson and Birch propose that behavior represents the manifestation of latent or unobservable motivational tendencies. The DOA is a theory of how tendencies grow and decay and, more important, how they interact to cause behavior. It is this interaction between latent tendencies that we observe as the phenomena of choice, persistence, latency, and intensity. By definition, ongoing behavior reflects the dominant resultant tendency which is assumed in the DOA to be the algebraic difference between approach tendencies and avoidance tendencies. In an achievement situation (one in which the outcome depends upon the subject's effort) approach tendencies are thought to reflect situational conditions such as incentives and the probability of success, as well as individual differences in the

"motive" to approach success. It is also assumed that the tendency to avoid such an achievement situation reflects situational conditions such as threat and the probability of failure, as well as individual differences in the motive to avoid failure, which is usually equated with trait anxiety or evaluation apprehension.

An important part of the DOA is that both approach and avoidance tendencies are changed over time as a consequence of perceived success and failure (Kuhl & Blankenship 1979; Revelle & Michaels 1976). Success in an achievement situation for a human is analogous to being given a pellet in an alley for a rat. That is, it has a short-term consummatory effect. Failure for the human, on the other hand, may be seen as the withholding of an expected reward for a rat. As we can see in Gray's model for a rat, such a lack of expected reward leads to an increase in running speed. In the low anxiety human, it leads to an increase in the intensity of performance. Revelle and Michaels (1976) have shown how the greater probability of failure in a difficult task in combination with the carry-over of motivation on failure trials leads to a change in task preference as a function of time on task. Kuhl and Blankenship (1979) have extended this analysis to show how anxious individuals take longer to switch their preferences to more difficult tasks. They interpret this result in terms of the buildup of avoidance motivation, but it may just as well be understood as due to the inhibitory effects of anxiety.

In the DOA, avoidance motivation reduces resultant motivation and thus events that increase avoidance motivation have an inhibitory effect upon behavior. Further evidence in support of this position may be found in a study by Weiner and Schneider (1971) who examined the paired associate learning task, which Gray finds hard to understand in *The Neuropsychology of Anxiety* (henceforth *Anxiety*). Weiner and Schneider showed that when success and failure are manipulated independently of task difficulty, it is feedback rather than difficulty that interacts with anxiety to determine rate of learning. In other words, highly anxious subjects did better when given success feedback than did low anxiety subjects, but low anxiety subjects did better with failure feedback than did highly anxious subjects. Contrary to the earlier studies which confounded difficulty with success and failure, in Weiner and Schneider's study, difficulty did not interact with anxiety.

If we extend these results to the general case of reversal learning which in *Anxiety* is cited as impaired by antianxiety drugs, we find that DOA predicts that high anxiety should facilitate changing from a well-learned but suddenly incorrect task to a less well learned but correct task. This results from the greater increase of avoidance motivation for the initial task on the part of the highly anxious subject. An interesting prediction from DOA is that the benefit associated with high anxiety in reversal learning should be reduced if the original learning task is made aversive before it becomes incorrect or if the reversal response is both rewarded (say through food) and punished (through shock).

A fruitful exercise which is beyond the scope of this commentary would be to see how many of the behavioral effects associated with the antianxiety drugs can be predicted by DOA. If the fit is as good as it seems on the basis of the analysis of reversal learning, then *Anxiety* will have made a major contribution to our understanding of the phenomenon of human anxiety.

An additional strength of *Anxiety* is the distinction Gray makes between behavioral inhibition and arousal. By providing evidence for two components of anxiety (see also Morris & Liebert 1970), Gray forces us to reject simpler, unifactorial, models of human anxiety.

Finally, *Anxiety* is very useful in that it shows how it is possible to study anxiety without resorting to self-report inventories. The scope of behaviors that have been shown to relate to lesions of the SHS or to antianxiety drugs (Précis, Table 1) is a challenge to the human experimenter who normally chooses from a much more limited experimental repertoire.

Does hippocampal theta tell us anything about the neuropsychology of anxiety?

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In our initial attempt to compose a commentary we tried to distill some of the major points of Gray's theory on the role of the septo-hippocampal system in anxiety. The following consist of direct quotations or paraphrases extracted from the accompanying Précis:

The behavioural inhibition system remains a *psychological* model, not itself to be directly identified with any brain structures or systems. ("The behavioural effects of antianxiety drugs," para. 3)

Activity in the behavioural inhibition system constitutes anxiety. ("The behavioural effects of antianxiety drugs," para. 3)

The kinds of information-processing that go on in the septo-hippocampal system are of central importance in the state of anxiety. (Précis, para. 1).

[The theory] does not claim that the septo-hippocampal system . . . mediates only anxiety, nor that anxiety depends solely on that system. (Précis, para. 1)

The septo-hippocampal system is sensitive to the same stimuli which operate the behavioural inhibition system. ("Relationship to previous theories," para. 3)

Activity in certain brain systems under certain conditions is equivalent to activity in the behavioural inhibition system. ("The behavioural effects of antianxiety drugs" para. 3)

Since most of the theoretical part of Gray's book consists of statements such as those listed above, we soon realized that any kind of comprehensive critique would be an exercise in futility (not to mention the fact that we couldn't possibly match Gray's eloquent use of the English language). Therefore, we will restrict our comments to two subjects we know something about: (1) hippocampal rhythmical slow activity (theta), and (2) sex.

Theta frequency. Although it is not particularly clear what role hippocampal theta plays in Gray's theory, it is obvious it plays an important one. Much of Gray's own work has concerned the selective effects of antianxiety drugs on a narrow frequency band (approximately 7.7 Hz) of the hippocampal theta-driving curve (Gray 1970b; Gray & Ball 1970). This 7.7 Hz theta is considered functionally unique because it "represents a transition point between Vanderwolf's atropine-sensitive (low-frequency) and atropine-resistant (high-frequency) theta" (*The neuropsychology of anxiety*, henceforth *Anxiety*, p. 192). Gray's emphasis on low frequency, high frequency, and 7.7 Hz frequency theta points out a general feature of his theory, at least as it pertains to hippocampal slow-wave activity, and that is that theta frequency is important in defining which hippocampal system is active at any one time. Gray states, "different frequencies of theta might determine, not only the rate at which the septo-hippocampal system processes information, but also *which* information is processed (*Anxiety*, p. 293, his italics). For example, it is proposed that atropine-resistant theta is involved in checking motor programs, and its frequency varies with the rate at which information is received. The atropine-sensitive system, on the other hand, is brought into play when "smooth motor programming is interrupted by a mismatch" (pp. 292–93). Supposedly this lower-frequency theta allows a more detailed analysis of information. Unfortunately, there is no discussion of why antianxiety drugs should selectively influence that frequency of theta which represents a transition from "just checking" to "control."

In contrast to Gray's emphasis on theta frequency we will argue that frequency *cannot* be used to determine whether the atropine-resistant or atropine-sensitive system is active. This has been pointed out by Vanderwolf and his coworkers a number of times (Kramis, Vanderwolf & Bland 1975; Vanderwolf

1975; Vanderwolf, Kramis, Gillespie & Bland 1975). In a recent review article, Robinson (1980) stated that "these two systems are operationally defined on the basis of their relative sensitivity to drugs, but not on their relations to behavior or frequency spectra" (p. 84). The reason for this is that under some circumstances atropine-resistant theta may occur without movement and atropine-sensitive theta may accompany movement. Also, the frequency spectra of the two types of theta overlap considerably (see below).

There are a variety of situations in which high frequency (>8 Hz), atropine-sensitive theta may occur during complete immobility:

1. In normal rabbits the theta elicited by a sensory stimulus is usually of a relatively low frequency (<7 Hz) when it is averaged over many seconds. Average frequencies may be misleading, however, in that the frequency of individual waves can vary widely. When a stimulus is presented to a rabbit the first few waves of a train of immobility-related theta may rise as high as 10–12 Hz (Kramis et al. 1975; Vanderwolf et al. 1975). Very high-frequency immobility theta can be clearly seen in Fig. 2 of Kramis et al. (1975). Figure 1 in Kramis et al. (1975) shows clear theta of >8 Hz occurring in the absence of any change in EMG activity. This high-frequency immobility-related theta is abolished by atropine (Kramis et al. 1975). High-frequency sensory stimulation-induced, immobility-related theta has also been observed in guinea pigs (T. E. Robinson, unpublished experiments).

2. Another situation in which high-frequency theta accompanies complete immobility is during electrical stimulation of the midbrain reticular formation. Stimulation at some sites initially causes immobility, followed by vigorous walking or running (Vanderwolf 1975; Whishaw, Bland, Robinson & Vanderwolf 1976). Throughout the stimulation period 8–10 Hz theta occurs. Atropine abolishes the 8–10 Hz theta accompanying immobility, but not the 8–10 Hz theta occurring during movement. The frequency of theta elicited by this electrical stimulation is far greater than "the limiting frequency" of 7.7 Hz suggested by Gray (*Anxiety*, p. 193).

3. If the medial septum is stimulated with brief trains (or pulses) that repeat at a frequency of approximately 5–12 Hz, it is possible to elicit frequency-specific driving of hippocampal slow-wave activity. Kramis and Vanderwolf (1980) have used this electrical stimulation paradigm (which is very similar to Gray's) to show that the hippocampal slow waves generated by such pulse trains display the behavioral, pharmacological, and electrographic characteristics of atropine-sensitive and atropine-resistant theta. When normal rats are completely immobile, hippocampal slow waves can be driven at any frequency between 5.5 and 12 Hz. When an animal is walking, frequency-specific driving can be obtained at frequencies between 7 and 12 Hz. It is not possible to drive hippocampal slow waves at a lower frequency than that normally accompanying a given behavior, although it is possible to drive them at a higher than normal one. The thetalike waves produced during immobility by septal stimulation are abolished by atropine, regardless of their frequency (5.5–12 Hz). The thetalike waves of identical frequency, but occurring during walking, are not abolished by atropine. The movement-related hippocampal activity is abolished by urethane. This frequency-specific driving of the hippocampus can be elicited by sites in the lateral hypothalamus and midbrain reticular formation, as well as the medial septum and diagonal band (Kramis & Vanderwolf 1980). Thus, once again we see that sensitivity to drugs and not theta frequency defines which system is active.

4. One other experimental preparation in which high frequency (>8 Hz) theta occurs during immobility is the curarized rat (Whishaw et al. 1976).

There are some experimental situations where low frequency (<7 Hz) theta occurs during behaviors such as walking: (1) Frequency of theta is strongly related to body temperature; the

correlation between body temperature and theta frequency being +0.96 (Whishaw & Vanderwolf 1971). The average theta frequency accompanying a given behavior, such as walking, may vary between 3 and 10 Hz depending on the body temperature of the animal. The amplitude of theta is not altered by comparable manipulations of body temperature (Whishaw & Vanderwolf 1971). (2) The frequency of theta accompanying walking is also altered by treatment with barbiturate drugs. After an injection of 10 mg/kg of sodium pentobarbital, walking is accompanied by 5.2–6.8 Hz theta and struggling by 5.5–6.4 Hz theta (Vanderwolf 1975).

We can see from the above that theta frequency is *not* a reliable indicator of which system (atropine-resistant or atropine-sensitive) is active. However, the portion of Gray's theory dealing with hippocampal theta rests heavily on the assumption that different frequencies of theta determine both the rate and *nature* of information processing in the hippocampus (*Anxiety*, p. 293). We feel that Gray needs to address this discrepancy between his theory and the facts.

Are the two theta systems simultaneously active? Another problem with Gray's attempt to incorporate our knowledge of hippocampal theta into his theory involves the way he relates the "control" mode of hippocampal function to a state when low-frequency theta is generated. He says that when a *mismatch* occurs, "the septo-hippocampal system assumes direct behavioral control," which is characterized by an "active inhibition of motor behaviour and the execution of information-gathering behaviour" (*Anxiety*, pp. 264–65). During the "control" mode, "lower-frequency theta (not related to movement) is generated" (p. 293). In contrast, in the "just-checking" mode, "movement-related (atropine-resistant, high-frequency) theta is associated with the checking of motor programmes that are in the course of smooth execution" (p. 292). From Gray's description of these two modes of operation one must conclude that they could not be simultaneously active, since one is associated with behavioral inhibition (low-frequency theta) and the other with active movement (high-frequency theta). The problem with this (as it relates to the two theta systems) is that it is very probable that the inputs to the hippocampus which produce atropine-resistant and atropine-sensitive theta are simultaneously active during Type 1 behaviors.

If an experimenter presents a sensory stimulus to a rabbit and the rabbit remains immobile for a few seconds and then walks, a continuous train of theta will be observed (during both immobility and walking). The theta which occurs during immobility is atropine-sensitive and the one accompanying walking is atropine-resistant. From this kind of data, Vanderwolf and his coworkers (Kramis et al. 1975; Vanderwolf 1975; Vanderwolf et al. 1975; Vanderwolf, Kramis & Robinson 1978) have suggested that an atropine-sensitive input to the hippocampus may produce theta during immobility, and that a second, pharmacologically distinct input produces theta during Type 1 movements. This model does *not* state that during the transition from immobility to walking the atropine-sensitive system "shuts off" as the atropine-resistant system "turns on." In fact, it is more likely that the atropine-sensitive system remains active during Type 1 behaviors, working in conjunction with the now active atropine-resistant system (this point was previously discussed in Kramis et al. 1975; Leung & Vanderwolf 1980; and Vanderwolf & Robinson 1981).

Gray reviewed evidence that immobility-related theta is due to the release of acetylcholine (ACh) from septo-hippocampal fibers (see Szerb and Dudar 1981 for an alternative viewpoint). This idea is supported by Dudar, Whishaw, and Szerb (1979), who found that in urethanized rats sensory stimulation produces ACh release from the hippocampus as well as theta. However, Dudar et al. (1979) also found that walking on a treadmill produces the same level of ACh release as sensory stimulation. If atropine-sensitive theta is due to the release of ACh from septo-hippocampal fibers, then the above data obviously sug-

gest that the atropine-sensitive septo-hippocampal input is active during both sensory stimulation and walking. Ranck's (1975) report that single cells in the medial septum fire rhythmically in phase with theta during behaviors such as walking supports this conclusion.

More recently Vanderwolf and Leung (1981) have provided even stronger evidence that the atropine-sensitive input to the hippocampus is active during Type 1 behaviors. They report that complete lesions of the entorhinal cortex abolish atropine-resistant theta in rats. After an entorhinal cortex lesion, rats continue to show theta during behaviors such as walking and struggling; the average frequency of theta accompanying walking ranges from 6.4 to 7.1 Hz, and that accompanying struggling from 6.0 to 7.8 Hz. However, in entorhinal cortex lesioned rats atropine sulfate totally abolishes all theta activity. Thus, the atropine-resistant input which normally produces movement-related theta is abolished by an entorhinal cortex lesion. The theta which accompanies walking in the entorhinal cortex lesioned rat is by definition atropine-sensitive. It is interesting that an entorhinal cortex lesion abolishes atropine-resistant theta while a medial septal lesion abolishes both atropine-resistant and atropine-sensitive theta. This suggests that a septal input to the hippocampus plays a permissive role in the production of atropine-resistant theta by an entorhinal input to the hippocampus.

The evidence cited above, which suggests that the atropine-sensitive and atropine-resistant inputs to the hippocampus are *both* active during Type 1 behaviors, is by no means conclusive. However, we know of *no* evidence to suggest that the atropine-sensitive system "shuts off" during Type 1 behaviors. Since Gray's theory demands that the two systems *not* be simultaneously active, it is incumbent upon him to show that they in fact are not.

Sex. As mentioned above, a major line of evidence used by Gray to relate the septo-hippocampal system to anxiety is the 7.7 Hz minimum in the theta-driving curve. A very interesting aspect of this phenomenon is that it occurs only in male rats. Female rats (and castrated males) do not show the 7.7 Hz minimum. Gray states that these findings support the relation of "the 7.7 Hz minimum in the theta-driving curve to anxiety . . . since female rats are in general less fearful than males" (*Anxiety*, p. 190; it should be noted that the terms fearfulness and anxiety are used interchangeably by Gray in this discussion). Whether female rats are less "fearful" than males is debatable. If they are not less fearful, then Gray's theory can at best account for only 50% of the population. Since rats do not normally take anti-anxiety drugs, one would hope that Gray's theory would generalize to the species that does. However, an examination of the literature on humans does not support the conclusion that females are less anxious than males. If anything, women are reported as being *more* anxious than men.

Women are considerably more likely than men to experience and express emotional problems, particularly those characterized by anxiety (Shepard, Cooper, Brown & Kalton 1964; Gove & Tudor 1973; Cooperstock 1976; Mellinger, Balder, Manheimer, Cisin & Parry 1978). For numerous reasons, women are also more likely than men to be diagnosed as anxious or neurotic (Broverman, Broverman, Clarkson, Rosenkranz & Vogel 1970; Lennane & Lennane 1973; Leeson & Gray 1978; McGroarty 1980) and to be prescribed psychotropic drugs, such as barbiturates and tranquilizers (Prather & Fidell 1975; Badiet 1976; Cooperstock 1976; Mellinger et al. 1978; Chambers & White 1980). In addition, recent trends show an increase in alcoholism among women (Badiet 1976; Homiller 1980), which has been related to a specific desire to reduce anxiety (Mulford 1977; Eddy 1979).

The sex differences in hippocampal activity reported by Gray and his colleagues are very interesting, especially since the hippocampus is known to be a target for many steroid hormones. It is possible that the septo-hippocampal system is organized

quite differently in males and females. Since Gray's theory on the role of the septo-hippocampal system in anxiety very much involves the effects of antianxiety drugs on the 7.7 Hz theta minimum, he must deal with these sex differences in a less superficial way.

Inferring anxiety and antianxiety effects in animals

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For many reasons, Gray's *The neuropsychology of anxiety* is an important contribution to the neurosciences. This work masterfully emphasizes how selected animal data (behavioral, pharmacological, electrophysiological, and anatomical – biochemistry is a less fortunate guest) can be confronted in order to explore the "black box" in its functioning. Animals are increasingly being compared to humans; now they experience anxiety too! This could be one of the main lessons of neuropsychopharmacology.

Gray's *Précis* appears to be quite comprehensive and attractive, and the greedy reader becomes frustrated when some particular new aspects have been omitted by the author.

The reader's frustration. One regrets that the possible role of opiate systems in anxiety has been neglected: opiates or endorphins exert a powerful influence on the hippocampal formation (Zieglängsberger, French, Siggins & Bloom 1979) and naloxone attenuates various antianxiety effects of benzodiazepines in animals (see reference in Birk & Nobel 1981).

More relevant data than those considered in Gray's *Précis* exist to suggest that an interaction between benzodiazepines and GABA (gamma-aminobutyric acid) processes could be a crucial mechanism for anxiolytic activity (Cananzi, Costa & Guidotti 1980). New potent anticonflict drugs (triazolopyridazines, quinoline derivatives) are not considered (see references in Le Fur, Mizoule, Burgelin, Ferris, Heaulme, Gauthier, Gueremy & Uzan 1981). Since these drugs seem to have no clear efficacy on GABA-related tests, the central point, concerning whether GABAergic systems are a necessary step in the antianxiety effects of benzodiazepines, is unfortunately dodged. No mention is made regarding an attractive new field: the putative endogenous ligands of benzodiazepine receptors, and their possible role in the genesis of anxiety. Still, recent work by Polc, Ropert, and Wright (1981) on β -carboline may support Gray's theory of a pivotal role of the hippocampal formation in anxiety-related states.

These are minor points with regard to the theory developed. Let us focus, then, on what is actually presented rather than on what is omitted.

A behavioral inhibition system (BIS) for anxiety? In the paradigms used, animals' anxiety is "recognized" through an inhibition of behavior. However, in many cases inhibition is really not the prominent behavioral characteristic of the animals subjected to "anxiogenic" situations. Others are:

Faster running speed (with hesitations, I agree) or increased food-intake (Simon & Soubrié 1979) as anticipatory responses to frustration.

Increased avoidance responses when a punishment signal is delivered during an avoidance task.

Substantial amounts of ambulation (exploratory activity?) in animals introduced to novel surroundings. Halliday (1966) has speculated that anxiety is necessary for exploratory behavior to occur.

One wonders whether these reactions can be explained only in terms of an increment in arousal or attention. Probably not, since benzodiazepines (agents decreasing arousal and attention,

according to Gray's theory) rather facilitate ambulation or may increase the heightened emission of avoidance responses in the presence of a punishment signal. One can therefore hypothesize that inhibition of ongoing behavior is not a sign that a BIS operates but that inhibition or facilitation of ongoing behavior can be initiated by novelty, punishment, or nonreward, depending on the exact nature of the current relationship between the organism and the environment. The same assumption could also be valid for the direction (facilitation, inhibition, no change) of the effects of antianxiety drugs. Why is active avoidance generally not altered by antianxiety drugs at doses which dramatically impair (in terms of errors of commission) passive avoidance? I cannot accept that this point can only be solved in terms of anatomical (septal) circuitry. What, in fact, is the entire neuropsychopharmacological picture hidden here? This question leads to another crucial one concerning the SHS and benzodiazepine activity.

Alleviation of anxiety or impairment of its expression? The septo-hippocampal system (SHS) and its noradrenergic and serotonergic afferents are regarded as the neural substrate of anxiety and as the elective target system for benzodiazepine-induced anxiolytic effects. This is based on the remarkable parallel that exists between the behavioral effect of lesions of the SHS and those caused by benzodiazepines in animals subjected to novelty, punishment or nonreward. However, it would be of interest to know whether SHS lesions were able to alter (potentiate or abolish) benzodiazepine effects. Furthermore, is the SHS itself, or the changes in its functioning caused by benzodiazepine, involved in the control of anxiety or only in the ability of the organism to translate anxiety into behavioral inhibition (or facilitation in some cases)? Note that benzodiazepines selectively impair passive avoidance and not active responses; also that drugs known to block serotonergic transmission attenuate punishment-induced inhibition in animals. Interestingly, serotonin lesions or serotonin blockers are known to cause excitation, potentiation of stereotyped behavior, and reduction of the ability of neuroleptics to induce catalepsy; and these drugs, in fact, have never been claimed to be effective in relieving human anxiety.

This implies that, based on *animal models of anxiety*, changes in behavior do not necessarily parallel changes in anxiety. Therefore Gray's implicit strategy suggesting that a relevant criterion for the assessment of the validity of tests of anxiety is their sensitivity to anxiolytics seems to me circular – unless one believes that the only neuropsychological factors that benzodiazepines can affect are anxiety-related ones.

A critical analysis of the various tests used and a fine dissection of the factors putatively altered by benzodiazepine, alcohol, and barbiturates are therefore needed. What is masked by the term "rate-dependency," and what are the features that ensure the selective sensitivity to benzodiazepines either of passive avoidance responses or of punishment elicited by contingent shocks? To find out, one must set aside the knowledge that drugs are antianxiety agents in humans and (perhaps) also in animals. In animals, this is the point to be demonstrated; it should not be taken as a premise.

One therefore wonders whether "a description of the neuropsychological processes which are altered by the antianxiety drugs" is an appropriate or an approximate description of the neuropsychology of anxiety.

The triangulation strategy. Gray uses the term "triangulation" when he is faced with the necessity of unravelling the specific neuropsychopharmacological role played by the antianxiety action of anxiolytic compounds from their other actions. This triangulation can be accomplished for a given drug by comparing the effects of various dosage regimens or different lengths of treatment (Gray remains rather silent on this point). Triangulation can be continued by analyzing the actions of different classes of drugs, with only some limited set of common properties. This elegant and fruitful strategy can have many pitfalls and

can be only used with caution. Alcohol is portrayed as having no anticonvulsant effect, yet it does protect animals against various types of experimentally induced seizures (Boissier, Simon & LeBourhis 1969). Is seizure susceptibility then part of anxiety? Lesions of the SHS have been found to potentiate barbiturate-induced "sleeping" time (Harvey, Heller, Moore, Hunt & Roth 1964). Is anaesthesia part of anxiety? What happens if we apply this triangulation strategy to benzodiazepine receptors? Triazolopyridazines preferentially release conflict behavior (producing minimal ataxia). Can any area of the brain where they displace benzodiazepines from their receptor sites therefore be regarded as an anxiety area? Unfortunately, triazolopyridazines preferentially affect cerebellar, but not hippocampal, benzodiazepine binding (Young, Niehoff, Kuhar, Beer & Lippa 1981). What is the meaning of this rather surprising finding? Perhaps only that an inherent gap exists between *in vitro* and *in vivo* assays?

If this commentary appears unjustly critical, it must be taken as a sign that Gray's work really offers food (and reward) for thought. Indeed, it is evident that Gray's theory of the "neuropsychology of anxiety" projects a remarkably valuable light on the countless data provided by hundreds of separate experiments dealing with benzodiazepines, hippocampus, behavior, electrophysiology, learned helplessness (a really winsome chapter), binding, and clinical data.

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Substrates of anxiety: But if the starting point is wrong?

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But what is the odds! It frequently happens
That very unusual methods of starting
Lead to the most original outcome.

Henrik Ibsen, *Peer Gynt*, 1867

Jeffrey Gray has produced an extensive volume with a very coherent and penetrating defense of his hypothesis. But, at least to me, the very starting point of his argument is difficult to accept. Gray defines anxiety as the psychological dimension which is reduced by "anxiety-reducing" drugs, and he consistently defines this "group" as consisting of benzodiazepines, barbiturates, and even alcohol. He claims that this heterogeneous group of drugs has the same site of action, at least for the anxiety effect, regardless of what is known about the specific receptors for benzodiazepines and their localization (Squires, Benson, Braestrup, Coupet, Klepner, Myers & Beer 1979). He simply claims, since it does not agree with his theory, that for the effect of the drugs he is discussing, the receptor sites are of no relevance. The site of action he is interested in is the limbic system, or rather, the septal area and the hippocampus.

Anxiety is a hard term to handle from a behaviorist point of view, and in most animal experimental situations we use the concept of fear, and the subcategories of flight and freezing. Gray does not seem to be comfortable with this tradition, and bypasses the literature with his pharmacological trick. It is a most interesting attempt to get around some of the problems with the term "fear," which admittedly involves a jump from human feelings to interpretations of animal behavior. But the anxiety jump is no less of a jump, even if the jump is regarded as a pharmacological one. There are also data that Gray admits weigh against any anxiety hypothesis for septal and hippocampal

function. In addition to the findings Gray discusses there are endocrinological and ethological data which seem incompatible with the anxiety position (Seggie & Brown 1976; Ursin, Coover, K hler, DeRyck, Sagvolden & Levine 1975; Ursin, Blanchard, Blanchard & Ursin 1981). As a matter of fact, I think a case may be made for an "Inverted Papez Theory of Emotion": Papez seems to have managed to mention all limbic structures that are unimportant for emotional behavior. The only one Papez missed, the amygdala, is undoubtedly involved in fear and defense; Gray does not pay much attention to it either.

Anxiety is by no means a well-defined pharmacological unit, or a well-defined psychiatric or psychological term. Gray relates his definitions to the factor-analytic concepts of Eysenck (Eysenck & Eysenck 1969) (although his own version of these). The essential part of his concept is that anxiety depends on a "behavior inhibition system," and it is this system that depends on the septo-hippocampal region.

The behavior inhibition system has three outputs, the behavioral inhibition itself, an increment in arousal, and an increment in attention. This clustering seems surprising at first, but it is related to the anxiety model. Freezing would involve all three categories. There is, therefore, an arousal element in this theory which seems to be essential to an understanding of the links between it and previous models of CNS (central nervous system) function. Gray discusses at length the brain stem influence on his system, but he uses the arousal concept with hesitation and only when he has to; it has no real status in his theory. This seems puzzling, since there would be less controversy if he based his neural hypothesis on concepts that are generally accepted. It would seem more parsimonious and less controversial to state that the three groups of psychoactive substances – tranquilizers, barbiturates, and ethanol – all reduce arousal or activation, but through different mechanisms. This should not make it necessary to change the essential elements in his behavior-inhibition theory. However, Gray makes it clear that the reason he does not use the arousal concept is that it has no status in learning theory. To many of us this may appear to be a problem for learning theory rather than for brain science. However, it is essential to understand that this is a real problem for Gray, since one of the most valuable contributions of the book is that it offers a line of argument which is based consistently on learning theory. [See also Vanderwolf & Robinson: "Reticulo-cortical activity and behavior" *BBS* 4(3) 1981.]

The assumption of one behavioral inhibition system linked to parts of the limbic structures is, of course, not new, and Gray makes proper reference to those preceding him. However, the Kaada-McCleary (Kaada 1951; McCleary 1966) model did not involve the hippocampus. The basis for this celebrated model was the neurophysiological finding that electrical stimulation produced strong somatomotor inhibition from two distinct zones, one in the septal/subcallosal area, one in the orbital area. Hippocampus did not give any such responses. The only "inhibition" elicited by this technique is the shortlasting arrest which is accompanied by EEG (electroencephalogram) desynchronization, and orienting responses when the area is stimulated in unanesthetized subjects. This type of inhibition is probably a specialised mechanism unrelated to the other behavioral inhibitory phenomena Gray deals with. It seems distinct from a physiological, anatomical, and behavioral point of view (Kaada 1960). There is also factor-analytic work demonstrating a distinction between the arrest produced by the orienting response and the response inhibition produced by the conditioned suppression technique (Svendsr d & Ursin 1974). Also in humans it is possible to discriminate between the orienting response and a conditioned emotional response by factor analysis of galvanic skin responses ( hman 1972).

In my opinion the anxiety/arousal and the arrest dimensions of the Gray behavioral inhibition concept are unnecessary complications and should be discarded from his theory. We would then be left with the best available version of the septo-hippo-

campal inhibition dimension available. To some of us it may appear that we obtain it ten years too late; the heyday of limbic inhibition may be over. On the other hand, since the memory theories of hippocampal function are back, Gray is probably ahead of us in reinstating the inhibition theory. There is no way this hypothesis is going to explain all functions of the septo-hippocampal complex, even if Gray seems to be trying to do so. However, Gray also states that it is almost a commonplace that the behavioral syndromes produced by lesions in this area cannot be explained by one factor. Attempts to investigate how many factors are required to explain the variance after large septal lesions have not been encouraging for simple concept theories, be these Gray's or anyone else's (Ursin, Dalland, Ellertsen, Herrmann, Johnsen, Livesey, Zaidi & Wahl 1978). But there is no doubt that Gray has furnished us with a thorough and systematic synthesis of all the behavioral data supporting the inhibition hypothesis, and this he has presented in a way which makes it possible to see the general principles that for many of us may be lost when we face the very large and complex literature on this part of the brain.

The septo-hippocampal system and behavior: Difficulties in finding the exit

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I shall limit my commentary to Gray's theory of septo-hippocampal function, making no reference to anxiety, or the antianxiety drugs. The principal strength of Gray's theory derives from the inclusion of data from micro- and macroelectrode recording experiments, brain stimulation studies, and brain lesion experiments. Gray has attempted to use data from these several approaches more completely and systematically than have other theoreticians. Moreover, he has compared the various extant theories of septo-hippocampal function and has made a conscious effort to subsume essentially every term of prior theories. In doing so, he has engaged more than others in "theory reduction" as it is practiced in the philosophy of science.

"Reduction" does not entail developing an explanation of a physical process based on the smallest units of analysis in the system. It means logically replacing several theories by one theory with a corresponding reduction in the number of terms. However, the construction of an empirically testable theory of brain-behavior function will probably require specifying the neural units involved in the function. Gray has also attempted to do this in his careful analysis of the dynamic functioning of the intrinsic circuitry of the hippocampus and its relation to the subiculum, entorhinal cortex, and septal nuclei. This analysis is certainly specific enough to generate experiments to test his hypothesis about the neurophysiological activity of these areas and its relation to the behavioral inhibition system. I find, then, that Gray's theory is logically sound and empirically heuristic. This book should be provocative reading for anyone interested in the role of the septo-hippocampal system in behavior.

Although he presents an adequate account of the efferent hodology of the septo-hippocampal system, Gray's attempt to relate anatomic outflow to the functional effects of septo-hippocampal activity does not equal his analysis of the inputs and intrinsic workings of the system. This weakness is due, in part, to the inherent difficulty in specifying the pathways and mechanisms whereby the septo-hippocampal system gains "control" over the motor system. It has long been recognized that the various structures comprising the limbic system, including the septum and hippocampus, are not anatomically positioned so

that they can influence directly either the sensory or motor systems (e.g., Thomas, Hostetter & Barker 1968). Gray seems to recognize this difficulty, as indicated by his statement that "the overwhelming impression is of a system superbly well equipped for talking to itself – loops abound, exits are few" (p. 75). However, despite the difficulty in describing any direct septo-hippocampal mechanisms for controlling motor patterns, the manner in which Gray states his theory requires that the septo-hippocampal system have the ability to affect such patterns very specifically. For example, according to Gray one "form of action undertaken by the septo-hippocampal system when mismatch is declared is the initiation of specific exploratory and investigatory behaviour designed to answer particular questions that arise from the operation of the comparator loops" (p. 280). However, he provides only very sketchy information as to the nature of the interface between the subicular comparator and the specific exploratory and investigatory behaviors it presumably produces.

Gray attempts to provide some information regarding the anatomical regions where the interface might exist. Two such places would be the cingulate gyrus and the hypothalamus. There is a problem with including the cingulate gyrus. While, on anatomic grounds, this region may seem appropriate because it receives input from both the subiculum and the mammillary bodies, it is not clear exactly how its actions relate to those of the septo-hippocampal system, especially since ablation of the supracallosal portion of this area has effects on avoidance behaviors (McCleary 1966; Lubar & Perachio 1965) and unlearned defensive behavior (Woodruff, Baisden & Douglas 1981) which are precisely the opposite of those produced by lesions of the septum, hippocampus, or fimbria-fornix.

The hypothalamus seems a reasonable candidate for the "organizer" of behavioral "action patterns" (p. 294, Fig. 10.8) since it is well known that electrical stimulation of the hypothalamus can elicit many behaviors. In addition, electrical stimulation of the hippocampus modulates hypothalamically induced attack in the cat (Flynn 1967); but even here there are difficulties for Gray's theory. Flynn reports that dorsal hippocampal stimulation inhibits hypothalamically elicited attack, while ventral hippocampal stimulation facilitates it. These data do not fit with Gray's tendency to treat the hippocampus as a functionally homogeneous structure. In addition, it is difficult to envision how hippocampal effects on hypothalamic activity lead to the memory deficits reported in the clinical literature.

Gray's theory of septo-hippocampal function is weakest where virtually all other theories concerning the function of this system in behavior are weak: in the specification of the mechanisms whereby the system exerts its influence. I admit that I cannot provide this specification either. However, I shall suggest that the nature of the effects produced by this system is a subtle change in the physiological state of the organism. In situations of meaningful environmental uncertainty the septo-hippocampal system, especially the hippocampal component, is involved in setting the physiological parameters of neural and hormonal systems so that prior behavioral strategies (either learned or innate) can be efficiently and clearly recalled. The end result of activity in the septo-hippocampal system, then, is to set the occasion for recall and subsequent behavior by means of a change in state of the organism. Such a change in state may well be very slight and could include alterations in neural activity, hormonal levels, and autonomic and somatic tone. Feedback from the latter three systems would constitute the state change. This idea provides little help in tying down the exact nature of the output of the hippocampal formation. It is not proposed toward that end. It is intended to suggest that the actual influence of the septo-hippocampal system on other neural systems (such as the neocortex) will most probably be detected as an exceedingly subtle change in ongoing activity and will not emerge as some obvious "engramatic" representation of experience.

Leaping up the phylogenetic scale in explaining anxiety: Perils and possibilities

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The number of theorists who believe that individual differences in human personality have an evolved biological basis is small. Those of us who approach the problem from the human level have mostly restricted our studies to psychophysiological and correlational methods of investigation. The concepts of "arousal" and "optimal level of arousal" have been important for a number of personality theorists (e.g., Eysenck 1967; Zuckerman 1969), as they were for Gray (1964) in the 1960s. But characterizing brain function by its general cortical arousal in response to stimulation is somewhat like describing the workings of an internal combustion engine by saying "when I press on the accelerator the motor runs faster." Gray has lifted up the hood and looked inside. To do this one must work with species other than humans, with all the risks attendant on leaping up the phylogenetic scale. Fortunately, the limbic structures involved in Gray's theory of anxiety have not changed much in the course of mammalian evolution and may serve some of the same functions in man as they do in rats. But there are still problems in identifying anxiety in man with behavioral patterns in other species.

Anxiety at the human level is a construct identified by three kinds of responses: (1) verbal reports that provide a clue to the phenomenal experience; (2) psychophysiological responses that, while not specific to anxiety, provide an objective estimate of the "arousal" dimension (or dimensions) of anxiety; (3) behavior in the form of approach or avoidance responses, or conditioning to aversive stimuli. Those, like Gray, who work almost entirely with rats cannot use the first type of responses and have not used the second type much, presumably because of technical problems. Most of their behavioral definitions must depend on the third type of data. But ethical constraints prevent investigators at the human level from using the kind of pain stimuli and conflict situations employed by comparative psychologists to study "fear." One consequence is that behavioral models, derived from work with other species, do not adequately explain the origins and course of human phobias (Klein 1981).

By identifying the trait of anxiety in rats and man with a behavioral inhibition system (BIS) I believe that Gray has overgeneralized from one behavioral expression of the trait. While temporary inhibition of behavior may be one expression of anxiety, one cannot work backwards and say that any stimulus that results in behavioral inhibition is an anxiety stimulus. The fact that the BIS is involved in the orienting reflex (OR) to novel stimuli does not make these a class of innate fear stimuli.

Our research (Neary & Zuckerman 1976) has shown that the strength of the electrodermal OR is positively related to a "sensation-seeking" trait (Zuckerman 1979), is *not* related to anxiety trait, and is *negatively* correlated with anxiety state at the time of the experiment. Lader (1979) says that anxiety neurotics respond to an initial (novel) stimulus with an OR that is *weaker* than that of controls. The human data simply do not support the extrapolation from Gray's theory of a positive association between the psychophysiological response to a novel stimulus (OR) and anxiety; if anything, the correlation is a negative one.

Gray sees behavior in novel situations as the outcome of an interaction between two traits: exploratory (which he calls "impulsiveness" and I term "sensation seeking") and anxiety. After the first exposure to the open field, ambulation in the field would be a positive function of the exploratory trait and negatively related to the anxiety trait. However, antianxiety drugs decrease defecation generally but not specifically in the open field where defecation is used as an index of anxiety. Furthermore, antianxiety drugs reduce the exploration of novel stimuli

and spontaneous alternation whereas according to the theory they should increase these forms of activity. The problem may be that Gray has chosen to define fully the mechanisms of anxiety before dealing with those of impulsivity (or sensitivity to signals of reward). The two mechanisms must interact in the response to novel stimuli and the neglect of impulsivity may account for the inconsistencies in the evidence on anxiety.

Gray maintains that the forebrain ascending noradrenergic (NA) system is part of the anatomical basis for the BIS and that its activity is involved in anxiety. An opposite viewpoint is found in the work of Stein (1978), who claims that this NA system is the basis for nonspecific reward effects. The two points of view will be presented in a forthcoming volume (Zuckerman, in preparation). Apart from the self-stimulation studies (not accepted by Gray as an appropriate methodology for studying reward) that find this phenomenon to be blocked by depletion of NA, studies showing greater vulnerability to learned helplessness in NA-depleted animals, and the link between NA and some forms of depression in humans, suggest that it is not an excess but a deficit in NA that underlies anxiety. Gray uses these data to suggest that NA depletion is related to depression of the endogenous type, but NA excess is typical in the neurotic when anxiety and depression are indistinguishable. He further speculates that NA depletion may represent the exhaustion phase of a prolonged NA response (anxiety) to stress. However, it is not typical that an endogenous depressive passes through a prolonged period of generalized anxiety before moving into depression; nor is it true that anxiety neurotics typically end up as depressives. Furthermore, recent findings show that tricyclic antidepressants, which are believed to activate the NA system, are very effective agents in treating the spontaneous panic attacks in agoraphobia as well as depression. In a recent correlational study of normal volunteers at the National Institutes of Mental Health (Ballenger, Post, Jimerson, Lake, Lerner, Bonney & Goodwin 1981) *negative* correlations were found between test measures of both depression and anxiety traits and MHPG, a metabolite of NA obtained in this case from both cerebrospinal fluid and plasma. [Note added in proof, not seen by J. A. Gray: However, further analyses showed that both CSF and plasma DBH, another indicator of the activity of the NA system, correlated negatively with anxiety trait, but positively with anxiety state measured just before the stressful lumbar puncture procedure. CSF MHPG also correlated positively with anxiety state at this time.]

Since many studies show that central NA is released at high rates during stress, what are we to make of these findings suggesting that trait anxiety is negatively related to turnover of NA and that drugs which release NA reduce the more severe forms of anxiety found in panic attacks? I would suggest that the function of the NA system is antianxiety in that it works to arouse the organism from the behavioral inhibition characteristic of anxiety. In nonthreatening conditions this NA and dopamine release may be rewarding or pleasurable as suggested by Stein (1978). The neglect of the dimension of phenomenal experience in Gray's review results in his ignoring the fact that drugs such as D-amphetamine and cocaine typically produce positive affect in their users (Carroll & Zuckerman, in press). It is true that chronic use of these stimulants may produce states of hyper-alertness and anxiety, but this may be due to the exhaustion of NA reserves or the development of supersensitivity in the receptor cells.

The construct of an "optimal level of cortical arousal" may be more profitably applied to the NA system arousal: moderate levels of NA activity may be intrinsically pleasurable and reduce the effects of stress, while extremely high levels may disorganize behavior and lead to anxiety of a paranoid type. Low levels of NA may create a vulnerability to either anxiety or depression because of the inability to counteract the effects of stress (helplessness).

The time is ripe for human studies aligning biological and

personality trait dimensions and looking for the biological correlates of the psychological variables (e.g., Zuckerman, Buchsbaum & Murphy 1980). While such studies must depend on less direct indicators of the brain's neurochemistry, and can only approach causation through techniques such as path analysis, they are necessary if Gray's elegant theoretical structure is to prove applicable to the phenomenon of anxiety-proneness in humans.

Author's Response

On mapping anxiety

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The *Précis* summarised an attempt to provide a cognitive map of anxiety, a map specified in terms of psychological, physiological, and anatomical coordinates. Objections to the map have been raised at a number of different levels. I shall take the most general first. Table 1 lists the sections into which I have broken my reply, and the commentators who figure in each section.

Where should we start? There is an old English joke, rediscovered, it seems, by Ursin. Citydweller, lost, to country man: "How do I get to Little-Chipping-in-the-Marsh?" Country man: "If I were going there, I wouldn't start from here." Lyons, Panksepp, and Ursin would not start where I started if they wanted to get to anxiety.

This point was put most forcibly by Lyons. He (see also Soubrié) demonstrates that there is nothing logically *compelling* in the following two arguments: (1) If the antianxiety drugs, in reducing anxiety, suppress psychological process, P, then P amounts to a psychological profile of anxiety. (2) If the antianxiety drugs reduce anxiety by altering brain region Z, then anxiety amounts neurologically to Z. I do not question this demonstration. Indeed, if this were all the theory developed in *The neuropsychology of anxiety* (henceforth *Anxiety*) rested upon, I would retire chastened from the field. But, although these arguments are certainly important in *Anxiety*, they serve as starting assumptions, not as logical proofs.

Consider the parallel case of antipsychotic drugs. As is well-known, these appear to have the general property of blocking dopaminergic receptors. But I would not care to defend on this basis (though others do) the hypothesis that schizophrenia consists in overactivity in dopaminergic neurons. My hesitation stems from the fact that there is no evidence that dopaminergic overactivity gives rise to behaviour that maps well onto the core symptoms of human schizophrenia (thought disorder, hallucinations, delusions, etc.). What dopaminergic overactivity (on the evidence from animal experiments) might be expected to produce is either stereotyped motor behaviour (Iversen 1977) or malfunctioning in reward processes (Wise 1978). [See also Wise: "Neuroleptics and Behavior" *BBS* 5(1) 1982.] But one can interpret schizo-

Table 1. *Issues discussed in commentaries and response.*

Section	Commentator
Where should we start?	Lyons, Panksepp, Soubrié, Ursin, Pribram & McGuinness
Do we have a map at all?	Iversen, Woodruff, Gabriel, Gold & Fox, Kimble, McNaughton, Rawlins, Olton, Bohus, Nadel & Morris
Have we come to the right place?	Lyons, Zuckerman, Panksepp, Crider & Solomon, Ursin, Gold & Fox, Iversen
Have we been here before?	Iversen, Pribram & McGuinness, Zuckerman
The antianxiety drugs	Nadel & Morris, McNaughton, Rawlins, Soubrié, Ursin, Zuckerman, Revelle, Leslie
The septal area	Olton, Pribram & McGuinness
The hippocampal formation	Gabriel, James, Nadel & Morris, Crider & Solomon, Kimble, Lyons, Iversen, Ursin, Rawlins, Woodruff, Pribram & McGuinness
The Papez circuit	Gabriel, Woodruff, Pribram & McGuinness
The hippocampal theta rhythm	James, McNaughton, Robinson & Therrien
The dorsal ascending noradrenergic bundle	Bohus, Gold & Fox, Pribram & McGuinness, Zuckerman, Mason, Leslie
The ascending serotonergic pathways	Iversen, Soubrié, Gold & Fox
Areas not on the map: Opiates, hormones	Panksepp, Ursin, Pribram & McGuinness, Gold & Fox, Soubrié, Zuckerman, Bohus
Clinical issues	Soubrié, Iversen, Pribram & McGuinness, Zuckerman, Olton, Panksepp, Gold & Fox, Bohus
Personality	Revelle, Pribram & McGuinness, Robinson & Therrien

phrenic symptoms along either of these lines only by arguments that are tortuous and implausible. Thus, while one might consider arguments (1) and (2) above as starting points for the construction of a theory of schizophrenia, they are not at present promising.

The situation with respect to anxiety and the antianxiety drugs is quite different. First, and most important, the psychological processes that appear to be impaired by the antianxiety drugs in animals are themselves plausible candidates for the psychological processes that make up human anxiety (though this point too is questioned by some commentators: see below). Were the antianxiety drugs to impair reward processes or motor behaviour in animals, I would not have regarded them as useful in the

construction of a theory of anxiety. But the fact that they reduce an animal's reactivity to threats of punishment or failure (which has face validity as a description of reduced anxiety in people) increases my confidence that argument (1) above, while it is not logically compelling, is empirically correct.

Second, arguments (1) and (2) are not (as Lyons's counterargument supposes) independent of each other. In constructing a theory of the neural basis of anxiety, I specifically did *not* look simply for regions of the brain upon which the antianxiety drugs act. Had I done so, the theory would have needed to allot functions to (inter alia) the spinal cord and the cerebellum, both rich in benzodiazepine receptors (Möhler & Okada 1977; Braestrup & Squires 1977), just as a purely "dopamine" hypothesis of schizophrenia is forced to allot a function to the corpus striatum. What I looked for instead was regions of the brain upon which the antianxiety drugs act, and which *on other grounds* appear to subserve functions that are relevant to anxiety. Had there been no way in which the antianxiety drugs could affect the functioning of the septo-hippocampal system (SHS), the argument for including this system in the neural substrate of anxiety would have been much weakened; but had the SHS appeared to have the function of increasing muscle tension, it might have been a plausible candidate for the muscle-relaxant effects of the antianxiety drugs, but not for their antagonism of anxiety. Thus it is the *conjunction* of psychological and neuropharmacological arguments that points to the SHS as an important node in the neural basis of anxiety.

The book, in short, is an exercise in empirical argument, not in philosophical logic. The theory it contains is more complex and more intricately built than Lyons's counterarguments assume. The antianxiety drugs provided the starting point for constructing this theory, but in its final form it rests upon many other evidential bases. Furthermore, the theory explicitly provides for the possibility that some features of anxiety should be attributed to regions of the brain on which the antianxiety drugs do *not* act. Thus, in discussing the role of the prefrontal cortex and the therapeutic effects of prefrontal and cingulate lesions (*Anxiety*, p. 419), I suggested that some of the symptoms of obsessional neurosis are due to activity in pathways that afford descending neocortical control over the SHS and that are unaffected by the antianxiety drugs (since these, *ex hypothesi*, act principally by way of the ascending monoamine pathways). This postulate accounts for the fact that prefrontal and cingulate lesions can give therapeutic benefit to patients who are resistant to pharmacotherapy (Laitinen & Vilkkii 1972). "In this way, though the behavioural effects of the anti-anxiety drugs provided the corner-stone of the theory built in this book, it may be possible to remove it without pulling down the entire edifice" (*Anxiety*, p. 419).

It is probable, in fact, that Lyons would accept much of this reply, since he does not deny that my "strategy corners neurological mechanisms essential to anxiety," but only that I can be sure that I have "cornered *all* or even most of what is peculiar, neurologically, to anxiety." But even against this weaker point I have a defence. I did *not* rely on the argument from the antianxiety drugs to corner "all of what is peculiar to anxiety." On the contrary, I drew on all sources of information that seemed

relevant and reliable. These included, for example, the data on the effects of psychosurgery mentioned above, which netted the prefrontal and cingulate cortices into the neurology of anxiety. Similarly, I drew on descriptions of the symptoms of the obsessive-compulsive neurosis (Rachman & Hodgson 1979) to net into the psychology of anxiety a number of essentially verbal phenomena (*Anxiety*, chapter 14), upon which drug studies have so far thrown no light.

Panksepp, Soubrié, and Ursin also criticise my choice of the antianxiety drugs as a direction finder for anxiety; the points they make are covered by the above reply to Lyons. However, these authors in addition propose (Panksepp directly, Ursin indirectly) an alternative starting point: the behavioural effects of brain stimulation, especially in the hypothalamus. Panksepp (1982, this issue) defines "fear" in terms of flight behaviour and "panic" in terms of separation distress. He seems uncertain how to define anxiety itself, at one time treating it as excessive fear (Panksepp 1982, Figure 5), at another as a messy amalgam of fear and panic (in his commentary). Armed with these behavioural criteria (which are similar to Ursin's "ethological data"), Panksepp is able to study the brain mechanisms subserving emotion by mapping the points from which electrical stimulation will elicit the appropriate behaviour.

There is, however, a major problem with this approach: The behavioural criteria chosen to define the emotions are arbitrary. So, of course, is the choice of the antianxiety drugs in my own approach. But there is an important difference between the two cases. Generalising from animal to human data is enormously aided where the drugs are concerned, because identical substances are used in the laboratory and the clinic. No such bridge is available to cross the gap dividing the two sets of data where brain stimulation is concerned. To be sure, there are some observations of the effects of brain stimulation in man; but these are necessarily few and ill-controlled. Thus we are forced to rely on direct interpretation of the effects of brain stimulation in animals. But how can we judge that flight in a rat is the equivalent of fear or anxiety in a man? It may be. But so may freezing behaviour or distress calls. Panksepp himself evidently has difficulty in deciding between flight and distress calls; and I prefer freezing to either. Unaided intuition is incapable of deciding between these alternatives, since human anxiety is not sufficiently similar in terms of *overt* behaviour to any of these proposed animal equivalents. Nor can we decide not to decide between them, for quite different brain structures appear to mediate the three kinds of behaviour (Panksepp 1982, this issue, for flight and distress calls; and Graeff & Silveira Filho 1978, for freezing).

Simply as a starting point, then, Panksepp's proposed use of brain stimulation is inadequate. But this does not mean that the data from brain stimulation studies are not of value in mapping the emotions. I accept Panksepp's criticism that I made insufficient use of such data in *Anxiety* (except insofar as they relate to the septal area and the hippocampus). Fortunately, this lacuna is to a large extent filled by his own review of these data (Panksepp 1982, this issue). However, I do not find Panksepp's interpretation of the data from brain stimulation experiments fully acceptable. In particular, for reasons stated briefly elsewhere (Gray 1982b), I do not find that he has

made out a sufficiently good case for distinguishing between separate systems mediating flight and fight behaviour, respectively; and I believe that a better case can be made for defining fear/anxiety in terms of behavioural inhibition (freezing) than in terms of flight behaviour.

This, of course, is a central issue. If anxiety is best approached by asking which brain structures subserve flight (or flight-and-fight), then Ursin is right in his criticism that, like Papez (1937) before me, I left out the most important structure of all: the amygdala. But, if the right direction finder consists in the antianxiety drugs, this criticism is misplaced. There are few similarities between the behavioural effects of the antianxiety drugs and those of lesions to the amygdala, and the effects of stimulating the amygdala are not the opposite of those of the antianxiety drugs; moreover, in both these respects the amygdala is unlike the SHS. In particular, amygdaloid lesions (usually) reduce fight and flight behaviour, and amygdaloid stimulation can elicit these forms of behaviour. These are the reasons for Ursin's belief that the amygdala deserves a central role in the analysis of anxiety; and they fit neatly with Panksepp's treatment of the hypothalamic sites (to which the amygdala projects) where similar effects are obtained. But the antianxiety drugs do not specifically antagonise escape behaviour when conventional reinforcement is used, nor do they consistently alter aggressive behaviour (Gray 1977). Thus, by accepting the Panksepp-Ursin analysis, we create a puzzle: If flight is a good animal analogue of human anxiety, and if antianxiety drugs do not specifically affect flight, why do they reduce anxiety? In the absence of an answer to this question, it is better to abandon the assertion that flight is an analogue of anxiety.

Pribram & McGuinness are also, though for different reasons, unhappy about my neglect of the amygdala, a neglect that (as they point out) extended even to studies involving both hippocampal and amygdaloid lesions. The reasons for this neglect should now be clear. My intention was to write a book about anxiety, not about the limbic system. Having decided that the amygdala does not have a crucial role to play in anxiety (which is not to say that it plays *no* role), I saw no point in reviewing data relevant to this region of the brain. It is in any case not exactly clear what role Pribram & McGuinness themselves wish to allot to the amygdala in *Anxiety*. As to studies in which both the hippocampus and the amygdala were lesioned, these do not lend themselves to ready interpretation with respect to purely hippocampal function; and it was this that I wished to establish.

Do we have a map at all? A map is a device used for prediction. Iversen and Woodruff feel that my map of anxiety can indeed be used in this way; and Gabriel, Gold & Fox, and Woodruff comment favourably on the degree to which I have succeeded in reducing the number of previous terms on the map by showing that they make similar predictions. But several other commentators (Kimble, McNaughton, Rawlins) make the charge that my theory of anxiety cannot easily be used predictively, or regret the absence in the book of a list of specific new predictions. This charge has some force, and I shall not try to rebut it in full; but, in its most general form, it can, I believe, be rejected.

McNaughton puts the charge in a manner that I find

correct: "both theories" (i.e., mine and O'Keefe & Nadel's 1978 cognitive mapping theory) "are sufficiently detailed to be wrong, neither is sufficiently formal to be unambiguous." The complexity of the information-processing functions and anatomical wiring diagrams postulated in both these theories has reached the point at which only a formalised version (couched perhaps in terms of a computer model) could determine whether they are internally consistent or guarantee that they yield unambiguous predictions for most experimental conditions. I very much hope that this BBS treatment will bring the theory developed in *Anxiety* to the attention of scientists better qualified than I am to develop such a formal model. [See also BBS multiple book review of O'Keefe & Nadel, *BBS* 2(4) 1979.]

To accept the charge of lack of predictivity in this form, however, does not mean that the theory fails to make *any* specific new predictions (a point with which McNaughton agrees). Indeed, it makes a great many such predictions; but they are so heterogeneous in nature and depend so much on precise descriptions of experimental conditions that a list of them would be unwieldy. Furthermore, at several points in the book (e.g., pp. 271, 287, 293, 370), alternative versions of parts of the theory are discussed; and, in certain experimental conditions, different predictions would follow depending on which version was adopted. Nonetheless, it is clear that sections of the theory are sufficiently well specified for a variety of detailed predictions to be made.

Let us take an example. Suppose we could record noradrenaline (NA) release continuously in the hippocampus during a behavioural experiment (a technical accomplishment that is coming closer with the development of voltammetric recording methods: Conti, Strobe, Adams & Marsden 1978). The theory predicts that: (a) during presentation of a stimulus associated with a punishing footshock (Pun-CS) there would be an increase in the release of NA in the hippocampal formation, and especially in the hilus of the dentate gyrus; (b) the intrahippocampal neuronal response to the Pun-CS would be greater than to a stimulus not associated with reinforcement; (c) antianxiety drugs would reduce the magnitude of effects (a) and (b); (d) more fearful rats (defined in terms of increased susceptibility to behavioural inhibition) would show effects (a) and (b) more than less fearful rats. These are highly specific predictions, which can be directly derived from the relevant parts of *Anxiety* (see in particular Fig. 11.4 and the accompanying discussion) and not (so far as I know) from any other theory. As it happens, Kovacs, Versteeg, de Kloet, and Bohus (1981), in a study cited in Bohus's commentary, have reported data that are consistent with prediction (d): animals that showed good passive avoidance, relative to animals poor at passive avoidance, had increased NA turnover specifically in the dentate gyrus.

A second example concerns the anatomy and physiology of neurons in the subicular area. The theory developed in *Anxiety* (p. 266 *et seq.*) requires that there exist cells in the subicular region that receive inputs from both CA1 (or possibly CA3) and the anteroventral thalamus and/or the cingulate cortex and/or the prefrontal cortex; that these cells are sensitive to their joint or several inputs and function either as "or" or "and" gates; that, if they are "or" gates, they project directly or indirectly to motor

systems; that, if they are “and” gates, they project to the entorhinal cortex; that both “or” and “and” gate cells should be found and that all the inputs they receive should be facilitated simultaneously at a particular phase of the theta rhythm. To my knowledge there are no data against which these predictions can yet be evaluated; but there is no reason why such data should not be obtained.

There are many other detailed predictions that could be derived from the theory; but to derive them here would detain us too long. One final case, however, will be considered, since it was specifically raised by Olton.

Olton asks what prediction I would make for the effects of antianxiety drugs in a procedure devised by Olton and Papas (1979) in which, in a radial-arm maze, there is a baited set of arms, each with one pellet of food in it, at the beginning of a given trial, and an unbaited set, in which food is never placed [see also Olton et al.: “Hippocampus, Space and Memory” *BBS* 2(3) 1979.]. In the Olton and Papas experiment, lesions of the fornix-fimbria severely disrupted “working memory” (i.e., the rat’s capacity to remember which arms in the baited set it had already entered within a given trial) but caused only transient disruption in “reference memory” (i.e., the rat’s capacity to choose arms in the baited rather than the unbaited set). (Parenthetically, I accept Olton’s comment that the concepts of working and reference memory are orthogonal to those of recognition and associative memory.) Let me confess at the outset that this is not a task for which the theory makes firm predictions as to the effects of the antianxiety drugs.

Consider first the task of choosing the baited rather than the unbaited set of arms. This is a complex version of a simultaneous discrimination. In general, antianxiety drugs do not impair simultaneous discriminations (Gray 1977). This is attributed to the fact that the need to inhibit nonrewarded responses plays little or no role, since “the subject can correctly solve the problem by learning simply that the positive stimulus is associated with reward” (*Anxiety*, p. 35). At first blush, then, one might predict that antianxiety drugs will fail to affect the reference memory component of the Olton and Papas paradigm. However, this paradigm introduces a complication into the usual simultaneous discrimination: There is a multiplicity of negative stimuli. This might increase the need for behavioural inhibition, as it seems to have done in an analogous experiment on the effects of hippocampal lesions (Han & Livesey 1977; *Anxiety*, p. 163). Thus one cannot rule out the possibility that antianxiety drugs will cause some degree of impairment (as did fornix-fimbria lesions in Olton & Papas’s experiment) in the reference memory component of the task.

Next, consider the task of choosing previously unentered arms in the baited set within a trial (working memory). The prediction my theory makes for this depends critically on the importance of the role played by the ascending monoaminergic inputs to the SHS (since it is these whose functioning is thought to be impaired by the drugs). The analysis of Olton’s working memory model given in *Anxiety* (p. 284) lays greater stress on the internal workings of the septo-hippocampal-subicular-entorhinal circuitry (and particularly on the projections from the subicular region to the septal area) than on the monoamine inputs to this circuitry; but a role for the latter inputs is certainly not excluded. Thus, again, the

theory cannot predict whether there will be some (probably slight) impairment or none in animals given antianxiety drugs. It does predict, however, that whatever the effects of the antianxiety drugs turn out to be, those of damage to the ascending monoamine pathways should be similar; and, since this kind of working memory task is almost certainly dependent more on sensory processing than on motor inhibition, the critical monoamine is likely to be NA rather than serotonin (see *Anxiety*, p. 370). (The theory also predicts that working memory would be severely disrupted if one could selectively section the connections between the subicular and septal areas.)

As it happens, we have data relevant to the working memory component of the Olton and Papas (1979) task, data obtained, however, in a simple radial-arm maze with only a baited set. We have twice examined the effects of 5 mg/kg chlordiazepoxide in this maze. In the first experiment, with a 16-arm maze, the drug produced a level of performance nonsignificantly lower than that of placebo-injected controls, but also nonsignificantly different from chance (Brookes 1981). In the second, with an 8-arm maze, the drug significantly impaired acquisition, but the drugged animals reached a final level of performance indistinguishable from that of controls (Rawlins, Preston & Gray, unpublished observations). Thus the wishy-washy prediction is matched by a wishy-washy outcome. Whether that is a triumph or a failure for the theory, the reader must judge.

Three particular ways in which the theory may lack predictive power were suggested by different commentators. Rawlins pointed to the problem of partial overlap (*Anxiety*, p. 299): “it is not clear to what extent the data derived from drug and lesion studies must agree in order for the theory to stand.” This is indeed a problem; although, as the discussion of the role of the monoamine inputs to the SHS in working memory (see above) perhaps makes clear, it is not totally intractable. The theory postulates that the effects of the antianxiety drugs are mediated by an impairment in the functioning of the ascending monoaminergic pathways; but these include other targets besides the SHS, some of which (notably the prefrontal cortex and hypothalamus) have been allotted particular functions within the overall theory. Thus the theory predicts certain definite patterns in the data: If (say) a prefrontal cortex lesion has an effect that can be interpreted as a disruption in the behavioural functions allotted to this region of the forebrain, and if the same effect is produced by disruption of the dopaminergic input to the prefrontal cortex, then the theory strongly predicts the same effect for the antianxiety drugs; and similarly for the hippocampus and its noradrenergic and/or serotonergic afferents, as in the working memory example above.

Kimble suggests that the distinction between the two postulated modes of operation of the SHS (“just checking” and “control” mode) poses a particular difficulty for predictive purposes. I accept this point. But the distinction is forced upon me (and must, I think, be forced on other theorists) by the data; it does not arise from other theoretical considerations necessarily linked to anxiety. As summarised in the *Précis*, there are now numerous instances of behaviour patterns during which the hippocampus (as shown by recording experiments) is highly active, yet which are not disrupted by hippocampal

lesions. It is perhaps inevitable that this kind of dissociation between recording and lesion data should pose problems for the construction of a predictively powerful theory. In the case of my own theory, these problems are at least circumscribed in the following manner. The theory requires that such lesion/recording dissociations should occur only in tasks in which the animal is normally able to carry out its motor programmes without disruption or mismatch. Thus the overall pattern of the data can be predicted from the theory, which in this way remains open to empirical test.

Rawlins, as well as Nadel & Morris, feel that the predictive power of the theory is weakened by reliance on the concept of interference. Nadel & Morris put this point in the following way: "[Gray] offers as an 'explanatory' alternative to the spatial theory the idea that certain deficits are due to the fact that the tasks employed were 'sensitive to interference effects.' Of course, this merely redescribes the defect." I have no quarrel with this point; indeed I make it myself (*Anxiety*, p. 252). In the *Précis* I indicated that one must seek the causes that underlie the heightened sensitivity to interference found in human amnesic and hippocampally damaged animals alike; and I suggested as possibilities Butters and Cermak's (1975) hypothesis of an impairment in multidimensional stimulus analysis, and Hirsh's (1974) and Winocur's (1982) ideas concerning the role played by contextual cues. Thus I agree with Nadel & Morris that talk of susceptibility to interference is by itself inadequate.

Nadel & Morris also ask me "by virtue of what property or capacity of the septo-hippocampal system . . . is the organism protected from interference effects?" The answer to this question is given in *Anxiety* (pp. 279–83). It calls upon the process of hypothesis generation and testing suggested by Kimble (1975), that of multidimensional stimulus and response analysis suggested by Butters and Cermak (1975), and that of contextual labelling and storage suggested by Hirsh (1974). Here we are in the heart of the information-processing aspect of the theory, a heart that (as McNaughton points out, and as O'Keefe & Nadel 1979, p. 524, predicted) beats closely in unison with the heart of cognitive mapping theory, if this is first stripped of its specifically spatial elements. And it is here, as I have admitted above, that both theories require more formal specification if they are to be unambiguous in their predictions (McNaughton, Rawlins).

Rawlins and Nadel & Morris make a further criticism which is harder to deal with. They point out that spatial learning, far from being particularly susceptible to interference effects in normal animals, as one might expect from my analysis, is relatively insensitive to such effects. Should this view be sustained by further data (and Nadel & Morris allow that it is still an unresolved issue), it would indeed pose a problem for the theory developed in *Anxiety*. It seems clear that spatial learning is particularly sensitive to hippocampal damage (e.g., Morris, Garrud & Rawlins 1981). If this special sensitivity is not due to disruption of the same processes that underlie the hippocampal deficits in other, nonspatial tasks, then the present writer is unable to account for it. But if it is due to the specifically spatial character of the relevant tasks (as proposed by O'Keefe & Nadel 1978), then it is difficult to make sense of the many deficits produced by hippocampal damage in clearly nonspatial tasks (see the *Précis*).

Resolution of this dilemma, therefore, will be particularly important for the further development of theory in this field.

Have we come to the right place? Several commentators (Lyons, Zuckerman, Panksepp, Crider & Solomon) wonder whether my model of anxiety captures the essential features of the human emotion that goes by this name; and even whether I have not in error constructed a model of some quite different emotion.

Again it is Lyons who puts this objection most forcibly. He constructs an example of a person who perceives a novel stimulus, which inhibits his ongoing motor behaviour and causes him to experience increased levels of arousal and increased levels of attention to the novel stimulus; and he comments (accurately) that "while this profile fulfills Gray's conditions for the person being in a state of anxiety, it also fits with that person being in a state of awe or excitement or admiration."

This mode of argument has its dangers. It assumes that to every name of an emotion (and there are many hundreds) there corresponds a distinct biological (as opposed to linguistically discriminable) state. If this were so, we should probably have to give up entirely using animals to search for the biological bases of the different human emotions. But there is no need to assume that it is so. As I have pointed out before (Gray 1972b) one may with greater plausibility suppose that there are only a limited number of biologically fundamental emotions, each of which may be given many names depending on the particular kinds of event that give rise to them. Still other emotion names may refer to different degrees of activation of the same fundamental emotions, or to different blends between them. These, of course, are moves that have also been made by other biologically inclined theorists in the field of emotion (e.g., Panksepp 1982, this issue). They are similar in essence to the moves that have successfully reduced the bewildering variety of colour experiences to combinations of outputs from a very small number of distinct chromatic channels. If we do not make some such set of moves, we shall open up an unbridgeable gap between the emotional life of man and that of the other animals; and this gap is unacceptable on general Darwinian grounds which (I imagine) Lyons would accept.

Nonetheless, while I do not in general find much force in arguments of the kind deployed here by Lyons, I accept that the particular example he has constructed poses a real problem for the theory. I might just imagine (to take his example further) that the first sight of a golden eagle in flight causes some degree of anxiety; but Mozart? No, not Mozart. So where does the problem lie? It turns particularly, I believe, on the choice of novelty as the eliciting stimulus. It is true, of course, that novelty is one of the kinds of event to which I have attributed the property of activating the behavioural inhibitions system (BIS); so Lyons's example is fair. But I doubt that he could construct a similar example using the other kinds of stimuli to which I have also attributed this property (innate fear stimuli, conditioned punishing stimuli, and conditioned frustrative stimuli).

Zuckerman also cites human data that pose problems for my model of anxiety, and he too concentrates on responses to novelty. This is not surprising. The role of

novelty has always been the weakest link in the chain of argument by which the concept of the BIS was built up. A tighter theory could undoubtedly be produced if one simply deleted novelty from the list of stimuli that activate the BIS, as Zuckerman appears to recommend. There are at least four reasons for not doing this. First, novel stimuli elicit in animals (though usually to a milder degree) the same general patterns of behaviour as the other stimuli that activate the BIS (Gray 1975). Second, antianxiety drugs affect responses to novelty in a manner consistent with the concept of the BIS (Gray 1977). Third, exposure to novel stimuli has been widely and effectively used in studies of individual differences in fearfulness in animals (Broadhurst 1960). Fourth, even in human beings, *extreme* degrees of novelty can undoubtedly elicit anxiety.

Novelty, then, has to remain on the list of anxiogenic stimuli. But it will continue to give rise to problems of the kind posed by Lyons and by Zuckerman. This is because novelty has mixed effects on behaviour. It simultaneously elicits approach and avoidance tendencies (Gray 1971a), whereas other anxiogenic stimuli will not normally elicit approach behaviour unless they are combined with another, positive reinforcer. This mixture of approach and avoidance makes good evolutionary sense: It is essential for animals to explore novel stimuli so that they can achieve sufficient familiarity with their environment; but if they do so incautiously they may meet an untimely death. It is probably this kind of interaction between different survival pressures that gives rise, say, to the pattern of mobbing behaviour in the chaffinch, described by Hinde (1954), in which approach and avoidance are exquisitely balanced, the former gradually coming to dominate as the bird learns that an initially novel object is not dangerous.

Ursin also points to a problem with responses to novelty. He remarks that "in humans it is possible to discriminate between the orienting response and a conditioned emotional response by factor analysis of galvanic skin responses"; he cites Öhman (1972) in support of this statement. But this is an overgeneralisation from Öhman's observations. Since Öhman used an orthogonal factor solution, his method of analysis did not permit him to discover whether or not responses to the conditioned stimulus itself (orienting responses) and to the conditioned stimulus qua signal of shock (conditioned emotional responses) were correlated. Perusal of his correlation matrix prior to factorisation makes it quite clear that they were. Thus this is a false problem.

Zuckerman correctly identifies the problem posed by the mixed effects of novel stimuli: "Gray has chosen to define fully the mechanisms of anxiety before dealing with those of impulsivity (or sensitivity to signals of reward). The two mechanisms must interact in the response to novel stimuli." And he is probably right in his belief that my neglect of the second of these mechanisms has led to the kind of inconsistency on which he and Lyons both comment. But, if so, the need is not for more theorising but for more data, especially at the animal level. There is very little sound evidence on which to base a careful analysis of the mode of interaction of the approach and avoidance tendencies elicited by novel stimuli, or of the way in which this interaction is biased by the antianxiety drugs; and the existing data base contains

many contradictions (see the discussion in *Anxiety*, p. 40 *et seq.*). If such an analysis were conducted in parallel with appropriate research on human subjects, one might gain some insight into the nature of the at-present rather obscure emotional states that Lyons summons in evidence. The "awe" felt by his birdwatcher seeing for the first time a golden eagle in flight might well turn out to be a mixture of emotions from which anxiety (as defined in terms of the BIS) is not entirely absent. And we might also find some understanding of the emotional mix that (as pointed out by Gold & Fox) leads people to seek out activities (such as skydiving or downhill skiing) that cause intense anxiety. At present, our models of emotion derived from the animal laboratory are probably incapable of handling these specifically human emotional vagaries; but there is no reason they should not eventually be able to do so.

Lyons points to another set of specifically human features of anxiety. He believes that a proper understanding of anxiety requires an analysis of the way in which an anxious person evaluates future events along sets of internal rating scales – "the certain/uncertain, demanding/undemanding, and important/unimportant scales for a start." It is not clear whether he regards such evaluation as part of the way in which situations come to evoke anxiety or as part of anxious behaviour once it is evoked. In either case the point is either largely irrelevant to an understanding of the biological basis of the emotions, or it is already largely taken account of in the theory developed in *Anxiety*. It is largely irrelevant if Lyons supposes that his postulated evaluations take place in the verbal mode, even if the subject addresses himself rather than engaging in public dialogue. For, if anxiety depends critically on events that take place in the verbal mode, animals do not experience it. Not only would this conclusion be implausible on general Darwinian grounds, it would also require one to explain the apparent continuity (documented in *Anxiety*) in the effects of antianxiety drugs in animals and man. If, on the other hand, Lyons regards his evaluations as taking place in a nonverbal manner, then they are precisely the kinds of evaluation that give rise to the formation of the conditioned aversive stimuli to which the BIS reacts. However, to document this claim in detail would take us too far afield, as several different learning processes would require discussion (see, e.g., Dickinson 1980; Gray 1975; and *Anxiety*, pp. 392–93).

Much the same can be said about a further claim by Lyons: that anxiety is characteristically not about the present but about the future. If this statement refers to nonlinguistic organisms, the only grasp *they* have of the future stems from their experience of the past; and reactions to conditioned aversive stimuli reflect both this past experience and the anticipation of the future to which Lyons refers (Dickinson 1980). Human beings, to be sure, owing to their possession of language, can anticipate a future that does not stem directly from their own previous experience; thus we can be afraid of cancer or atomic war without having had direct experience of either. But if anxiety as such depends on that kind of faculty, we are again cut off from our animal cousins. It is more reasonable to suppose that anxiety remains basically the same in linguistic and nonlinguistic mammals; but that the possession of language leads to a number of

special features in human anxiety. This formulation applies with particular force to the obsessive-compulsive neurosis; and I have suggested in some detail (*Anxiety*, chapter 14) how, without leaving the confines of my general theory, one can account for the peculiarly cognitive and verbal symptoms of this condition.

Panksepp's concern is that there are two quite separate places that my map names "anxiety." One is properly called (in his terms) "fear"; it manifests itself in animals as flight behaviour and clinically as "anticipatory anxiety" (Klein 1981). The other is properly called "panic"; this manifests itself in animals as separation distress and clinically as the panic attacks experienced by agoraphobics (Klein 1981). The distinction Panksepp makes is one that I accept. Indeed, it would be strange if I did not, since it is based in large measure on the same kind of pharmacological dissection that has given rise to my own theory. In this instance the critical observation is that anticipatory anxiety is responsive to benzodiazepines but not to imipramine, whereas panic attacks show the reverse pattern of response (Klein 1981). This finding alone must strongly incline me to include Klein's (1981) and Panksepp's "anticipatory anxiety" as part of my "anxiety" and to treat panic attacks as something else. The most obvious something else is depression, since imipramine is most commonly used in the treatment of this condition; and separation distress (which Panksepp proposes as an animal analogue of panic attacks) has often been regarded as a biological prototype of depression (Lewis, Young, McKinney & Kraemer 1976). However, Klein's (1981) data suggest that the effect of imipramine on clinical panic attacks is independent of its effect on depression (though there are some contrary data: McNair & Kahn 1981). [See also Anisman & Zacharko: "Depression: The Predisposing Influence of Stress" *BBS* 5(1) 1982.]

In fact, a solution to this problem is not hard to find, and it is one that fits naturally into the theoretical framework of *Anxiety*. The analysis of agoraphobia offered by Klein (1981) in effect treats the spontaneous panic attack as an unconditioned response to unknown unconditioned stimuli; this then acts as the reinforcement to give rise to anticipatory anxiety by standard Pavlovian conditioning mechanisms. A similar analysis is suggested by Gold & Fox, in their discussion of the panic attacks that sometimes accompany mitral-valve prolapse (Gorman, Fyer, Glicklich, King & Klein 1981). But a distinction between responses elicited by unconditioned and conditioned aversive stimuli, respectively, has been fundamental to the concept of the BIS since it was first developed (Gray 1971a; 1980). The BIS is thought to be specialised to respond to conditioned but not to unconditioned aversive stimuli. Thus, if the panic attack is an unconditioned response, it should be (as it is) insensitive to benzodiazepines (which *ex hypothesi* act on the BIS). Of course, in the absence of knowledge of the stimuli that in fact elicit a given patient's panic attacks, this solution may be purely verbal; although it seems safe to apply it to the cases of mitral-valve prolapse raised by Gold & Fox, since the panic attacks these patients experience are presumably directly due to malfunctioning in cardiovascular mechanisms. It is in addition a solution that says nothing positive about the nature of panic attacks – merely that they are not part of (though they seem to generate) anxiety.

Crider & Solomon, while not rejecting the possibility that I have correctly modelled anxiety, suggest that I may (also?) have modelled schizophrenia. They base this suggestion in part on my treatment of the attentional role of the BIS, and in part on their own work on attentional dysfunctions in animals with hippocampal lesions (Solomon 1979) and in human schizophrenics (Crider 1979). They go on to say that "it is entirely possible that the selective attention deficit seen in schizophrenia will also be found among the anxiety disorders." But, given that there is little in common between these two conditions (Eysenck 1960), such an outcome would be surprising. Nor, I think, does their own analysis really predict it. For they see a similarity between the behaviour of human schizophrenics and animals with *damaged* hippocampi; whereas my theory of anxiety treats this as *heightened* activity in *normal* hippocampi. Thus, if anything, schizophrenics and patients with anxiety disorders ought to display opposite patterns of attentional function. More specifically, provided stimuli that activate the BIS are used (*Anxiety*, pp. 456–57), the anxious individual should selectively attend to them *more* than unanxious individuals; whereas (on the Crider–Solomon analysis) schizophrenics should be *less* selective than normal in their attention to all stimuli.

Activity in the BIS, then, cannot provide a model for both anxiety and schizophrenia; but it is possible that hyperactivity in the BIS constitutes anxiety, while some kind of malfunctioning in the hippocampal formation gives rise to some of the symptoms of schizophrenia. Given the evidence for overactivity in dopamine transmission in schizophrenia, such malfunctioning may involve the dopaminergic afferents to the entorhinal cortex and/or the lateral septal area. Mason (1981), in an analysis similar to that offered by Crider & Solomon, has suggested that an impairment in the noradrenergic input to the SHS might produce the attentional dysfunctions seen in schizophrenia. But this proposal has the disadvantage that, allied to my own treatment of anxiety, it would place schizophrenia and anxiety at the opposite poles of a single continuum; yet the evidence is that anxiety and schizophrenia lie on orthogonal dimensions relative to normality (Eysenck 1960). A further suggestion has been made by T. J. Crow (pers. comm. 1981), namely, that confusion between the internally generated stimuli that constitute a prediction in the Papez loop and the external stimuli entering the SHS via the entorhinal cortex might underlie some features of schizophrenic hallucinations or delusions. But this suggestion is at present highly speculative.

Iversen is of the opinion that I have not clearly shown the connections between the frontal lobes and the other places on my map, or indeed shown good cause for them to be on the map at all. It is certainly not the case, as she suggests, that the prefrontal cortex is included in the map on anatomical grounds. On the contrary, this structure has figured in the theory, on behavioural and clinical grounds, since 1970 (Gray 1970a); if anything, I have ever since been searching for anatomical justification for its inclusion (this fortunately exists). I thought the connection with the rest of the theory – that the prefrontal cortex (a) provides information to the SHS about motor programmes, and (b) allows in man for verbal control over the activities of the SHS – was tolerably clear. But, since

Iversen does not say in what way my treatment of the prefrontal cortex was “largely unintegrated” with the rest of the book, there is nothing much I can add.

Iversen’s similar comment on my treatment of memory is, I feel, more justified. As I stated in the *Précis*, “it is still not clear whether the human amnesic syndrome . . . genuinely results from damage to the hippocampus and related structures, or whether the involvement of the hippocampus has been incidental . . . This ambiguity presents obvious difficulties for the would-be theorist needing to know what he is to theorise about.” Thus, although I believe the theory satisfactorily incorporates Olton’s model of working memory (*Anxiety*, p. 284), based as it is on data from animal experiments, I am by no means confident that it encapsulates whatever features of human amnesia may eventually turn out to depend on damage to the hippocampus.

Have we been here before? The feeling that they have seen all this before was expressed by Iversen and Pribram & McGuinness. Pribram & McGuinness claim, in fact, that they anticipated my theory, especially in their 1975 paper on “effort.” This construct, they say, is “often called ‘chronic arousal’ in the psychophysiological literature and ‘stress’ and ‘anxiety’ in the clinical literature.” But this is the only clue they give as to exactly *how* their “effort” theory is essentially “identical” to my theory of anxiety; and rereading Pribram and McGuinness (1975) failed to clarify this issue for me. I could not reread Pribram and Isaacson’s (1975) summary of the Isaacson and Pribram (1975) multiauthor treatise on the hippocampus, since (to my shame) I had not read it before. Having now done so, I agree that many of the key ideas of *Anxiety* are contained in the Pribram and Isaacson (1975) paper. But this is hardly surprising, since they, like I, were reacting to the concepts and data presented by the contributors to the treatise; and I certainly made no secret of the debt my theory owes to many of these (e.g., Vinogradova, Kimble, Weiskrantz, and Warrington).

Iversen’s feeling of *déjà vu* is more general. The book, she says, raises no new issues; theoretically it is a “disappointment”; and, in its application of experimental findings to the clinical phenomena of anxiety, it “raises largely the same issues” as my earlier *Psychology of Fear and Stress* (Gray 1971a). I can only suppose that Iversen has a very high threshold for the detection of novelty (Zuckerman). For where, I wonder, has she seen before the suggestions (to take just a few examples) that the Papez loop mediates the generation of predictions; that the subicular region acts as a comparator; that the hippocampal formation, in conjunction with its noradrenergic and serotonergic afferents, serves as a device for the selection of stimuli to be analysed within the comparator; that the relative involvement of the ascending noradrenergic and serotonergic pathways depends on the relative proportions of sensory analysis and motor inhibition respectively required by a given task; or that the SHS can function in both a “checking” and a “control” mode? These ideas may be bad; but they are (or so I thought) both new and my very own. As to the application of the theory to clinical phenomena, far from being the same as in Gray 1971a, it is in one important respect quite different. The earlier book treated obsessive-compulsive symptoms as a form of active avoidance behaviour. In

Anxiety (chapter 14) I specifically criticise this hypothesis and substitute a new one which treats obsessive-compulsive behaviour largely as a form of excessive checking.

I have so far dealt with rather general and fundamental objections to *Anxiety*. In the remaining sections I deal with more specific issues.

The antianxiety drugs. Given the central importance of the antianxiety drugs in the construction of my theory of anxiety, it is fitting that we should start with them.

Nadel & Morris raise a general issue concerning my interpretations of drug effects: Why, they ask, do I assume that the antianxiety drugs impair (rather than perhaps facilitate) the neural structures they affect? This is a fair point. Although it is more likely that, like electrical stimulation of the brain, a drug will impair the delicate balance of function of the brain, it is certainly possible (again as with electrical stimulation) that under certain conditions it could in some way boost normal function. Caffeine, for example, seems to work in this way (Revelle, Humphreys, Simon & Gilliland 1980). Had the antianxiety drugs acted consistently in a manner opposed to septo-hippocampal lesions, I would simply have stood the argument on its head; the resulting hypothesis would have attributed anxiety to decreased functioning in the SHS. I ignored this possibility because these substances *consistently* (and that is the chief point) act like septo-hippocampal lesions.

This is not to say that the similarities between the effects of the antianxiety drugs and those of septo-hippocampal lesions are without exception, as noted by both McNaughton and Rawlins; and, indeed, much of the research from my own laboratory has been concerned to discover at just what point these similarities break down. But, given the current version of the relevant hypothesis (i.e., that the antianxiety drugs impair septo-hippocampal function, not directly, but by way of a reduction in the noradrenergic and serotonergic inputs to the SHS), we would expect them to break down somewhere, as the discussion of Olton and Papas’s (1979) experiment, above, has made clear. Of course, this is not to say that, at present, we have an understanding of all the detailed differences between the effects of septo-hippocampal lesions and those of the antianxiety drugs. I accept McNaughton’s point that these are the kinds of detail with which a theory *should* be able to cope. But we are not yet at that stage with respect to either the spontaneous alternation paradigm he discusses or (in spite of a considerable research effort) the partial reinforcement extinction effect (PRE) discussed by Rawlins. Whether progress now depends on more data or better theorising is a moot point; but I believe that there are many empirical issues needing clarification before theory can develop further. If, however, we disregard these discrepancies between the drug and lesion effects, the overall pattern of similarity displayed in Table 1 of the *Précis* is difficult to explain except on the hypothesis that, in some way, the antianxiety drugs indeed impair the functioning of the SHS.

Soubrié raises a series of points. Some of these concern matters that could not have figured in my book because they were not established at the time the book was written. The pace of research in biochemical pharmacology (from which most of Soubrié’s points stem) is much

more rapid than in physiological psychology; perhaps that is why psychologists are more prone than biochemists to the malady of book writing. Happily, what Soubrié takes with one hand he mostly gives back with the other.

Consider the hypothesis that the mode of action of the antianxiety drugs involves critically an interaction with the inhibitory transmitter γ -amino-butyrate (GABA). Soubrié states that this has recently received support from the demonstration (Cananzi, Costa & Guidotti 1980) that the GABA agonist, muscimol, reduces the effects of punishment when it is delivered intraventricularly. But he also points out that new classes of drug with this same behavioural effect – the triazolopyridazines (which, incidentally, were considered in *Anxiety*; see p. 305) and quinoline derivatives – do not appear to affect GABA mechanisms. He might also have added that the picture has in any case been complicated by the discovery of a second kind of GABA receptor, which is insensitive to muscimol or the classical GABA antagonist, bicuculline, but binds the GABA agonist, baclofen (Bowery, Hill, Hudson, Doble, Middlemiss, Shaw & Turnbull 1980; see *Anxiety*, p. 310). Since there is evidence that the baclofen receptor is situated on noradrenergic and serotonergic terminals in the forebrain and regulates the release of these transmitters, this is a potentially important link in the chain of evidence linking the primary neurochemical action of the antianxiety drugs with their behavioural effects. Recent observations made by S. Quintero in my laboratory strengthen this link, since we have found that baclofen and muscimol given together alleviate punishment-produced suppression of barpressing in a manner characteristic of antianxiety drugs.

This discussion, and the corresponding parts of both the *Précis* and *Anxiety* (pp. 302–12), ought to be enough to answer Ursin's criticism that "since it does not agree" with my theory I treat the receptor sites upon which the antianxiety drugs (sc. benzodiazepines) act as "of no relevance." On the contrary, I regard this issue as of sufficient importance for it to guide much of the current research in my laboratory. Ursin seems unwilling, however, to accept that the antianxiety drugs *defined behaviourally* (Gray 1977) include more than the benzodiazepines; yet there is no way in which a psychological concept like anxiety can be primarily defined in any other terms, be they chemical or neurochemical. Thus the only mode of neurochemical action that can be essential to the understanding of anxiety (as distinct from the pharmacology of a particular group of drugs) is one that is common to *all* the anxiolytics (Gray 1979a). This point appears to be accepted now by biochemical psychopharmacologists themselves (e.g., Olsen 1981). Taken seriously, it precludes the inference that occupancy of the benzodiazepine receptor is critical for antianxiety action (since none of the other classes of anxiolytic – the barbiturates, ethanol, or meprobamate – binds to this receptor). But it is not yet excluded that facilitation of GABA-ergic transmission (in consequence of interaction with a number of different, but closely related, receptors: Olsen 1981) is the common step in anxiolytic action. However, even in this case, it is almost certain that such action will depend on facilitated GABA-ergic transmission only at specific points in the brain, not everywhere (Gray 1979a), and it remains to establish which points these are. Thus any final map of anxiety will need to be constructed from

two complementary points of view: the way the antianxiety drugs act neurochemically; *and* the way the separate brain mechanisms on which each drug acts converge on a common behavioural exit.

Another omission in my treatment of antianxiety drug action mentioned by Soubrié concerns the putative endogenous ligand for the benzodiazepine receptor. But, as discussed in *Anxiety* (p. 303), there are several candidates for this ligand and none is yet firmly established (Braestrup & Nielsen 1980); and, if the ligand exists, it is as likely to be a naturally occurring antianxiety as proanxiety agent. This state of affairs has not radically changed, so far as I know, since the book was written. The substance ethyl β -carboline-3-carboxylate (β -CCE; Braestrup, Nielsen & Olsen 1980) is certainly a strong candidate for the role of endogenous ligand; and I am grateful to Soubrié for drawing my attention to the report by Polc, Ropert, and Weight (1981) that this drug antagonises the action of both the benzodiazepines and GABA in the hippocampus. Data from our own laboratory (S. Quintero & D. Nutt, pers. comm.) agree with this report, since we find that β -CCE blocks the effect of chlordiazepoxide on the theta-driving curve. Thus, if β -CCE is a close relative of the putative endogenous ligand, the latter seems likely to be an endogenous proanxiety agent. But, even assuming this tentative conclusion to be correct, it is not yet clear what further inferences flow from it.

Unlike Ursin (see above), Soubrié comments favourably on my use of a triangulation strategy, that is, the strategy in which one seeks the *common* action of the different classes of antianxiety drug. However, he identifies an error in the argument (*Anxiety*, p. 16) by which I segregated anticonvulsant from anxiolytic action. This argument supposed that alcohol is free of anticonvulsant effects. But Soubrié cites Boissier, Simon, and LeBourhis (1969) as showing that alcohol *does* protect animals against experimentally induced seizures. I am grateful to Soubrié for drawing my attention to this error; and even more, for the promptness with which he offered me a way out of the problem it creates. For another paper cited by Soubrié (Le Fur, Mizoule, Burgelin, Ferris, Heaulme, Gauthier, Gueremy & Uzan 1981) demonstrates a dissociation between anticonflict and anticonvulsant activity in the case of two new quinoline derivatives; thus the triangulation strategy is restored, and convulsions are not part of anxiety. As for anaesthesia (another problem raised by Soubrié in the strategy of triangulation), considerations of dose (which, I agree, received little attention in *Anxiety*) can eliminate this as part of antianxiety action: The behaviourally relevant effects of the antianxiety drugs are obtained at doses lower than those necessary for anaesthetic (or even general sedative) effects (Gray 1977).

A final issue raised by Soubrié concerns the triazolopyridazines and the Type I benzodiazepine receptor (Lippa, Coupet, Greenblatt, Klepner & Beer 1979) to which they bind. The triazolopyridazine receptor is particularly strongly present in the cerebellum. Thus these compounds pose a real problem, discussed in some detail in *Anxiety* (pp. 304–308). My own view is that any hypothesis that predicts a major role for the cerebellum in anxiety can be ruled out on that count alone; thus the significance of the special properties of this class of drugs is still obscure.

Soubrié and Zuckerman both apparently misunderstand how my theory couples behavioural inhibition (in the narrow sense) and anxiety. I neither see anxiety as necessarily leading to the overt inhibition of movement (Soubrié), nor the overt inhibition of movement as necessarily indicating the presence of anxiety (Zuckerman); nor do I state that antianxiety drugs will necessarily increase motor activity (by disinhibition). These issues are discussed in some detail in *Anxiety* (especially pp. 46–48) and Gray (1977) with respect to drug action, and in Gray (1975) and Gray and Smith (1969) with respect to general learning theory. Examples I give in *Anxiety* of cases in which anxiety increases motor activity (reflecting *ex hypothesi* an increment in level of arousal: Gray & Smith 1969) and in which antianxiety drugs reverse this increase include: blockade by both benzodiazepines and barbiturates of the potentiated startle response to a secondary punishing stimulus (M. Davis 1979; Chi 1965); blockade of the partial reinforcement acquisition effect by several antianxiety drugs (e.g., Gray 1969); blockade by chlor-diazepoxide of conditioned acceleration on a baseline of DRL (differential reinforcement of low rates of response) (Sanger & Blackman 1976); and blockade of increased locomotion to an air-puff by diazepam (Simon, Fraisse, Tillement, Guernet & Boissier 1968). Conversely, if conditioned suppression of bar pressing is produced by a stimulus associated, not with punishment but with reward, antianxiety drugs do *not* alleviate this motor inhibition (Miczek 1973). Thus no generalisation can be made in terms of the direction of change in motor behaviour. The correct generalisation is that the antianxiety drugs “counteract whatever is the effect of a secondary aversive or a novel stimulus: if the stimulus suppresses behaviour, the anti-anxiety drugs facilitate it; if the stimulus facilitates behaviour, the anti-anxiety drugs suppress it” (*Anxiety*, p. 47).

The reverse bias is shown in Ursin’s comment on my interpretation of the effects of the antianxiety drugs. He believes that it would be “more parsimonious and less controversial to state” that these drugs “all reduce arousal or activation.” This move might indeed be more parsimonious; but it would be virtually devoid of predictive power (Revelle is in agreement on this point). Anxiety is accompanied by *both* an increase in level of arousal *and* an increased probability of motor inhibition (Gray & Smith 1969). If Ursin believes that variation in one underlying process can account for all the findings in this field, both behavioural and psychopharmacological (Gray 1971a; 1975; 1977), then let him produce a theory that will achieve this.

Leslie questions my interpretation of the data on the effects of antianxiety drugs on conditioned suppression (Gray 1977; *Anxiety*, pp. 23–28). The conclusion I reached was that while these agents impair both punishment-produced and on-the-baseline conditioned suppression, they do not in general reduce off-the-baseline conditioned suppression. Leslie believes that in arriving at this conclusion, I erroneously lumped together two different off-the-baseline procedures, of which one (Estes & Skinner 1941) is sensitive to the antianxiety drugs but the other (Leaf & Muller 1965) is not. I hope Leslie is right. I agree with him that if he is, considerable theoretical simplicity is gained; and it is for this reason that, in my initial formulation of the concept of the BIS (Gray 1971a),

I adopted the stance that Leslie now recommends. Nor would it be necessary, if he is right, to change much in the theory developed in *Anxiety*; although one of its (post hoc) predictions (p. 296) would now become redundant. The change required would consist in the assumption that mismatch, as well as tagging as “faulty” motor programmes in progress at the time mismatch is detected, also causes stimuli associated with mismatch to acquire the property of subsequently disrupting any motor programme. But it remains, in my view, an open empirical question whether off-the-baseline suppression is indeed as sensitive to the effects of antianxiety drugs as Leslie claims. In addition, if Leslie’s analysis is correct, some account will be required of *why* the antianxiety drugs have inconsistent effects in the Leaf & Muller (1965) procedure. I agree, however, that Leslie’s position (as he neatly shows) brings a nice theoretical bonus in integrating Kamin, Brimer, and Black’s (1963) experiment on two-way avoidance with the drug literature. Thus this remains a fruitful field for psychopharmacological research.

Soubrié asks why the antianxiety drugs do not generally alter active avoidance at doses that impair passive avoidance. The answer within my overall theory is clear. These two kinds of behaviour depend on different underlying processes (Gray 1975; 1977): Passive avoidance depends on the BIS, upon which the drugs act; active avoidance depends (largely) on the same processes that give rise to rewarded behaviour and so, like the latter, it is insensitive to the action of the drugs.

Soubrié also asks whether lesions to the SHS alter the behavioural effects of the antianxiety drugs. This is an issue to which my laboratory has devoted some attention. As summarised in *Anxiety* (pp. 300–302), the results of such experiments have yielded ambiguous results, and this is one reason I abandoned the hypothesis of a direct action of the anxiolytics on the SHS (Gray 1970b) in favour of an action on the ascending monoamine inputs to the SHS (Gray 1977); although it is possible that both sites of action are involved (see the *Précis*, section entitled “How do the antianxiety drugs reduce anxiety?”).

McNaughton asks how my model predicts that sodium amylobarbitone will specifically affect the control of spontaneous alternation by body turns (as opposed to odour trail avoidance). Alas, it does not predict this. But it does predict (contrary to Zuckerman’s opinion) that antianxiety drugs should block spontaneous alternation in some way or other. Spontaneous alternation, in terms of the BIS model, is an increase in orienting-investigative behaviour caused by exposure to novelty; thus it should be reduced by the antianxiety drugs. The situations in which the model might predict the opposite are those in which the amount rather than the direction of motor behaviour is measured. Under these conditions activation of the BIS may reduce exploration by direct motor inhibition, and this should be alleviated by the drug (see *Anxiety*, pp. 40–44, for a fuller discussion). The other points raised by Zuckerman, concerning defecation and ambulation in the open field, are indeed problems for the theory (*Anxiety*, pp. 40–44); but the problem with defecation may simply be one of measurement (the antianxiety drugs may have two effects, the nonspecific inhibition of defecation obscuring the specific inhibition of fear-related defecation).

The septal area. Only two comments specifically concerned the septal area.

Olton believes that the septum should be considered important in memory processing. I agree that there are no data to rule out a role in memory for the septal area similar to that of the hippocampus, and that some data suggest that these two regions function jointly in memory as in other behavioural processes; though exactly what role either part of the SHS plays in memory as such remains, I believe, uncertain. It is a welcome development that Olton's group is applying the same types of behavioural analysis to the effects of septal lesions as to those of fornical and entorhinal lesions (Mitchell, Rawlins, Steward & Olton, in press).

Unlike **Olton**, who wishes to stress still further the role of septo-hippocampal connections, **Pribram & McGuinness** believe that I have overemphasised them. To illustrate this point they state that "the effects of septal and of hippocampal lesions are not always the same," citing the case of aggressive behaviour as an example. Certainly, this is correct. But the intention of my review of the lesion literature was to delineate a septo-hippocampal syndrome on the basis of those effects of septal and hippocampal lesions (and they turn out to be the great majority) that *are* the same. Naturally, given this aim, I paid less attention to the anatomical basis of those features of the septal and hippocampal syndromes that differ from each other (but see *Anxiety*, pp. 176–80). No doubt it would be possible to determine which features of the septal syndrome owe something to the close connections of the nucleus accumbens to the basal ganglia and amygdala, as recommended by Pribram & McGuinness. But this was not necessary for the argument I was pursuing, especially since the septo-hippocampal syndrome itself maps so well onto the effects of the antianxiety drugs (Précis, Table 1).

The hippocampal formation. The idea that the hippocampal formation functions as a comparator, which is central to the theory developed in *Anxiety*, met with approval from several commentators (**Gabriel, James, Nadel & Morris, Crider & Solomon, Kimble, and Lyons**) and no explicit opposition. Given "the maddening confusion that is the literature on the septo-hippocampal system" (Nadel & Morris), this must be counted as an advance, one foreshadowed in the Ciba Foundation symposium (Elliott & Whelan 1978) on the SHS. Somewhat surprisingly, no one commented on the specific form in which I developed this concept, namely, by allocating the main comparator function to the subicular area, with Vinogradova's (1975) hippocampal circuits relegated to the role of selection of input to the comparator.

The remaining points concerning the hippocampus form a rather heterogeneous list.

I am grateful to **James** for his fascinating speculation on the possible evolutionary development of the comparator function of the hippocampal formation.

Iversen suggests that the hippocampal formation may mediate both anxiety and memory by "independent use of its neural connections." This suggestion is congruent (though not necessarily identical) with my conclusion that the bidirectional relations between the septo-hippocampal system and neocortical language areas . . . allow it to function under certain conditions independently of its ascending inputs. It is this interaction

between language areas and the septo-hippocampal system which permits the latter to play a major role in human cognitive function even when this is not emotionally coloured. (*Anxiety*, p. 460)

One can make a similar analysis of the performance of animals in **Olton's** radial-arm maze (see above).

Ursin questions the assumption that electrical stimulation of the hippocampal formation can produce somatomotor inhibition. I accept his correction that such an effect was not demonstrated in Kaada's (1951) original experiments. But it was subsequently demonstrated by Bland and Vanderwolf (1972) among others (see *Anxiety*, p. 215, for further references). [See also Vanderwolf & Robinson: "Reticulo-Cortical Activity and Behavior" *BBS* 4(3) 1981.]

Rawlins points to a contradiction between two ways in which, in separate places, I deal with the prediction from the spatial mapping theory (Black, Nadel & O'Keefe 1977) that the disinhibitory effects of hippocampal lesions should be greater, the further the animal is from a goal in which it receives both reward and punishment (or presumably reward and nonreward). In one I count it as evidence against the spatial theory when this prediction is disconfirmed (e.g., *Anxiety*, pp. 124, 159); in the other I show how the same prediction can be derived from my own hypothesis that the BIS is concerned with reactions to stimuli that warn of punishment, not with reactions to punishment itself (*Anxiety*, p. 297). The point is well taken: I cannot take on the spatial theory's prediction without its baggage of disconfirmations. On balance, there are more confirmations than disconfirmations (*Anxiety*, p. 124), so both theories are in poor but not desperate straits where this prediction is concerned. Further progress is likely to rest upon an analysis of the conditions that favour confirmation and disconfirmation respectively. The two theories can be differentiated by the use of nonspatial tasks, which would cancel the cognitive mapping prediction but leave mine unchanged.

Nadel & Morris caution against taking hippocampal involvement in nonspatial classical conditioning as necessarily evidence against the spatial mapping theory. I accept their point that spatial context plays a significant role in some conditioning tasks that are themselves not spatial (as demonstrated also by Odling-Smee 1975). But it will be necessary to show in detail how this fact and the spatial theory can combine to account for the effects of hippocampal lesions in classical conditioning tasks (e.g., Solomon 1977). It is not easy to see how this can be done.

I also accept the point made by **Nadel & Morris** concerning the interpretation of experiments using lesions of the fornix-fimbria. But I do not know of any alterations in the conclusions of my review of the literature that would in consequence be necessary (I have already admitted that the special sensitivity of place learning to hippocampal damage poses a problem for my theory).

Another criticism I accept is made by **Woodruff**, namely, that I have not clearly specified how the SHS exerts its influence on behaviour. But, as Woodruff also points out, this is where "all other theories concerning the function of this system in behavior are weak." The reason for this weakness is not hard to find: The known anatomy of the connections of the SHS (*Anxiety*, chapter 3) offers very few clues as to how this system outputs to regions of the

brain that control motor behaviour. Until this anatomical lacuna is filled theorising is bound to remain difficult.

Nadel & Morris raise two points of detail that I fail to understand. The first concerns the study by Isaacson, Olton, Bauer, and Swart (1966), in which rats run from a shaking elevated alley to descend into shock. This, they say, involves escape (from the alley) as well as passive avoidance (of the grid floor delivering the shock). So it does. But it does not involve an escape component in the sense defined by Black, Nadel, and O'Keefe (1977) as being necessary for the production of a hippocampal deficit in passive avoidance: that is, escape first *from* the place that subsequently has to be passively avoided (which, according to Black et al. 1977, gives rise to a requirement for spatial analysis otherwise absent from step-down passive avoidance). Thus I stand by my treatment of this study (*Anxiety*, p. 125).

The second point concerns the effects of prehabitation to a conditioned stimulus (CS) in one- and two-way active avoidance tasks in Olton and Isaacson's (1968) experiment. This procedure retarded the learning of one-way avoidance in hippocampal but not in normal animals, and it improved the learning of two-way avoidance in normal but not in hippocampal animals (*Anxiety*, p. 138). **Nadel & Morris** state that "within spatial map theory such a difference develops naturally from the differing roles played by place, and other, strategies in the two situations." Unfortunately, they do not explain why this is so. Assuming that hippocampal animals are in both cases more dependent than normal animals on nonspatial cues (including the CS), I can understand the one-way result (the CS loses salience and the hippocampal animals do not have space to use instead), but not the two-way result. For, in the two-way case, should not the spatial hypothesis predict that, after prehabitation to the CS, normal animals will be even worse relative to hippocampal animals than usual (the CS loses salience, so the normal animals rely on an inappropriate spatial strategy that the hippocampal animals do not have)?

Pribram & McGuinness make three points against the hypothesis that the hippocampus is involved in the inhibition of motor behaviour.

1. I am unable to agree with their formulation of the literature on distraction after hippocampal lesions, which they see as incompatible with this hypothesis. Since they do not cite any particular experiments in support of their view, I can only fall back on my own summary of the relevant experiments (*Anxiety*, p. 147):

In predicting the effect of novel stimuli on hippocampal animals, one has to take account of the animal's dominant activity or "set." The principle which appears to operate is this: the lesioned animal displays the dominant motor activity with greater vigour than controls. If this activity is that directly elicited by the novel stimulus, the lesioned animal displays this directly elicited response to a more marked degree. But if some other behaviour is dominant, this is performed with greater vigour by the lesioned animals, and the directly elicited response is weakened.

This is fully compatible with a behavioural inhibition hypothesis (albeit a complex one) of hippocampal function.

2. With respect, however, to go/no-go alternation, I agree with **Pribram & McGuinness** that the data from the

relevant experiments constitute a serious challenge to the behavioural inhibition hypothesis (*Anxiety*, p. 168); though it is also the case that they do not fit easily with other hypothesis of hippocampal function.

3. **Pribram & McGuinness's** final point concerns Berger and Thompson's (1978a) evidence that a "neuronal model" of the rabbit's nictitating response is formed in the hippocampus prior to the "occurrence (not the inhibition) of that response." But this neuronal model plays no role in the control of the nictitating membrane response (since this is unaffected by hippocampal lesions). Thus I do not see the relevance of this point to the behavioural inhibition hypothesis.

Crider & Solomon also raise an objection to the hypothesis that the hippocampus is involved in behavioural inhibition. If so, they ask, why do hippocampal lesions have no effect on Pavlovian conditioned inhibition (Solomon 1977)? The answer to this question is straightforward: Pavlovian conditioned inhibition is a different process from behavioural inhibition (Gray 1975). To see that this is so, consider an experiment on conditioned suppression of bar pressing with an aversive unconditioned stimulus. This is an example of behavioural inhibition (suppression of a motor response by a stimulus associated with an aversive event). If we now set up a conditioned inhibitor (by standard Pavlovian procedures), this will *alleviate* conditioned suppression. Thus conditioned inhibition cannot be the same process as behavioural inhibition. The puzzle in the literature, therefore, is not the failure of hippocampal lesions to reduce conditioned inhibition, but rather its failure to reduce conditioned suppression (*Anxiety*, pp. 133–34; and see the discussion of Leslie's commentary, above).

Woodruff questions my treatment of the hippocampus as a "functionally homogeneous structure" in the light of Flynn's (1967) report that dorsal hippocampal stimulation inhibits hypothalamically elicited attack while ventral hippocampal stimulation facilitates such attack. However, I did recognise that the ventral part of the subicular complex (which may well have been stimulated in Flynn's experiments) "gives rise to a pattern of outputs that is very different from the projections of the remainder of the subiculum" (*Anxiety*, p. 72). I was largely able to ignore this complication in my interpretation of the data on the effects of hippocampal lesions because rather few of the relevant experiments have damaged principally the ventral part of the hippocampal formation. But I agree that any final theory – when there are sufficient data – will have to give a better account of the role played by the ventral subiculum.

The Papez circuit. The part of the theory dealing with the Papez circuit (subicular area-anteroventral thalamus-cingulate gyrus-subicular area) is the most speculative. It is also the least well specified, a fact not unconnected with the paucity of basic anatomical and physiological information concerning this circuit (as compared to the hippocampal formation itself). Where maps are most sketchy, imagination can reign most free. And **Gabriel's** criticism that I attributed homuncular properties to the subicular comparator is well deserved. But, as Gabriel admits, it is not impossible that the subiculum *can* do all of the things I attribute to it. What is needed if one is to attempt to show *how* it might do these things is much more data on

the interactions between the different classes of neurons, and their inputs and outputs, that go to make up the subicular region. Only then will theorising at the level of detail attempted for the hippocampus and dentate gyrus by O'Keefe and Nadel (1978) and myself become possible also for the subiculum.

Gabriel suggests that I might have done better to reverse my suggestions that the anteroventral thalamus is a point of access for response-independent environmental regularities and the cingulate gyrus one for response-dependent regularities. His own data, I agree, are consistent with such a possibility. Let me emphasise again that these suggestions (*Anxiety*, p. 269) were made very much in the spirit of speculation and with no hard data to back them up. I hope they will provide a heuristic spur to the design of experiments. If the Papez loop turns out to have a predictive function in the general manner described in *Anxiety*, I shall count that a great success; but it would be surprising indeed if, with so much unknown, I have got the details correct. Let us hope that other groups will soon follow Gabriel's own pioneering efforts in this field and begin to reduce the area of the unknown.

For similar reasons I have no ready answer to the problem raised by Woodruff concerning the effects of cingulate lesions (generally opposite to those of septal or hippocampal lesions). This is an area of the literature that I wrongly omitted to review in *Anxiety*. But it is one that is difficult to interpret for lack of anatomical data. It is still not clear, at least for that favourite (*pace* Pribram & McGuinness) laboratory animal, the rat, which are the appropriate regions to attack. Much of what used to be regarded as cingulate cortex, for example, was transposed into the rodent equivalent of prefrontal cortex as a result of Leonard's (1969) studies, which postdate some of the experiments cited by Woodruff. At this time I do not see any easy way in which I can predict that disruption of the Papez circuit at *any* point should have effects on behaviour opposite to those seen after hippocampal lesions; but if one knew exactly which points (in terms of their connections) were critical for this kind of effect, theorising would at least become easier.

The hippocampal theta rhythm. Comments on the theta rhythm were made by James, McNaughton, and Robinson & Therrien.

James remarks on my statement, following Bennett, French, and Burnett (1978), that the observed differences between the rat, on the one hand, and the cat and rabbit, on the other, in theta frequency may "reflect species-specific modes of exploratory behaviour" (*Anxiety*, p. 188). He states that this is "not a full explanation of why hippocampal theta frequencies should vary so widely across species when hippocampal anatomy appears relatively constant." I agree. But it was not intended to be a full explanation. For the more clear-cut species difference appears to be between rats and rabbits (somewhat higher theta frequencies), on the one hand, and cats and dogs (somewhat lower) on the other (*Anxiety*, p. 79). For this difference I did not offer an explanation. In view of the hypothesis that theta frequency is related to the passage of information around the hippocampal and Papez circuits (*Anxiety*, p. 291 *et seq.*), it is possible that the determining factor is not the gross conformation of hippocampal anatomy but the actual lengths of the neuronal

pathways concerned and the time taken for signals to traverse them. This is a testable hypothesis, but I know of no relevant data.

McNaughton makes several points. The most substantial is his derivation from my theory that medial septal lesions (which abolish the theta rhythm) "should reproduce all of the behavioural effects of hippocampal lesions," since loss of theta should disrupt temporal quantising of the flow of information around the hippocampal and Papez circuits. And, since medial septal lesions reproduce only some of the effects of hippocampal lesions, this derivation is empirically incorrect. I accept this point. To repair the theory I would have to suppose that accurate quantising of the temporal flow of information is more important for some tasks than for others; or that, for some tasks but not others, there are alternative sources of timing (e.g., environmental cues). I see no easy way of doing either of these things. Thus I see McNaughton's derivation as the most damaging single criticism in the commentaries. As he notes, however, my theory is not alone in its inability to account for the failure of medial septal lesions fully to duplicate the effects of hippocampal lesions.

A second point made by McNaughton concerns the interpretation of the shape of the theta-driving curve, that is, the plot of threshold septal driving current (eliciting hippocampal theta) against stimulation frequency. McNaughton sees the minimum in this curve at 7.7 Hz as reflecting a general lowering of threshold in the middle of the theta frequency range, which is simply most pronounced at 7.7 Hz, rather than a notch more directly related to this frequency. This may be so. However, since all reported effects of drugs on the theta-driving curve have been obtained in experiments with only five sampling frequencies of stimulation (e.g., McNaughton, James, Stewart, Gray, Valero & Drewnowski 1977), it is not at present possible to choose between the two interpretations. In favour of the notch view is the evidence that serotonergic afferents to the hippocampus inhibit septal elicitation of theta at every one of the five sampling frequencies other than 7.7 Hz (McNaughton et al. 1977; McNaughton, Azmitia, Williams, Buchan & Gray 1980).

McNaughton's third point is that one cannot assume that the same processes that affect the threshold for septally elicited theta also affect spontaneous theta; and he cites examples of differences between the effects of various treatments on the two kinds of theta. The point is valid; although there are also demonstrations of important similarities between spontaneous and septally elicited theta (Rawlins, Feldon & Gray 1979; Kramis & Vanderwolf 1980). The discrepancies to which McNaughton points may arise because septal stimulation is able to reveal properties of the septo-hippocampal circuitry controlling theta that escape other methods. This would be particularly likely if the 7.7-Hz minimum in the theta-driving curve is in part due to feedback from the hippocampus to the septal area, as suggested by the fact that fimbrial section eliminates the minimum (Rawlins et al. 1979). If so, direct stimulation of the medial septal pacemakers for theta has the effect (at least in part) of opening a servoloop. This might reveal properties of the loop that are also operative when the loop is closed, but that escape analysis under those conditions.

McNaughton also comments that "even when one

might wish to attribute an action of the minor tranquillisers to an effect on theta mechanisms, there is no necessity to relate this to changes in noradrenergic systems." Indeed there is not. Since these drugs reduce the serotonergic input to the SHS, and since this input also plays an important role in the control of (septally elicited) theta, one would at least wish to relate the effects of the drugs to an alteration in both monoaminergic inputs to the SHS. Reduction in ascending cholinergic mechanisms might also play a role (Graeff, Quintero & Gray 1980).

The biggest onslaught on my interpretation of the data on hippocampal theta comes from **Robinson & Therrien**. I hope they will forgive my putting their views into the general context of the Vanderwolf school, with which Robinson has been closely associated. For the remarkable thing is the degree to which, in attacking my position, they have radically changed their own, as expressed in previous papers by Vanderwolf and Robinson.

Vanderwolf and Robinson have as recently as 1981 reiterated the lesson that Vanderwolf has been stressing for over a decade: Close observation of behaviour is primary; this and only this can determine what behavioural function we should attribute to the electrical activity of the brain. On this basis Vanderwolf (1971) first proposed that hippocampal theta is related to movement. Subsequently, he modified this position to take account of the fact that theta is sometimes correlated with movement, but sometimes with immobility. He then postulated two types of theta, one related to movement, the other not, supporting this distinction by showing that it was related to two other distinctions (e.g., Vanderwolf, Kramis, Gillespie & Bland 1975), one to do with theta frequency, the other pharmacological. But now, according to **Robinson & Therrien**, citing Robinson (1980), the two Vanderwolfian theta "systems are operationally defined on the basis of their relative sensitivity to *drugs*, but not on their relations to behavior or frequency spectra" (my italics).

I cannot but welcome this somersault, for my own dissection of theta has also been based to a considerable degree on pharmacological criteria (e.g., Gray & Ball 1970). Furthermore, we appear to be in agreement that a dissection based on the effects of drugs on septally elicited theta (McNaughton et al. 1977) is as valid as one based on the effects of drugs on spontaneous theta or theta elicited by reticular or hypothalamic stimulation (Kramis & Vanderwolf 1980, cited approvingly by **Robinson & Therrien**). Thus, since we agree on criteria, one would have thought that we might also agree that the classification of theta systems (if indeed there is more than one) should take account of the data gathered by both Vanderwolf's group and my own. This is what I attempted to do in *Anxiety*, as well as in earlier publications (e.g., Gray 1977). Robinson & Therrien, however, following in the traditions of their school (e.g., Kramis & Vanderwolf 1980), prefer to rely exclusively on their own data. Thus their comment makes no attempt to fit into their scheme of things (from which theta frequency is now excluded) the considerable accumulation of data from my own laboratory demonstrating that the effects of drugs, hormones, and lesions on septally elicited theta are highly frequency dependent (e.g., McNaughton et al. 1977; 1980; Gray, McNaughton, James & Kelly 1975; Mellanby, Gray, Quintero, Holt & McNaughton 1981; Drewett,

Gray, James, McNaughton, Valero & Dudderidge 1977; Valero, Stewart, McNaughton & Gray 1977). But why, I wonder, if we are now both using pharmacological criteria, should those based on cholinergic blockers be more significant than those based on other drugs?

Even within their own terms I find **Robinson & Therrien's** position hard to understand. They abandon theta frequency as a criterion of the two Vanderwolfian types of theta and present this as a move of old standing. Yet Vanderwolf and Robinson (1981) still state that "the atropine-sensitive and anaesthetic-sensitive components of RSA [rhythmic slow activity, i.e. theta] differ in mean frequency although there is some overlap in the two frequency distributions. In both rats and rabbits the RSA accompanying Type 1 behaviour [roughly, movement] ranges in frequency from about 7–12 Hz while the RSA which sometimes accompanies waking immobility or other Type 2 behaviour usually has a frequency of about 6 Hz." Somehow Robinson & Therrien find a major difference between this formulation and my own summary (*Anxiety*, p. 186):

"Movement" theta is resistant to systemic injections of antimuscarinic drugs (e.g., atropine, scopolamine), is abolished by anaesthetics (e.g. urethane, ether), and is normally of relatively high frequency (in the rat, more than 7–8 Hz). "Non-movement" theta is abolished by systemic treatment with antimuscarinics, it can occur in anaesthetized animals, and it is normally of relatively low frequency (in the rat, less than 7–8 Hz).

Notice the little word "normally." It means precisely what much of Robinson & Therrien's commentary says, that there are exceptions to these generalisations. If, however, Robinson & Therrien mean that there are now *no* generalisations to be made about theta frequency, then Vanderwolf and Robinson (1981) should have been more careful not to make them.

In fact, **Robinson & Therrien** would probably be willing to agree that *in general* atropine-sensitive theta is of lower frequency than atropine-insensitive theta. (If they are not willing to agree, they will need to account for the findings that once led some of us to believe that this was so; e.g., O'Keefe & Nadel 1978, Tables A6a and c.) Thus the critical question is whether the exceptions to the generalisations are an embarrassment for my interpretation of the function of theta. In other words, must I postulate *fixed* frequency bands for control- and checking-mode theta? I see no need to do so. The relevant discussion in *Anxiety* (pp. 292–93) is as follows:

The frequency of theta seen during this checking activity would be expected to vary with the rate at which items of information controlling the behaviour are received. This gives a natural explanation (similar to the one advanced, in the context of their own theory, by O'Keefe and Nadel 1978, p. 226) for the observed positive correlation between the frequency of movement-related theta and the speed of movement. . . . If, however, smooth motor programming is interrupted by a mismatch, lower-frequency theta (not related to movement) is generated, allowing slower, and thus more detailed analysis of the information passing at each "theta sweep" around the septo-hippocampal system.

There is nothing in this formulation that *precludes* the occurrence at times of high-frequency control-mode

theta, if the kinds of information processing occurring demand it; nor indeed the occurrence of low-frequency checking theta. Only experimental test, of course, can determine whether the actual exceptions to the (erst-while?) Vanderwolfian correlations between movement and theta-frequency can be predicted from this analysis; but if these correlations are to be of any use, some account of the exceptions to them will need to be given.

I have no quarrel with some of the particular points made by **Robinson & Therrien** in their demotion of theta frequency as a distinctive feature of atropine-resistant versus atropine-sensitive theta. Indeed, one of the observations they cite in support of this move was originally made in my own laboratory (McNaughton et al. 1977), namely, that the effect of systemic anticholinergics on septal driving of theta is insensitive to stimulation frequency but depends entirely on whether or not the animal is moving. We interpreted these observations, however, to mean that if there is a relation between theta frequency and Vanderwolf's distinction between cholinergic and noncholinergic theta, it depends on mechanisms afferent to the septal area. This is a possibility that Robinson & Therrien ignore; I return to it below.

It is easier to interpret the effects of septal stimulation (since much is known about the anatomy and physiology of septal control of theta) than those of stimulation at other sites. In particular, correlations between behavioural and electrophysiological effects simultaneously produced by electrical brain stimulation must be interpreted with caution. For example, Graeff et al. (1980) elicited behavioural immobility and theta from the same electrode (in the median raphe); but the former resulted from activation of a serotonergic pathway, the latter from activation of a cholinergic pathway. In the experiments cited by **Robinson & Therrien**, can one be sure that the simultaneously elicited high-frequency theta and immobility (e.g., from reticular sites) are not each attributable to activation of functionally independent pathways which simply happen both to lie near the electrode tip? Why should we rely on these observations, rather than on those, say, of Graeff et al. (1980), in which theta remained at 7.7 Hz or below and the animal was totally immobile in response to raphe stimulation? In the latter case, we showed considerable similarities between the electrophysiological, behavioural, and pharmacological profiles elicited by, on the one hand, raphe stimulation and, on the other, a stimulus associated with punishment (Graeff et al. 1980). This strengthens the relevance of these particular effects of brain stimulation to the theoretical framework developed in *Anxiety*. Thus I am not persuaded by the fact that reticular stimulation sometimes causes immobility together with 8–10 Hz theta that I need abandon the notion that 7.7-Hz theta is (usually) a limiting frequency for control-mode theta. If it is not, some other account must be offered that will link our pharmacological observations with those of Vanderwolf's group.

Somewhat surprisingly, **Robinson & Therrien** state that I gave "no discussion of why antianxiety drugs should selectively influence that frequency of theta which represents a transition from 'just checking' to 'control,'" that is, 7.7-Hz theta. As discussed exhaustively both in the *Précis* and in the book, the answer to this question is: because they reduce the noradrenergic input to the SHS.

A further argument pursued by **Robinson & Therrien** is that my theory cannot allow the hippocampus simultaneously to function in control and checking modes (which is true); and that the data demonstrate that hippocampal theta can simultaneously be of the atropine-sensitive and atropine-resistant types. If that demonstration had in fact been made, I would need to give up the hypothesis that these two types of theta correspond to the two modes of functioning which my theory postulates for the SHS. But the demonstration has not been made. Robinson & Therrien's arguments to the contrary depend upon a probably false assumption: namely, that the two types of theta depend upon separate cholinergic and noncholinergic inputs to the hippocampus.

Let me deal with the reasons for rejecting this assumption. There is good evidence that the medial septal projection to the hippocampus is cholinergic; and the loss of theta after septal lesions corresponds well with the loss of staining for acetylcholinesterase in the relevant portion of the hippocampus (Rawlins, Feldon & Gray 1979). It cannot be excluded that there is also a noncholinergic projection from the medial septal area to the hippocampus (Lynch, Rose & Gall 1978; C. Köhler, pers. comm.); but there is no evidence to suggest that, if it exists, it has anything to do with theta. Both Vanderwolf's types of theta are abolished by medial septal lesions, and both are accompanied by hippocampal release of acetylcholine (Dudar, Whishaw & Szerb 1979), as would be expected if both depend on activation of a single, cholinergic, final common path from septum to hippocampus. Furthermore, systemic atropine, which abolishes only immobility-related theta, has no effect on the intrahippocampal release of acetylcholine, whether this is elicited by movement or sensory stimulation (Dudar et al. 1979). I conclude, as do Szerb and Dudar (1981), that the difference between cholinergic and noncholinergic theta must depend on a cholinergic mechanism afferent to the septal area.

Although **Robinson & Therrien** are aware of these data (indeed, they cite them), they ignore their implications. Instead, they offer an analysis in which Vanderwolf's cholinergic theta is dependent on the septo-hippocampal cholinergic projection and atropine-resistant theta on a novel entorhinal-hippocampal projection. For the latter they rely on data obtained by Vanderwolf and Leung (in press) which I have not been able to consult. According to Robinson & Therrien, Vanderwolf and Leung have shown that lesions of the entorhinal cortex abolish atropine-resistant theta in rats. This may be so. But I doubt that the correct interpretation of their findings is that, to the well-established septal control of theta, we must now add a new, entorhinal control. Myhrer (1980) has recorded hippocampal theta after complete section of the angular bundle and it is unchanged; in particular, movement-related theta was clearly present after the lesion. Thus there must have been some aspect of Vanderwolf and Leung's procedure other than removal of the entorhinal-hippocampal projection that caused disruption of theta.

One possibility is that the septal projection cells themselves suffered retrograde degeneration. For Mitchell et al. (in press) have demonstrated (a) that there is a theta rhythm in the entorhinal cortex, (b) that it is controlled by the medial septal area, and (c) that the loss of entorhinal

theta after septal lesions correlates well with the loss of staining for acetyl-cholinesterase (as in the hippocampus). Their data, moreover, agree with a model according to which single septal theta-controlling cells send collaterals to both the hippocampus and the entorhinal cortex (Mitchell et al., in press); and C. Köhler (pers. comm.) has made anatomical observations that also agree with this model. Thus Vanderwolf and Leung's effects of entorhinal lesions may have arisen in consequence of retrograde degeneration in such bifurcating septal cells.

Whether or not this account of Vanderwolf and Leung's findings is correct, the balance of data is strongly in favour of the view that there is a single, cholinergic, final common pathway for theta, and that it originates in the medial septal area (Szerb & Dudar 1981). If the reader will now reexamine the section of Robinson & Therrien's commentary entitled "Are the two theta systems simultaneously active?" bearing this conclusion in mind, he will see that all the arguments by which they suggest that the answer to this question is "yes" fall to the ground.

The dorsal ascending noradrenergic bundle (DANB). My interpretation of the functions of the DANB met with general approval from Bohus, Gold & Fox, and Pribram & McGuinness. I am grateful to Bohus for drawing my attention to the data on NA turnover in the dentate gyrus which fit so well with this interpretation; and to Pribram & McGuinness for pointing out the similarities between my treatment of the ascending noradrenergic and serotonergic projections and their own (Pribram & McGuinness 1975). In contrast to these commentators, Zuckerman and Mason each offer alternative interpretations of the functions of the DANB: Zuckerman defends the view that this structure is concerned with reward processes (Stein 1978), Mason that it is concerned with selective attention (Mason & Iversen 1979). This is not the place to go again over the data that have served as the battleground for these differing views (see *Anxiety*, chapter 11; McNaughton & Mason 1980; Zuckerman, in press). I shall confine my remarks, therefore, to the specific points raised by Mason and Zuckerman.

Mason mounts a major attack on my theory of DANB function in part by the simple expedient of ignoring the actual structure of the theory. In this way he is able to cite as opposed to that theory data that are in fact irrelevant or even in some instances favourable to it. To begin with, he states that I incorporated "a role for the DANB in fear and anxiety as suggested initially by Redmond and associates (Redmond, Huang, Snyder & Maas 1976)." In fact, Redmond and I each proposed this hypothesis independently (see Gray et al. 1975; Gray 1976) and from quite separate data bases. This convergence is itself one source of strength for the hypothesis. I clarify this point not only in the interests of accuracy, but also because, by obscuring the origin of my own hypothesis, Mason is able to cite against it data that are not in fact in opposition to it. Thus he remarks that I ignore the evidence that lesions of the DANB do not impair one-way or two-way active avoidance. I do not ignore this evidence; it is carefully reviewed in *Anxiety*. It is simply not evidence against the theory. On the contrary, neither antianxiety drugs nor hippocampal lesions consistently impair one-way active avoidance, and both facilitate two-way active avoidance. These findings have been incorporated in my general

model of the BIS. Thus, if the DANB input to the SHS is an important component of the BIS, we must predict that lesions to this pathway will leave one-way avoidance unchanged and facilitate two-way avoidance. This is indeed the pattern of empirical findings (*Anxiety*, pp. 332–33), and Mason does not dispute this.

Other parts of Mason's attack are vitiated because he leaves out of account critical experimental parameters that, given the structure of the theory he is attacking, are known to be important. One example of this concerns the data on passive avoidance. Mason states roundly that this is not impaired by DANB lesions. The actual data are much more mixed (*Anxiety*, pp. 327–31). If we take into account the effects of septal and hippocampal lesions (*Anxiety*, pp. 119–32) and antianxiety drugs (*Anxiety*, pp. 20–22; Gray 1977), it is at once obvious that the great majority of studies have used an insensitive test, that is, step-down passive avoidance. We have recently used two tests that are, in contrast, sensitive to the effects of antianxiety drugs: suppression of bar pressing (1) in a variant of Geller and Seifter's (1960) conflict schedule, and (2) in Estes and Skinner's (1941) on-the-baseline conditioned suppression paradigm (see Leslie). In both paradigms we find a reliable release of response suppression in animals with lesions of the DANB produced by local injection of 6-hydroxydopamine (6-OHDA) (E. Tsaltas & Gray, in preparation; Response Figure 1). The result of the conditioned suppression experiment is in contrast to Mason and Fibiger's (1979e) failure to affect conditioned suppression. But they used an off-the-baseline paradigm which (*pace* Leslie) is usually less sensitive to antianxiety drug effects than is our on-the-baseline procedure (*Anxiety*, pp. 23–38).

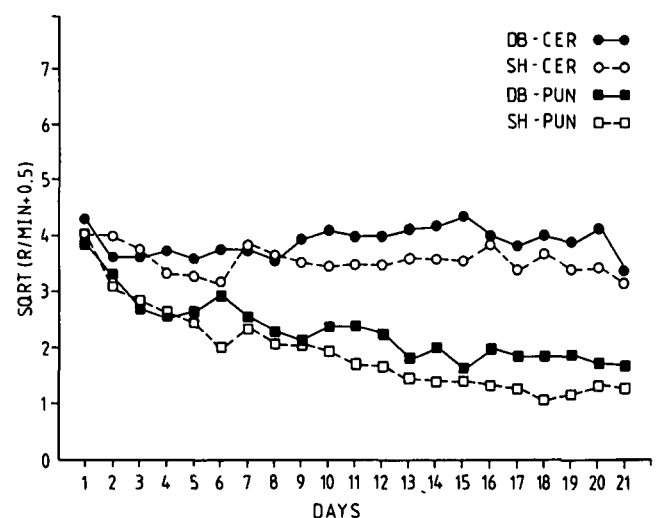


Figure 1. Development of conditioned suppression (CER) and punishment-produced (PUN) suppression of bar pressing in animals with lesions of the dorsal ascending noradrenergic bundle (DB) or sham operations (SH). Each curve is for a separate group of animals in the four experimental conditions, and displays response rates when footshock is signalled after these rates have been covaried on response rates during no-shock periods. The lesioned animals develop suppression of bar pressing more slowly (significant differences in linear trend) than controls in both CER ($P < 0.025$) and PUN ($P < 0.01$) conditions. This result is qualitatively similar to the effects of antianxiety drugs, though quantitatively smaller. From Tsaltas & Gray, in preparation.

A second example of Mason's omission of important experimental parameters lies in his discussion of the partial reinforcement extinction effect (PREE). He states that there have been three relevant experiments and that in only one of these did DANB lesions abolish the PREE in the manner predicted by the anxiety theory, namely by reducing resistance to extinction in the partial reinforcement condition as well as increasing resistance to extinction in the continuous reinforcement (CRF) condition. This is a travesty of the facts. Mason and Fibiger's (1978c) experiment had nothing at all to do with the PREE (see the detailed discussion of this experiment in Owen, Boarder, Gray & Fillenz 1982). There are three (not two) other experiments, all reported in detail by Owen et al. (1982). Two obtained the pattern of results predicted by the anxiety hypothesis of DANB function. (Note that this pattern is *not* predicted by the attentional hypothesis; according to this, the only way in which the PREE can be abolished is by increased resistance to extinction in the lesioned CRF condition – Mason & Iversen 1979.) The third found no effect of the DANB lesion on the PREE – but this negative result was *also* predicted by the anxiety hypothesis. Since this supposes that DANB lesions will affect the PREE in virtue of their deafferentation of the SHS, and since lesions of the septal area have been shown to block the PREE with 50 acquisition trials but not with 100 (Feldon & Gray 1979b), we predicted no effect of DANB lesions with 100 trials but blockade of the PREE with 50 trials. This is the result we obtained (Owen et al. 1982). It is consistent with the view that the DANB acts to *enhance* the animal's ability to adapt rapidly to change without necessarily providing *all* of the animal's capacity to adapt to change: a plausible view of the function of anxiety.

A further aspect of performance during extinction upon which Mason comments is the dorsal bundle extinction effect (DBEE), that is, the increased resistance to extinction initially reported by Mason and Iversen in DANB-lesioned rats trained on a CRF schedule of reward. On its own this finding is compatible with either the anxiety or the attentional hypotheses of DANB function (Mason & Iversen 1979). Mason states, however, that increased resistance to extinction is similarly seen with negative reinforcers, and that this finding is incompatible with the anxiety interpretation of the DBEE. But, although this argument is reasonable, the data cannot bear the interpretation Mason puts upon them. Most of the studies he cites are dealt with in *Anxiety* (pp. 340–41), to which the reader is referred. In addition, the Ashford and Jones (1976) study involved intraamygdaloid injection of 6-OHDA, which depleted dopamine (DA) as well as NA; this can hardly be relevant to a theory of the function of the NA input to the SHS.

A more substantial objection to the anxiety interpretation of the DBEE is the fact that the effect is not obtained if the lesion is made after acquisition (Mason 1979; Mason & Fibiger 1979b). On the anxiety view lesions to the DANB reduce responses to the omission of reward during extinction, so the DBEE should be obtained whether the lesion is made before or after acquisition (but see *Anxiety*, p. 353, for an alternative analysis). However, the experiments by Mason (1979) and Mason and Fibiger (1979b) did not control for state dependency; and the difference between the effects of the lesion made before and after

acquisition, respectively, depended on changes in the behaviour of the controls, not the lesioned animals. Thus, I am not persuaded by the published data that DANB lesions did in fact fail to alter resistance to extinction when made after acquisition. It is as likely that such an effect was obscured by other changes (see *Anxiety*, p. 340, for a more detailed discussion).

Mason raises a series of other points which I shall attempt to deal with briefly. Why (on the anxiety hypothesis), Mason asks, are DANB-lesioned rats more "distractable by irrelevant stimuli than controls?" Certainly, the anxiety hypothesis does not predict such an effect; if anything, it predicts the opposite. There is no evidence that such lesions reduce distractibility; so to that extent the anxiety hypothesis is weakened. But I do not agree that the existing data allow the conclusion that DANB lesions actually increase distractibility – they are far too inconsistent, and they can be fitted into the attentional hypothesis only by way of considerable post hoc acrobatics (*Anxiety*, pp. 334–6).

Mason also asks why lesions of the DANB impair the acquisition of visual and auditory discriminations. The answer, on the anxiety hypothesis, is because the relevant data have all been obtained in *successive* discriminations, which are impaired also by antianxiety drugs. I predict that, if the experiments were rerun using simultaneous discrimination procedures, the lesions would have no effect, just as they have no effect on simultaneous spatial discrimination. Reversal learning poses a more serious problem for the anxiety hypothesis, since this ought to be disrupted by DANB lesions. The impairment in the reversal of successive discriminations, cited by Mason, alas cannot be taken as evidence in favour of this prediction, since it could depend on the initial acquisition deficit in this paradigm; and the failure to obtain a reversal learning deficit when spatial (position) learning is used is a count against the anxiety hypothesis (*Anxiety*, p. 355). Other data that I accept as running counter to the anxiety hypothesis are the lack of effect of DANB lesions on Sidman bar-press avoidance (Mason & Fibiger 1979e) and in File's social interaction test of anxiety (Crow, Deakin, File, Longden & Wendlandt 1978). I am not impressed, however, by the report that DANB lesions fail to affect escape from a stimulus (S-) associated with nonreward (Mason & Iversen 1978a), any more than I am by the similar demonstration in the case of septal lesions (Henke 1976), since C. Preston has shown in my laboratory that, in intact rats, such behaviour is not normally controlled by the aversive properties of S-.

Finally, Mason cites several phenomena that provide direct support for the attentional hypothesis of DANB function. These experiments must certainly be taken seriously. But one of them, elimination of Kamin's blocking effect by DANB lesions (Lorden, Rickert, Dawson & Pellemounter 1979), can be accommodated within the theory developed in *Anxiety* (p. 284), though further specification of the exact role played by the noradrenergic input to the SHS will be required for this to be done. The same is probably true of the blockade by DANB lesions of latent inhibition (Mason & Fibiger 1979f;¹ see *Anxiety* pp. 259, 284, and 354). The requisite analysis would treat latent inhibition as analogous to the PREE (*Anxiety*, pp. 276–77). If so, blockade of latent inhibition by DANB lesions would presumably depend on the same param-

eters as blockade of the PREE (Owen et al. 1982). As we saw above, one important parameter in the latter case is the number of acquisition trials, as initially described for septal lesions by Henke (1974). E. Tsaltas is currently testing this prediction in my laboratory. The last specifically attentional phenomenon cited by Mason is the improvement in nonreversal shift produced by DANB lesions (Mason & Fibiger 1979f). I do not see how the anxiety hypothesis could handle this finding; but it is not clear that it can be replicated (Pisa & Fibiger 1980).

There is, it is clear, much about which Mason and I disagree; but we share the opinion that forebrain NA does not subserve reward (Mason & Iversen 1979). Zuckerman, however, follows Stein (1978) in claiming that it does. Of the points Zuckerman makes in defence of this claim I can see the attractions of the notion that NA release "works to arouse the organism from the behavioral inhibition characteristic of anxiety." But this notion must founder, I think, on the demonstration (Response Figure 1) that DANB lesions reduce punishment-produced and conditioned suppression of bar pressing (Tsaltas & Gray, in preparation). Zuckerman's hypothesis must surely predict the opposite. I can also see the force of the argument concerning the positively reinforcing effects of amphetamine and cocaine administration. But it is not possible to distinguish the roles of NA and dopamine (DA) respectively, in producing these effects; and there is at least as good a case to be made for DA as for NA as the "reward" transmitter (Wise 1978). The other matters that Zuckerman raises are best dealt with later, in the section on clinical issues.

The ascending serotonergic pathways. The data on the behavioural effects of DANB lesions and destruction of the ascending serotonergic pathways are used by Iversen to question the equivalence of stimuli associated with punishment and nonreward, respectively, postulated by the BIS model. But there is an error in her summary of these lesions effects. It is correct that DANB lesions have large effects on responses to nonreward and small (but not negligible: Response Figure 1) effects on responses to punishment; and it is correct that damage to ascending serotonergic fibres has a large disinhibitory effect on behaviour suppressed by punishment; but, contrary to Iversen's summary, there is little or no evidence of any effect of the latter lesion on responses to nonreward (N. M. Davis 1979; *Anxiety*, pp. 369–70). I agree with Iversen that this pattern of data is a real challenge to the parsimony of the punishment = nonreward hypothesis. But it is at least possible that the challenge can be met by the following postulates: (a) tasks involving punishment typically require more straightforward motor inhibition and less stimulus/response analysis than those involving nonreward; (b) the ascending serotonergic pathways facilitate motor inhibition; and (c) the DANB facilitates stimulus/response analysis. These postulates can handle the existing data, and they are relatively easy to test.

If the above analysis is correct, NA is a much more "cognitive" monoamine than serotonin. This might account for the fact, pointed out by Soubrié, that serotonin antagonists have never been claimed to be effective in the treatment of human anxiety, which typically has more to do with thinking than with freezing. In contrast, as pointed out by Gold & Fox (and see Redmond 1979),

there is much evidence that drugs that reduce NA function (e.g., clonidine) can alleviate anxiety. However, the specially useful feature of the benzodiazepines and barbiturates may consist precisely in the fact that they reduce both noradrenergic and serotonergic function and are thus able simultaneously to attack the cognitive *and* motor aspects of anxiety.

Areas not on the map: Opiates, hormones. Sharp limits between different systems exist neither in behaviour nor in the brain, but one must pretend that they do or analysis is impossible. Thus it is not surprising that several commentators remarked on the areas I left out of my map of anxiety. I have already noted Panksepp's concern about my neglect of the hypothalamus, and Ursin's and Pribram & McGuinness's about the amygdala; the last two authors also regret the absence of the basal ganglia and central adrenergic systems. I agree that the finished map must one day include these territories, or at least show the routes by which they communicate with the heartland of anxiety; but the same is true also of other structures, certainly, for example, the neocortex. My powers of analysis did not stretch so far, nor my year of sabbatical leave. But in any case it is probably premature to attempt to construct a truly complete map, since so much around the borders is still unknown.

One area, however, that can lay claim to inclusion in the very near future carries the flag "opiates." The relation between anxiety and the opiates, both exogenous and endogenous, has become much clearer in recent years owing to the work of Gold, Redmond, and their associates. I commented briefly on this work in *Anxiety* (pp. 361–63), and I welcome Gold & Fox's additional comments here. Pribram & McGuinness and Soubrié also gave reasons why the opiates should have figured more clearly on my map of anxiety, including the evidence that these agents act on the hippocampus. Note, however, that the phenomena reported by Zieglängsberger, French, Siggins & Bloom (1979), to which Soubrié refers, do not fit in any straightforward fashion with current views of how the anxiolytics work. These, as we have seen, generally facilitate GABA-ergic transmission, including that in the hippocampus (Haefely 1981). Yet Zieglängsberger et al. (1979) show that iontophoresis of opiates increases the firing of hippocampal pyramidal cells, and that this is probably due to the *blockade* of GABA-ergic inhibitory interneurons.

As an additional reason for including opiates among the agents that affect anxiety Soubrié points to the evidence that the opiate antagonist, naloxone, also antagonises some of the effects of the benzodiazepines. However, the only one of the relevant experiments (so far as I know) that used a behaviourally appropriate test of anxiety (as distinct from measures of food intake) was reported by Soubrié himself (Soubrié, Jobert & Thiébot 1980). Furthermore, in this experiment, although naloxone reversed the increase in punished bar pressing caused by diazepam, it did so only if the test situation included the punishing footshocks. If these were omitted, the disinhibitory effect of diazepam was not blocked by naloxone. Thus the apparent blockade of diazepam's antianxiety effect may in fact represent a direct increase by naloxone in the felt intensity of footshock. This interpretation would fit with older data that show that morphine does

not reduce the suppression of bar pressing in a Geller-Seifter schedule (Geller, Bachman & Seifter 1963; Kelleher & Morse 1964). Conversely, though the antianxiety drugs attenuate response suppression in the Geller-Seifter schedule, unlike opiates they do not reduce direct measures of pain. Thus, while I agree with Soubrié and Gold & Fox that there is likely to be a close link between the modes of action of the opiates and the classical anxiolytics, I do not yet see clearly what that link is.

The evidence from Gold's group pointing to a central role for opiate inhibition of the locus coeruleus in constituting that link is impressive; and I adopted in *Anxiety* (pp. 361–63) their hypothesis that the anxiety symptoms seen during opiate withdrawal reflect rebound hyperactivity in the projections of this nucleus. The additional suggestions made by Gold & Fox that the SHS may in some way activate endorphin-mediated inhibition of the locus coeruleus and so “turn itself off,” and that this process might be positively reinforcing, are of considerable interest and well worth following up. Note that they stand Stein's (1978; see Zuckerman) hypothesis on its head. Self-stimulation might now arise from *disruption* of activity in the locus coeruleus. This could account for the self-stimulation observed with electrodes implanted in this nucleus; but activation of such an electrode would now be seen as jamming rather than turning on normal locus coeruleus outputs.

Juxtaposition of the comments by Gold & Fox, Panksepp, and Zuckerman raises an important question. Are the panic attacks seen after opiate withdrawal, which are sensitive to clonidine treatment, the same as the panic attacks seen in agoraphobia, which are sensitive to tricyclic (but not benzodiazepine) treatment (Klein 1981)? If so, the combined psychopharmacology of these observations is difficult to comprehend; if not, we should purify the language of panic to remove the ambiguity.

Bohus and Pribram & McGuinness comment on my neglect of hormones and the central neuroregulators related to peripheral hormones. As with the opiates, I agree that this was a serious omission, but I did not (and still do not) see clearly where they fit in. Pribram & McGuinness point to the likely role of the adrenocortical hormones, especially since the hippocampus avidly binds corticosterone; but, as Bohus pertinently remarks, we have found it difficult to fit the behavioural effects of these agents into the behavioural scheme suggested by the effects of antianxiety drugs (Gray & Garrud 1977). I am grateful to Bohus for pointing to the way the data he and his associates have gathered on the effects of vasopressin “would fit excellently” my theory. While I was aware of some of these data (*Anxiety*, p. 330), I had not realised their full significance. The suggestion that the red nucleus may provide a way into the elusive area of motor control is an intriguing one. Indeed, all the issues raised in Bohus's commentary are well worth following up at both empirical and theoretical levels.

Clinical issues. Certain clinical issues, inevitably, have already come up in preceding sections. Additional comments follow.

Soubrié, Iversen, and Pribram & McGuinness comment favourably on my treatment of learned helplessness and the relation between depression and anxiety; and Pribram & McGuinness make the interesting suggestion

that individual differences in the susceptibility to anxiety or depression respectively (*Anxiety*, p. 402 *et seq.*) may be related to differences in the secretion of endogenous hormones or opiates.

Zuckerman, however, finds difficulties with my analysis of these issues. In particular, he takes the evidence (*Anxiety*, chapter 12) that learned helplessness is associated with NA depletion as indicating that “it is not an excess but a deficit in NA that underlies anxiety.” But this is so only if learned helplessness is a model of anxiety or the neurotic depression that is similar to anxiety. But, as argued in *Anxiety* (p. 405), a better case can be made for its being a model of psychotic depression, which is the polar opposite of anxiety. Zuckerman also finds a conflict between my analysis and the fact that anxiety neurotics do not “typically end up as depressives.” But I do not see why my analysis, postulating as it does that there are stable individual differences in the susceptibility of noradrenergic systems to functional exhaustion (*Anxiety*, p. 405), should predict the contrary. A more substantial objection is that “it is not typical that an endogenous depressive passes through a prolonged period of generalized anxiety before moving into depression.” I accept that my analysis predicts that this pattern of change should occur in some patients, though more probably in those who are *not* at the extreme psychotic pole of the neurotic-psychotic depressive continuum, who would be expected rather to pass rapidly into depression. But more clinical data than I know or Zuckerman cites would be needed to decide that such a development from prolonged anxiety to depression is in fact atypical. As to the evidence that tricyclics are effective in the treatment of panic attacks in agoraphobia (Klein 1981), I have already commented on the fact that these attacks are apparently different from other aspects of anxiety (and see Panksepp). In any case, the ambiguities concerning the neurochemical basis of the therapeutic effect of the tricyclics (*Anxiety*, pp. 399–401) are such that one cannot with any certainty infer from their action on panic attacks that these are due to decreased NA function rather than the reverse (as believed by Gold & Fox). The remaining datum cited by Zuckerman in support of the view that anxiety is due to a deficit in NA is the observation of negative correlations between measures of trait anxiety and NA metabolites in cerebrospinal fluid and plasma. This observation is certainly unexpected on my analysis; but such correlations are notoriously difficult to interpret.

Bohus comments on the possible role played by the interaction between vasopressin and central noradrenergic mechanisms in the maintenance of phobias and obsessive-compulsive rituals, given the evidence that this hormone facilitates the retention of both active and passive avoidance behaviour. This is an intriguing idea. My only reservation is that the theory developed in *Anxiety* (pp. 439–44) downgrades (though it does not eliminate) the role played in obsessional rituals by active avoidance behaviour.

Olton asks what role I see biofeedback as playing in the treatment of anxiety. I am not sufficiently familiar with this area to give a precise answer to the question. Certainly, there is every reason to suppose (as suggested by Gold & Fox) that feedback from peripheral autonomic manifestations of anxiety can itself maintain an anxiety state. Reduction in such feedback, for example, remains

the best account of the clinical effects of the beta-noradrenergic blocker, propranolol (*Anxiety*, p. 363). Even in rats, similar effects seem to occur. Thus P. Salmon, working in my laboratory, has shown that responding during the extinction component of a multiple variable-interval/extinction schedule of reinforcement (which is increased by antianxiety drugs) is similarly increased by both propranolol and a second beta-noradrenergic blocker, practolol, which barely crosses the blood-brain barrier. I would suppose, therefore, that biofeedback works in virtue of the reduction it produces in the peripheral manifestations of anxiety.

Personality. Personality was only briefly touched upon in the book; correspondingly, only Revelle devoted his commentary to this field. The parallels he draws between my treatment of the anxious personality and Atkinson and Birch's (1970; 1978) dynamics of action model are potentially of great importance. I welcome also his comment that those who work on human anxiety should consider extending their repertoire of behavioural (as distinct from questionnaire) methods in the light of the methods that have proved successful in studying anxiety in animals. Finally, I am grateful to Revelle for pointing out that Weiner and Schneider's (1971) findings dissolve the difficulty I had (*Anxiety*, p. 457) in accommodating the data on reversal learning in paired-associate tasks as a function of trait anxiety.

Both Pribam & McGuinness and Robinson & Therrien comment on the problem of sex differences, especially in the light of the fact that rats show a sex difference in the theta-driving curve (Drewett et al. 1977): The 7.7-Hz minimum which in male rats is abolished by antianxiety drugs is absent in the undrugged female rat, and this difference depends upon circulating testosterone. From this finding one can predict that, in general, female rats should be less anxious or fearful than males, as indeed they are (Gray 1971b). Robinson & Therrien regard this generalisation as "debatable." Certainly, it has been debated (Archer 1973); but I have recently answered previous objections, and it is now in my view firmly established (Gray 1979b). Robinson & Therrien take me to task for dealing with this sex difference in a "superficial" way, apparently because of the fact that the sex difference in human anxiety takes the opposite direction to that in rodents. But it is precisely because of this fact, which I pointed out myself (Gray 1971b), that I made no attempt to derive any particular inferences about the human sex difference from the rodent one. I am glad to see that the more recent references on the human sex difference cited by Robinson & Therrien continue to support my earlier conclusion. They should note that this is not the only type of behaviour in which it is difficult to generalise from rodent to human sex differences (Gray & Drewett 1977). Such changes in the direction of sex differences may reflect changes in the biologically based roles the two sexes play in the social organisation of different species (Gray & Buffery 1971).

Conclusion. The Précis drew fire at many different levels, from the most general to the most specific. This is not surprising, since the book it summarises is wide-ranging in its scope, yet offers at many points interpretations of

detailed bodies of data, interpretations often differing markedly from other strongly held points of view. Given the former characteristic, I was not surprised at the criticism that the book lacks predictive power (though I hope I showed this criticism to be ill-founded). Given the latter characteristic, it was to be expected that major attacks would be mounted by those who espouse the other points of view; and commentators such as Mason and Robinson & Therrien did not disappoint that expectation.

If there was a level of criticism I hoped for, but did not find, it lies between the most general and the most specific. *Anxiety* contains specific theoretical innovations which have implications for several different data areas (for example, the detailed working out of the way in which the septo-hippocampal system and the Papez circuit might jointly function as a comparator). These came in for surprisingly little direct comment (though see Gabriel and Woodruff). Conversely, if there was a kind of criticism I feared but was thankful to find largely absent, it was the reflex negative response to the idea that the septo-hippocampal system might have something to do with anxiety. Indeed, the commentators in general showed great willingness to confront and compare the ideas contained in the Précis with alternative hypotheses and other approaches to the functions of this mysterious part of the brain. In this, they followed what is already almost a tradition in the pages of this journal. For recent issues of BBS have shown clearly the vitality of the current debate about the septo-hippocampal system. It is a debate which augurs well for an eventual (though perhaps not speedy) resolution of the issues to which the Précis and many of the commentaries were addressed.

NOTE

1. Mason cites Mason and Lin (1980) as also demonstrating this; but, so far as I can judge, the data reported in this paper are identical to those reported in Mason and Fibiger (1979f), though there appears to be no acknowledgement of this fact in either paper.

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